Approximation and Representation

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2 Fourier Series

2.1 The Fourier Basis

Periodic and Trigonometric Functions

We begin by presenting the central definition of this ribbon:

Definition 2.1.1 (Periodic Functions)

The function $f\colon (a,b) \to \mathbb{R}$ is **periodic** with **period** $p \in \mathbb{R}$ if for any $x \in (a,b)$ and $n \in \mathbb{Z}$ such that $x + np \in (a,b)$, we have

$$f(x+np) = f(x).$$

Equivalently, f is periodic if

$$s - t = np \text{ for some } n \in \mathbb{N} \implies f(s) = f(t).$$

The most important periodic functions are sines and cosines — we have that

$$\sin(x) = \sin(x + 2\pi n) \text{ for any } x \in \mathbb{R},$$

so $\sin(x)$ is periodic on $\mathbb{R}$ with period $p = 2\pi$. Similarly, $\cos(x)$ is periodic on $\mathbb{R}$ since

$$\cos(x) = \cos(x + 2\pi n) \text{ for any } x \in \mathbb{R}.$$

Whenever we have a function f defined on a real interval (a,b), we're often interested in its *average value* — if $x_1, x_2, \dots, x_n \in (a,b)$, then the average value of $f(x)$ on (a,b) is approximately equal to

$$\frac{f(x_1) + \cdots + f(x_n)}{n}$$

This should become a better approximation whenever we take the increments $\Delta x_i := x_i - x_{i-1} \equiv \Delta x$ to be a constant for each i and let the number of such increments grow — in particular, if we multiply the above identically by $\Delta x/\Delta x$, then the average of f on (a,b) is equivalently

$$\frac{[f(x_1) + \cdots + f(x_n)]\Delta x}{n\Delta x}.$$

But now, since $x_1, \dots, x_n$ with $\Delta x_i = x_i - x_{i-1} \equiv \Delta x$ partitioning the interval (a, b) into n equal segments of length $(b-a)/n$, we have that

$$n\Delta x = n\left(\frac{b-a}{n}\right) = b - a;$$

this holds regardless of n and the common increment Δx. In particular, if we let $n \to \infty$ and $\Delta x \to 0$, the numerator

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x_i = \int_a^b f(x)\, dx,$$

and we have the following:

Definition 2.1.2 (Average Value of a Function)

Let $f\colon (a, b) \to \mathbb{R}$. Then, the

$$\text{Average of } f(x) \text{ on } (a, b) := \overline{f}_{(a,b)} = \frac{\int_a^b f(x)\, dx}{b-a}.$$

One immediate observation is that, if $f\colon (a, b) \to \mathbb{R}$ is a function for which

$$\int_a^b f(x)\, dx = 0,$$

then

$$\overline{f}_{(a,b)} = 0.$$

Now, connecting the two ideas we've discussed so far: if $f\colon (a, b) \to \mathbb{R}$ is a periodic function with period $p \in \mathbb{R}$, then if we denote the antiderivative of f by F and define

$$F_p(x) := \int_x^{x+p} f(t)\, dt,$$

the fundamental theorem of calculus gives

$$F_p'(x) = f(x+p) - f(x) = 0$$

since $f(x+p) = f(x)$ by periodicity of f. But notice that nowhere here did the argument $x \in (a, b)$ introduce a dependency; in particular, the above holds for any $x \in (a, b)$ such that $x + p$ is a valid point in the domain of the function (a, b).

We can recover a few more facts from this; first, since $s = x + kp$ and $t = x + (k+1)p$ implies $s - t = -p$, we have that $f(x + kp) = f(x + (k+1)p)$ for any valid $k \in \mathbb{Z}$, so by an inductive argument, we obtain

$$\int_x^{x+np} f(t)\, dt = \int_x^{x+p} f(t)\, dt + \int_{x+p}^{x+2p} f(t)\, dt + \cdots + \int_{x+(n-1)p}^{x+np} f(t)\, dt$$

$$= n \int_x^{x+p} f(t)\,dt.$$

Second, if g is a function such that

$$g(x) = f(x+c) \text{ for some } c \in \mathbb{R},$$

then we have

$$\int_x^{x+p} g(t)\,dt = \int_x^{x+p} f(t+c)\,dt = \int_{x+c}^{x+c+p} f(u)\,du = \int_x^{x+p} f(u)\,du.$$

In particular, what we have is that if g is a **phase-shifted** copy of the function f so that $g(x) = f(x+c)$ for any $x \in (a,b)$, then their integrals over any period are equal:

$$\int_x^{x+p} g(t)\,dt = \int_x^{x+p} f(t)\,dt.$$

We bundle our observations about periodic functions formally in the following theorem:

Theorem 2.1.3 (Integrals of Periodic Functions)

Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and periodic with period $p > 0$, so that

$$f(x) = f(x+kp), \quad \forall k \in \mathbb{Z}.$$

Then the following hold for all $a, b \in \mathbb{R}$, $x \in \mathbb{R}$, and integers $n \in \mathbb{Z}$:

(i) **(Translation of bounds)** Shifting the limits of integration by integer multiples of the period leaves the integral invariant:

$$\int_{a+np}^{b+np} f(t)\,dt = \int_a^b f(t)\,dt.$$

(ii) **(Independence of starting point)** Any integral of $f(x)$ over an interval of length p captures the same total area — in particular,

$$\forall x, y \in \mathbb{R}\colon \quad \int_x^{x+p} f(t)\,dt = \int_y^{y+p} f(t)\,dt = \int_0^p f(t)\,dt.$$

(iii) **(Scaling over periods)** Integrating over n consecutive periods captures the same area as n times the integral over a single period:

$$\int_x^{x+np} f(t)\,dt = n \int_0^p f(t)\,dt.$$

(iv) **(Stability of average value)** The average value of f over any interval spanning $n \neq 0$ complete periods is the same as the average value of f over a single period:

$$\overline{f}_{(x,x+np)} = \overline{f}_{(x,x+p)} = \overline{f}_{(0,p)};$$

or, in other words,

$$\frac{1}{np} \int_x^{x+np} f(t)\,dt = \frac{1}{p} \int_0^p f(t)\,dt.$$

(v) **(Symmetry and parity)** If furthermore we have that f is an **odd function** — i.e., $f(-t) = -f(t)$ for any $t \in \mathbb{R}$ — then the net area over any complete period is zero. In particular, we observe that

$$f \text{ is odd and periodic} \implies \int_{-p/2}^{p/2} f(t)\,dt = 0.$$

Proof:

For (i), letting $u = t - np$, we shift the integration variable by an integer number of periods. The limits of integration become a and b, respectively, yielding

$$\int_{a+np}^{b+np} f(t)\,dt = \int_a^b f(u+np)\,du = \int_a^b f(u)\,du,$$

where the final equality follows directly from the assumption that f is p-periodic.

For (ii), we utilize the additive property of definite integrals to split the domain at p:

$$\int_x^{x+p} f(t)\,dt = \int_x^p f(t)\,dt + \int_p^{x+p} f(t)\,dt.$$

Applying the translation property from (i) to the second integral shifts its bounds down by one period p, giving

$$\int_x^p f(t)\,dt + \int_0^x f(t)\,dt = \int_0^p f(t)\,dt.$$

That $\int_x^{x+p} f(t)\,dt = \int_y^{y+p} f(t)\,dt$ for any $x, y \in \mathbb{R}$ follows since the point $x \in \mathbb{R}$ was arbitrary in the argument above — in particular, both of those integrals equal $\int_0^p f(t)\,dt$.

For (iii), we handle the case where $n > 0$ by partitioning the interval $[x, x+np]$ into n contiguous periods:

$$\int_x^{x+np} f(t)\,dt = \sum_{k=1}^{n} \int_{x+(k-1)p}^{x+kp} f(t)\,dt.$$

By (ii), each integral in this sum equals $\int_0^p f(t)\,dt$, so the total area is exactly $n \int_0^p f(t)\,dt$. Considering the case when $n < 0$, let $m = -n > 0$. Reversing the limits of integration, we have that

$$\int_x^{x+np} f(t)\,dt = -\int_{x-mp}^{x} f(t)\,dt = -m \int_0^p f(t)\,dt = n \int_0^p f(t)\,dt.$$

The trivial case $n = 0$ evaluates to 0, confirming that

$$\forall n \in \mathbb{Z}\colon \quad \int_x^{x+np} f(t)\,dt = n \int_0^p f(t)\,dt.$$

Property (iv) is a direct consequence of (iii). Dividing both sides of the scaling property by the interval length np yields the stability of the average value:

$$\frac{1}{np} \int_x^{x+np} f(t)\,dt = \frac{1}{np} \left(n \int_0^p f(t)\,dt \right) = \frac{1}{p} \int_0^p f(t)\,dt.$$

For (v), we apply (ii) to center the integration window at the origin, allowing us to directly exploit the function's odd symmetry:

$$\int_x^{x+p} f(t)\,dt = \int_{-p/2}^{p/2} f(t)\,dt.$$

Splitting the integral at 0, we have

$$\int_{-p/2}^{p/2} f(t)\,dt = \int_{-p/2}^{0} f(t)\,dt + \int_0^{p/2} f(t)\,dt.$$

Substitute $u = -t$ into the first integral, meaning $du = -dt$. The bounds invert from $[-p/2, 0]$ to $[p/2, 0]$, so flipping them back using the negative introduced by the differential, we have

$$\int_{-p/2}^{0} f(t)\,dt = \int_{p/2}^{0} f(-u)(-du) = \int_0^{p/2} f(-u)\,du.$$

Because f is an odd function, $f(-u) = -f(u)$. The first integral is therefore the exact negative of the second:

$$\int_{-p/2}^{0} f(t)\,dt + \int_0^{p/2} f(t)\,dt = -\int_0^{p/2} f(t)\,dt + \int_0^{p/2} f(t)\,dt = 0.$$

Thus, for any periodic, odd function f with period $p > 0$, any integral over a complete period is zero:

$$\int_x^{x+p} f(t)\,dt = 0.$$

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As an example of the applications of the theorem above, consider the function $f(x) = \sin(x)$ for $x \in \mathbb{R}$. Then $\sin(x) = \sin(x + 2\pi k)$ for any $k \in \mathbb{Z}$, so by definition, $f(x) = \sin(x)$ is periodic on $\mathbb{R}$. By the theorem we just proved, we have that, for any $k \in \mathbb{Z}$, that

$$\forall x \in \mathbb{R}\colon \quad \int_x^{x+2\pi k} \sin(t)\,dt = k\int_0^{2\pi} \sin(t)\,dt = k[-\cos(t)]_0^{2\pi} = k[-1+1] = 0.$$

Of course, we could have realized that in addition to being periodic, we have that

$$\sin(-x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}(-x)^{2k+1} = -1 \times \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}x^{2k+1} = -\sin(x),$$

since for any $k \in \mathbb{N}$, $(-x)^{2k+1}$ is an odd power, so $(-x)^{2k+1} = -1 \times x^{2k+1}$. In particular, $\sin(x)$ is an odd function, so that

$$\int_x^{x+2\pi k} \sin(t)\,dt = \sum_{k=1}^{n} \int_{x+(k-1)p}^{x+kp} f(t)\,dt = 0,$$

since each of the integrals is equal to zero (cf. Part (v)).

What we also obtain from the analysis of $\sin(x)$ is a full suite of analogous results for $\cos(x)$. In particular, realize that these two functions are related via

$$\sin(x) = \cos\left(x - \frac{\pi}{2}\right), \quad x \in \mathbb{R}.$$

From this observation, we find that most of the results from the theorem hold automatically simply by the independence of starting point condition (ii). In particular, we have that

$$\int_x^{x+2\pi k} \cos(t)\,dt = \int_x^{x+2\pi k} \sin(t+\pi/2)\,dt = \int_{x+\pi/2}^{x+\pi/2+2\pi k} \sin(u)\,du = \int_0^{2\pi k} \sin(u)\,du = 0.$$

However, it is no longer the case that condition (v) from the theorem holds, since it is not the case that $\cos(x)$ is an odd function — we can observe directly that

$$\cos(-x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}(-x)^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^{2k} = \cos(x),$$

since $(-1)^{2k} = 1$ for every $k \in \mathbb{N}$. But this is not the same form as

$$\sin(-x) = -\sin(x);$$

in particular, we have the following.

Theorem 2.1.4 (Periodic Even Functions)

Suppose that a periodic function $g\colon \mathbb{R} \to \mathbb{R}$ with period $p > 0$ is **even**, so that

$$g(-x) = g(x), \quad x \in X.$$

Then

$$\int_{-p/2}^{p/2} g(t)\,dt = 2\int_0^{p/2} g(t)\,dt.$$

Proof: This follows almost immediately; we have that

$$\begin{aligned}
\int_{-p/2}^{p/2} g(t)\,dt &= \int_{-p/2}^{0} g(t)\,dt + \int_0^{p/2} g(t)\,dt \\
&= -\int_0^{-p/2} g(t)\,dt + \int_0^{p/2} g(t)\,dt \\
&= \int_0^{p/2} -g(-u)\,(-du) + \int_0^{p/2} g(t)\,dt && \Big(u := -t\Big) \\
&= \int_0^{p/2} g(u)\,du + \int_0^{p/2} g(t)\,dt && \Big(g(-u) = g(u) \text{ since } g \text{ is even}\Big) \\
&= 2\int_0^{p/2} g(t)\,dt,
\end{aligned}$$

since the labels on the indices u and t are arbitrary.

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Let us now apply these results to another set of functions — namely,

$$f_n(x) = \sin^2(nx), \quad g_n(x) = \cos^2(nx).$$

First, realize that both f_n and g_n are periodic functions with $p_n = \pi/n$ for each n. To see this, write

$$f_n(x) = \sin^2(nx) = \frac{1 - \cos(2nx)}{2};$$

since $\cos(2nx)$ has period $2\pi/2n = \pi/n$, it follows that $\sin^2(nx)$ is periodic with period $p_n = \pi/n$. The same holds for $\cos^2(nx)$ since

$$g_n(x) = \cos^2(nx) = \frac{1 + \cos(2nx)}{2}.$$

Next, observe that g_n is a *phase-shifted* copy of f_n for each n — in particular,

$$g_n(x) = \cos^2(nx) = \sin^2\left(nx + \frac{\pi}{2}\right) = \sin^2\left(n\left(x + \frac{\pi}{2n}\right)\right) = f_n\left(x + \frac{\pi}{2n}\right).$$

Thus, by the independence of starting point (cf. (ii) of the last theorem), the integrals of f_n and g_n over any complete period are equal: for each fixed $n \in \mathbb{N}$ and any $x \in \mathbb{R}$,

$$\int_x^{x+\pi/n} \cos^2(nt)\,dt = \int_x^{x+\pi/n} \sin^2(nt)\,dt.$$

In particular, if we let $x = -\pi/2n$, then

$$\int_{-\pi/2n}^{\pi/2n} \cos^2(nt)\,dt = \int_{-\pi/2n}^{\pi/2n} \sin^2(nt)\,dt.$$

In this way, it is clear that for each n, the interval $[-\pi, \pi]$ captures $2n$ periods of each function f_n and g_n. Since the integrals are equal over one such period, it follows that

$$\int_{-\pi}^{\pi} \cos^2(nt)\,dt = \int_{-\pi}^{\pi} \sin^2(nt)\,dt.$$

We use this observation to compute the integrals: in particular, since for each fixed n, the Pythagorean identity gives

$$\sin^2(nx) + \cos^2(nx) = 1,$$

we can integrate each side over the interval $[-\pi, \pi]$ to obtain

$$\int_{-\pi}^{\pi} (\sin^2(nt) + \cos^2(nt))\,dt = \int_{-\pi}^{\pi} 1\,dt = 2\pi.$$

But by linearity, we can expand the left side and use the equality of the integrals to write

$$\int_{-\pi}^{\pi} (\sin^2(nt) + \cos^2(nt))\,dt = \int_{-\pi}^{\pi} \sin^2(nt)\,dt + \int_{-\pi}^{\pi} \cos^2(nt)\,dt$$

$$= 2\int_{-\pi}^{\pi} \sin^2(nt)\,dt$$
$$\equiv 2\int_{-\pi}^{\pi} \cos^2(nt)\,dt.$$

Since the left side here is equal to 2π, it follows that

$$2\int_{-\pi}^{\pi} \sin^2(nt)\,dt = 2\pi \implies \int_{-\pi}^{\pi} \sin^2(nt)\,dt = \pi,$$

and similarly, that

$$2\int_{-\pi}^{\pi} \cos^2(nt)\,dt = 2\pi \implies \int_{-\pi}^{\pi} \cos^2(nt)\,dt = \pi.$$

Finally, since over the interval $[-\pi, \pi]$ spanning $2n$ complete periods of f_n and g_n we have

$$\frac{1}{2\pi}\cdot\int_{-\pi}^{\pi} f_n(t)\,dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} g_n(t)\,dt = \frac{1}{2},$$

it follows by the stability of the average value (iv) that on any interval spanning a whole number of complete periods,

$$\overline{f}_n = \overline{g}_n = \frac{1}{2}.$$

We give a generalization of the idea from the argument above as a theorem.

Theorem 2.1.5 (Integrals of $\sin^2(nx)$ and $\cos^2(nx)$)

Suppose that $(a, b) \subset \mathbb{R}$ is an interval for which $n(b-a)$, $n \in \mathbb{Z}$ is an integral multiple of π:

$$n(b-a) = k\pi \quad \text{for some} \quad k \in \mathbb{N}.$$

Then,

$$\int_a^b \sin^2(nt)\,dt = \int_a^b \cos^2(nt)\,dt = \frac{b-a}{2}.$$

Proof:

On (a, b), the functions $f_n(x) = \sin^2(nx)$ and $g_n(x) = \cos^2(nx)$ with periods $p_n = \pi/n$ complete $k \cdot p_n$ periodic revolutions on (a, b). In particular, since $n(b-a) = k\pi$ for some $k \in \mathbb{N}$,

$$kp_n = \frac{k\pi}{n} = \frac{n(b-a)}{n} = b - a.$$

so the interval (a, b) spans exactly k complete periods of both f_n and g_n. Since

$$n(b-a) = k\pi \implies b = a + \frac{k\pi}{n} = a + kp_n$$

and since $g_n(x) = f_n(x+\pi/(2n))$ is a phase-shifted copy of f_n, the independence of starting points (ii) gives

$$\int_a^{a+kp_n} \sin^2(nt)\,dt = \int_a^{a+kp_n} \cos^2(nt)\,dt \implies \int_a^b \sin^2(nt)\,dt = \int_a^b \cos^2(nt); \qquad b := a+kp_n.$$

By the Pythagorean identity, we have

$$\int_a^b \sin^2(nt)\,dt + \int_a^b \cos^2(nt)\,dt = \int_a^b 1\,dt.$$

The right side equals $b - a$, and since the integrals on the left are equal on the given interval,

$$\int_a^b \sin^2(nt)\,dt + \int_a^b \cos^2(nt)\,dt = 2\int_a^b \sin^2(nt)\,dt = 2\int_a^b \cos^2(nt).$$

Together, we have that for any interval $(a, b) \subset \mathbb{R}$ spanning some integral number of periods $b - a = kp_n$ of the functions $f_n(x) = \sin^2(nx)$ and $g_n(x) = \cos^2(nx)$,

$$\int_a^b \sin^2(nt)\,dt = \int_a^b \cos^2(nt)\,dt = \frac{b-a}{2}.$$

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The results derived above for $\sin^2(nx)$ and $\cos^2(nx)$ are in fact special cases of a more general question: what are the values of

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx, \quad \int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx, \quad \text{and} \quad \int_{-\pi}^{\pi} \cos(mx)\sin(nx)\,dx?$$

Whenever $n = m$ in the first two, we recover the familiar $\sin(nx)\sin(nx) = \sin^2(nx)$ and $\cos(nx)\cos(nx) = \cos^2(nx)$. But what about when $n \neq m$, and what about the integrals of the cross-terms $\cos(mx)\sin(nx)$?

It is a bit trickier to handle these integrals; to do so, we first need some additional trigonometric machinery.

Theorem 2.1.6 (Angle Addition)

For any $\alpha, \beta \in \mathbb{R}$, we have that

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta, \tag{A1}$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta. \tag{A2}$$

Proof:

We derive all four identities (two for each (A1) and (A2) based on the sign inside $\pm/\mp$) from the formula for $\cos(\alpha-\beta)$. Start by considering two points on the unit circle,

$$P = (\cos\alpha, \sin\alpha), \qquad Q = (\cos\beta, \sin\beta).$$

We can compute the squared (Euclidean) distance between P and Q in two ways, which we will then equate and use to solve for the desired expression. First, we have that in Cartesian coordinates,

$$|PQ| = \sqrt{(P_x - Q_x)^2 + (P_y - Q_y)^2} \implies |PQ|^2 = (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2.$$

Expanding the right,

$$(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = (\cos^2\alpha + \cos^2\beta - 2\cos\alpha\cos\beta) + (\sin^2\alpha + \sin^2\beta - 2\sin\alpha\sin\beta),$$

or

$$(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = (\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta).$$

Each of the $\cos^2\alpha + \sin^2\alpha = \cos^2\beta + \sin^2\beta = 1$ by the Pythagorean identity, so

$$|PQ|^2 = 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta).$$

Second, we can compute $|PQ|^2$ using the **law of cosines**: recall that for a triangle with sides a, b, c and angle C opposite side c, the law of cosines states

$$c^2 = a^2 + b^2 - 2ab\cos C. \tag{Law of Cosines}$$

Now, consider the triangle OPQ, where $O = (0,0)$ is the origin. Both $P = (\cos\alpha, \sin\alpha)$ and $Q = (\cos\beta, \sin\beta)$ lie on the unit circle, so the two sides emanating from the origin have

$$|OP| = |OQ| = 1.$$

The angle $\angle POQ = \angle QOP$ at the origin O subtended by the arc from Q to P is $\alpha - \beta$ (the difference between the angles that P and Q make with the positive x-axis). Applying the law of cosines with

$$a := |OP| = 1, \quad b = |OQ| = 1, \quad C = \alpha - \beta,$$

we obtain

$$|PQ|^2 = 1^2 + 1^2 - 2(1)(1)\cos(\alpha - \beta) = 2 - 2\cos(\alpha - \beta).$$

Equating the two expressions for $|PQ|^2$,

$$|PQ|^2 = 2[1 - \cos\alpha\cos\beta - \sin\alpha\sin\beta] = 2[1 - \cos(\alpha - \beta)],$$

and simplifying by dividing out the factor of 2, subtracting the factor of 1, and finally dividing by -1 gives the desired result:

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

To derive $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$, replace β with $-\beta$ in the identity we just showed — since $\cos(-\beta) = \cos\beta$ and $\sin(-\beta) = -\sin\beta$, we have that

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

Finally, to derive the analogous expressions for $\sin(\alpha \pm \beta)$, use the co-function identity

$$\sin\theta = \cos(\pi/2 - \theta), \quad \theta \in \mathbb{R}$$

to write

$$\sin(\alpha+\beta) = \cos\left(\frac{\pi}{2} - \alpha - \beta\right) = \cos\left(\left(\frac{\pi}{2} - \alpha\right) - \beta\right),$$

so that

$$\sin(\alpha+\beta) = \cos\left(\left(\frac{\pi}{2} - \alpha\right) - \beta\right) = \cos\left(\frac{\pi}{2} - \alpha\right)\cos\beta + \sin\left(\frac{\pi}{2} - \alpha\right)\sin\beta = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

after re-applying the co-function identity. Replacing β by $-\beta$ yields the final identity,

$$\sin(\alpha-\beta) = \sin\alpha\cos(-\beta) + \cos\alpha\sin(-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta,$$

again since $\cos(-x) = \cos(x)$ is even and $\sin(-x) = -\sin(x)$ is odd.

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Corollary 2.1.7 (Double-Angle and Power-Reduction Formulas)

Setting $\alpha = \beta$ in the angle addition formulas yields the **double-angle formulas**: for any $\theta \in \mathbb{R}$,

$$\sin(2\theta) = 2\sin\theta\cos\theta,$$

and, all equivalently, that

$$\begin{aligned}\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta.\end{aligned}$$

Solving for $\sin^2(\theta)$ and $\cos^2(\theta)$ in the double-angle formula for $\cos(2\theta)$, we obtain the **power-reduction formulas**:

$$\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}, \qquad \cos^2(\theta) = \frac{1+\cos(2\theta)}{2}.$$

Using the angle addition formulas, we can quickly prove a series of other useful trigonometric identities:

Theorem 2.1.8 (Sum-Product and Product-Sum Formulas)

For any $\alpha, \beta \in \mathbb{R}$, the following **product-to-sum** identities hold:

$$\cos\alpha\cos\beta = \frac{\cos(\alpha-\beta)+\cos(\alpha+\beta)}{2}, \tag{P1}$$

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta)-\cos(\alpha+\beta)}{2}, \tag{P2}$$

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}, \tag{P3}$$

$$\cos\alpha\sin\beta = \frac{\sin(\alpha+\beta)-\sin(\alpha-\beta)}{2}. \tag{P4}$$

Equivalently, for any $u, v \in \mathbb{R}$, the following **sum-to-product** identities hold:

$$\cos u + \cos v = 2\cos\left(\tfrac{u+v}{2}\right)\cos\left(\tfrac{u-v}{2}\right), \tag{S1}$$

$$\cos u - \cos v = -2\sin\left(\tfrac{u+v}{2}\right)\sin\left(\tfrac{u-v}{2}\right), \tag{S2}$$

$$\sin u + \sin v = 2\sin\left(\tfrac{u+v}{2}\right)\cos\left(\tfrac{u-v}{2}\right), \tag{S3}$$

$$\sin u - \sin v = 2\cos\left(\tfrac{u+v}{2}\right)\sin\left(\tfrac{u-v}{2}\right). \tag{S4}$$

Proof:

The product-to-sum identities follow from the angle addition formulas. For (P1), add the two cosine addition formulas in (A1) to obtain

$$\cos(\alpha-\beta)+\cos(\alpha+\beta) = [\cos\alpha\cos\beta+\sin\alpha\sin\beta]+[\cos\alpha\cos\beta-\sin\alpha\sin\beta] = 2\cos\alpha\cos\beta.$$

Dividing by 2 gives (P1).

For (P2), subtract instead:

$$\cos(\alpha-\beta)-\cos(\alpha+\beta) = [\cos\alpha\cos\beta+\sin\alpha\sin\beta]-[\cos\alpha\cos\beta-\sin\alpha\sin\beta] = 2\sin\alpha\sin\beta.$$

For (P3), add the two sine addition formulas:

$$\sin(\alpha+\beta)+\sin(\alpha-\beta) = [\sin\alpha\cos\beta+\cos\alpha\sin\beta]+[\sin\alpha\cos\beta-\cos\alpha\sin\beta] = 2\sin\alpha\cos\beta.$$

For (P4), subtract:

$$\sin(\alpha+\beta)-\sin(\alpha-\beta) = [\sin\alpha\cos\beta+\cos\alpha\sin\beta]-[\sin\alpha\cos\beta-\cos\alpha\sin\beta] = 2\cos\alpha\sin\beta.$$

The sum-to-product identities are simply the product-to-sum identities in reverse. Substituting

$$\alpha = \frac{u+v}{2}, \qquad \beta = \frac{u-v}{2},$$

so that $\alpha+\beta = u$ and $\alpha-\beta = v$ into each of (P1)-(P3) yields (S1)-(S4). For instance, (P1) becomes

$$\cos\left(\tfrac{u+v}{2}\right)\cos\left(\tfrac{u-v}{2}\right) = \frac{\cos u+\cos v}{2}$$

which, after multiplying by 2, is (S1).

$\|$

Notice that, in our analysis of $\sin^2(nx)$ and $\cos^2(nx)$, we never evaluated a single integral directly. In particular, the same analysis can be completed by a more involved argument where we use the power-reduction formula for $\sin^2(x)$ to write

$$\sin^2(nt) = \frac{1-\cos(2nt)}{2} \implies \int_{-\pi}^{\pi} \sin^2(nt)\,dt = \int_{-\pi}^{\pi} \frac{1}{2}\,dt - \frac{1}{2}\int_{-\pi}^{\pi} \cos(2nt)\,dt.$$

The second integral vanishes since $\int_{-\pi}^{\pi}\cos(2nt)\,dt = \frac{1}{2n}[\sin(2nt)]_{-\pi}^{\pi} = 0$. This leaves

$$\int_{-\pi}^{\pi} \sin^2(nt)\,dt = \int_{-\pi}^{\pi} \frac{1}{2}\,dt = \frac{2\pi}{2} = \pi,$$

which agrees with the result we obtained by evaluating the periodic and odd/even properties of $\sin(x)$ and $\cos(x)$. Nevertheless, the former argument is perhaps more revealing for our particular applications, for which we will now begin our pursuit.

2.1.1 Fourier Coefficients and Partial Sums

Computing Fourier Series

As alluded to in the previous ribbon, the current interest is in evaluating integrals of the form

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx, \quad \int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx, \quad \text{and} \quad \int_{-\pi}^{\pi} \cos(mx)\sin(nx)\,dx.$$

The product-to-sum formulas from the previous section allow us to reduce each of these products to sums of sines and cosines of integer multiples of x. In particular, realize that we already know how such functions behave under integration over $[-\pi,\pi]$: for any nonzero integer $k \in \mathbb{Z}$,

$$\int_{-\pi}^{\pi} \sin(kx)\,dx = -\frac{1}{k}\cos(kx)\Big|_{-\pi}^{\pi} = 0,$$

since we're evaluating the integral over the length of a complete period. Furthermore, when $k = 0$, we have

$$\int_{-\pi}^{\pi} \sin(0)\,dx = \int_{-\pi}^{\pi} 0\,dx = 0,$$

so that

$$\forall k \in \mathbb{Z}\colon \quad \int_{-\pi}^{\pi} \sin(kx)\,dx = 0. \tag{I1}$$

The analogous procedure for cosines reveals that

$$\int_{-\pi}^{\pi} \cos(kx)\,dx = \begin{cases} 2\pi & k = 0, \\ 0 & k \neq 0, \end{cases} \tag{I2}$$

where the case when $k = 0$ follows because $\cos(0 \cdot x) = 1$ for all $x \in [-\pi,\pi]$. (Exercise: compute the values of the integrals above without performing any actual integration — in particular, argue from the perspective of the periodicity and evenness/oddness of $\cos(x)$ and $\sin(x)$, respectively.)

We're now prepared to compute the integrals of interest.

Theorem 2.1.9 (Sine-Cosine Orthogonality Relations)

For nonnegative integers $m, n \geq 0$, we have that

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = \begin{cases} 0 & \text{if } m \neq n \text{ or if } m = n = 0, \\ \pi & \text{if } m = n \neq 0; \end{cases} \tag{I3}$$

that

$$\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = \begin{cases} 0 & \text{if } m \neq n, \\ \pi & \text{if } m = n \neq 0, \\ 2\pi & \text{if } m = n = 0; \end{cases} \tag{I4}$$

and that

$$\int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,dx = 0 \quad \forall m, n \geq 0. \tag{I5}$$

Proof:

We proceed by evaluating each integral in order; we begin with (I3). First, recall the sine power-reduction formula (P2), which says that for any $\alpha, \beta \in \mathbb{R}$,

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta) - \cos(\alpha+\beta)}{2} = \frac{1}{2}\cos(\alpha-\beta) - \frac{1}{2}\cos(\alpha+\beta).$$

Working with the exposed linearity of the latter form, take $\alpha = mx$ and $\beta = nx$ in the above and integrate with respect to x on the interval $[-\pi, \pi]$:

$$\begin{aligned} \int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx &= \frac{1}{2}\int_{-\pi}^{\pi} \cos(mx - nx)\,dx - \frac{1}{2}\int_{-\pi}^{\pi} \cos(mx + nx)\,dx \\ &= \frac{1}{2}\int_{-\pi}^{\pi} \cos\big((m-n)x\big)\,dx - \frac{1}{2}\int_{-\pi}^{\pi} \cos\big((m+n)x\big)\,dx. \end{aligned}$$

Whenever $m \neq n \geq 0$, $m - n \neq 0$ and $m + n \neq 0$, and by our result in (I2) both of the cosine integrals vanish. Therefore,

$$m \neq n\colon \quad \int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = 0.$$

If $m = n = 0$, then the integral is trivially $\int_{-\pi}^{\pi} \sin(0)\sin(0)\,dx = 0$. On the other hand, if $m = n \geq 1$, then $m - n = 0$ while $m + n = 2m \neq 0$, so starting from the lattermost form of the integral above,

$$\frac{1}{2}\int_{-\pi}^{\pi} \cos\big((m-n)x\big)\,dx - \frac{1}{2}\int_{-\pi}^{\pi} \cos\big((m+n)x\big)\,dx = \frac{1}{2}\int_{-\pi}^{\pi} 1\,dx - \frac{1}{2}\int_{-\pi}^{\pi} \cos\big(2mx\big)\,dx = \frac{1}{2}\cdot 2\pi - \frac{1}{2}\cdot 0 = \pi,$$

meaning that

$$\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = \pi.$$

(Of course, this also follows immediately from our previous discussion upon observing that $\int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,dx = \int_{-\pi}^{\pi} \sin^2(mx)\,dx$ when $m = n \geq 1$; nevertheless, we wanted to demonstrate this alternative method for establishing the same result.)

Now let us show the cosine identities in (I4). Recall the cosine power-reduction identity (P1),

$$\cos\alpha\cos\beta = \frac{\cos(\alpha-\beta)+\cos(\alpha+\beta)}{2} = \frac{1}{2}\cos(\alpha-\beta) + \frac{1}{2}\cos(\alpha+\beta).$$

Proceeding entirely analogously to the argument above, write

$$\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = \frac{1}{2}\int_{-\pi}^{\pi} \cos\Big((m-n)x\Big)\,dx + \frac{1}{2}\int_{-\pi}^{\pi} \cos\Big((m+n)x\Big)\,dx.$$

If $m \neq n \geq 0$, both $m - n \neq 0$ and $m + n \neq 0$, so that both of the cosine integrals vanish by (I2):

$$m \neq n \geq 0\colon \quad \int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = 0.$$

(notice that this is *exactly* the same as what happened in the sine case). If $m = n = 0$, then $m - n = 0$ and $m + n = 0$, so both of the integrands are identically $\cos(0) = 1$; in particular,

$$m = n = 0\colon \quad \int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = \frac{1}{2}\cdot 2\pi + \frac{1}{2}\cdot 2\pi = 2\pi.$$

If $m = n \neq 0$, then $m - n = 0$ while $m + n \neq 0$, so that

$$m = n \neq 0\colon \quad \int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,dx = \frac{1}{2}\cdot 2\pi + \frac{1}{2}\cdot 0 = \pi.$$

This establishes all three parts of (I4).

For (I5), we use the sine-cosine power-reduction identity (P3):

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2} = \frac{1}{2}\sin(\alpha+\beta) + \frac{1}{2}\sin(\alpha-\beta).$$

As in the above,

$$\int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,dx = \frac{1}{2}\int_{-\pi}^{\pi} \sin\Big((m+n)x\Big)\,dx + \frac{1}{2}\int_{-\pi}^{\pi} \sin\Big((m-n)x\Big)\,dx.$$

For any integers $m, n \geq 0$, each integrand on the right is $\sin(kx)$ for some integer k. But by (I1), these integrals are zero for any $k \in \mathbb{Z}$ — in particular, both of the integrals on the right vanish for any $m, n \in \mathbb{Z}$, so that

$$\forall m, n \geq 0\colon \quad \int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,dx = 0.$$

‖

Let us now take a step back and review exactly what the theorem above gives us: given the infinite list of functions taking the form $\cos(nx)$ or $\sin(nx)$ for some nonnegative integer $n \geq 0$ — i.e., those functions of the form

$$\mathcal{T} := \{1,\ \cos(x),\ \sin(x),\ \cos(2x),\ \sin(2x),\ \cos(3x),\ \sin(3x),\ \ldots\ \}$$

what we *know* to be true is that, if we take *any two distinct* members of $\mathcal{T}$, multiply them together, and integrate their product along $[-\pi, \pi]$, the integral will vanish no matter what — for instance, the integrals of

$$\cos(mx)\cos(nx), \quad \sin(mx)\sin(nx), \quad \cos(mx)\sin(nx), \quad \cos(mx)\cdot 1, \quad \sin(nx)\cdot 1,$$

and so on, all vanish. In particular, the *only* integrals resembling this form that survive are the ones where we multiply a function by itself — in such a case, we get that

$$\int_{-\pi}^{\pi} 0\cdot 0\,dx = 0,$$

(cf. $\sin(0)\sin(0) = 0$), that

$$\int_{-\pi}^{\pi} 1\cdot 1\,dx = 2\pi,$$

(cf. $\cos(0)\cos(0) = 1$), and that

$$m = n \geq 1\colon \quad \int_{-\pi}^{\pi} \cos^2(nx)\,dx = \int_{-\pi}^{\pi} \sin^2(nx) = \pi.$$

The situation may begin to feel familiar, perhaps with some additional prodding. Recall from linear algebra that two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** (or **perpendicular**, in which case we occasionally write $\mathbf{u} \perp \mathbf{v}$) if their inner product vanishes:

$$\mathbf{u} \perp \mathbf{v} \iff \langle \mathbf{u}, \mathbf{v}\rangle_{L_2} := \sum_{i=1}^{n} u_i v_i = 0.$$

(We use the notation $\langle\cdot,\cdot\rangle_{\mathbb{R}^n}$ with $\mathbb{R}^n$ in the subscript as shorthand for denoting the Euclidean inner product).

Now, try picking two arbitrary *distinct* members from $\mathcal{T}$ and label them ψ and φ. What the orthogonality relations are saying is that

$$\int_{-\pi}^{\pi} \psi(x)\varphi(x)\,dx = 0.$$

To see how this is analogous to Euclidean orthogonality, imagine sampling both functions at representative points $x_1, \ldots x_N \in [-\pi, \pi]$ with common spacing $\Delta x = 2\pi/N$. Then the integral is, by definition, the limit of the Riemann sum

$$\int_{-\pi}^{\pi} \psi(x)\varphi(x)\,dx = \lim_{N\to\infty} \sum_{i=1}^{N} \psi(x_i)\varphi(x_i)\Delta x.$$

But before taking the limit, look at the sum itself: if we form the pointwise Euclidean vectors

$$\boldsymbol{\psi} = \begin{pmatrix} \psi(x_1) \\ \vdots \\ \psi(x_N) \end{pmatrix} \in \mathbb{R}^N, \qquad \boldsymbol{\varphi} = \begin{pmatrix} \varphi(x_1) \\ \vdots \\ \varphi(x_N) \end{pmatrix} \in \mathbb{R}^N,$$

then

$$\sum_{i=1}^{N} \psi(x_i)\varphi(x_i)\Delta x = \Delta x \sum_{i=1}^{N} \psi(x_i)\varphi(x_i) = \frac{2\pi}{N}\langle \boldsymbol{\psi}, \boldsymbol{\varphi}\rangle_{\mathbb{R}^N}.$$

That is, the Riemann sum defining the integral is, up to a positive scaling factor $\Delta x = 2\pi/N$, the Euclidean inner product of $\boldsymbol{\psi}$ and $\boldsymbol{\varphi}$ in $\mathbb{R}^n$.:

$$\int_{-\pi}^{\pi} \psi(x)\varphi(x)\,dx = \lim_{N\to\infty} \frac{2\pi}{N} \langle \boldsymbol{\psi}, \boldsymbol{\varphi} \rangle_{\mathbb{R}^N}.$$

Now, in this light, consider what it means for $\psi, \varphi \in \mathcal{T}$ to imply

$$\int_{-\pi}^{\pi} \psi(x)\varphi(x)\,dx = 0.$$

From the expression above, this means that

$$\lim_{N\to\infty} \frac{2\pi}{N} \langle \boldsymbol{\psi}, \boldsymbol{\varphi} \rangle_{\mathbb{R}^N} = 0,$$

and since $2\pi/N > 0$ for every $N \in \mathbb{N}$, this is equivalent to saying that

$$\langle \boldsymbol{\psi}, \boldsymbol{\varphi} \rangle_{\mathbb{R}^N} = \sum_{i=1}^{N} \psi(x_i)\varphi(x_i) = 0 \to 0 \quad \text{as} \quad N \to \infty.$$

In other words, the sampled vectors $\boldsymbol{\psi}$ and $\boldsymbol{\varphi}$ from $\mathcal{T}$ become *orthogonal in* $\mathbb{R}^N$ as we sample finer and finer partitions $(x_1, x_2, \ldots)$ of $[-\pi, \pi]$.

All of these considerations motivate the following definition.

Definition 2.1.10 (The Space L^2)

Let $L^2[-\pi, \pi]$ denote the space of all real-valued functions $f\colon\ [-\pi, \pi] \to \mathbb{R}$ that are **square-integrable** on $[-\pi, \pi]$ — that is, for which

$$\int_{-\pi}^{\pi} [f(x)]^2\,dx < \infty.$$

For $f, g \in L^2[-\pi, \pi]$, define the L^2 **inner product**

$$\langle f, g \rangle_{L^2} := \int_{-\pi}^{\pi} f(x)g(x)\,dx,$$

and the corresponding L^2 **norm**

$$\|f\|_{L^2} := \sqrt{\langle f, f \rangle} = \left(\int_{-\pi}^{\pi} [f(x)]^2\,dx \right)^{1/2}.$$

Then, equipped with the L^2 inner product, the space $L^2[-\pi, \pi]$ is a **Hilbert space** — an **inner product space** that is **complete** (recall: every Cauchy sequence in $L^2[-\pi, \pi]$ converges to a limit in the same space) in the metric

$$d(f, g) := \|f - g\|_{L^2}.$$

It is not hard to see that the notion of $L^2[-\pi, \pi]$ can be extended relative to any bounded interval $[a, b] \subset \mathbb{R}$ just by reconciling the square-integrability condition as appropriate.

What we intuitively imagine is that L^2 spaces are the natural infinite-dimensional analogues of Euclidean spaces $\mathbb{R}^n$ in the limit as $n \to \infty$. The crucial feature of $L^2[-\pi, \pi]$ in the current context is its inner product: it gives us a rigorous notion of defining *orthogonality between functions*, and the orthogonality relations tell us that

$$\mathcal{T} := \{1,\ \cos(x),\ \sin(x),\ \cos(2x),\ \sin(2x),\ \cos(3x),\ \sin(3x),\ \dots\}$$

is an **orthogonal set** in $L^2[-\pi, \pi]$:

$$\psi \neq \varphi \in \mathcal{T} \implies \langle \psi, \varphi \rangle_{L^2} = 0,$$

with squared norms

$$\|1\|_{L^2}^2 = \int_{-\pi}^{\pi} 1^2\, dx = 2\pi; \qquad \|\cos(nx)\|_{L^2}^2 = \|\sin(nx)\|_{L^2}^2 = \pi, \quad n \geq 1.$$

In linear algebra, we gravitate toward orthogonal bases of $\mathbb{R}^n$ — in particular, if $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is an orthogonal basis for $\mathbb{R}^n$, then any $\mathbf{v} \in \mathbb{R}^n$ can be decomposed as a linear combination of the basis vectors in $\mathcal{B}$:

$$\mathbf{v} = \sum_{i=1}^{n} c_i \mathbf{e}_i \quad \text{for some} \quad (c_1, \dots, c_n) \in \mathbb{R}^n.$$

Supposing we're interested in extracting the contribution c_k of the kth basis vector $\mathbf{e}_k$ to the vector $\mathbf{v}$, we can use orthogonality: we have that $\langle \mathbf{e}_i, \mathbf{e}_j \rangle_{\mathbb{R}^n} = 0$ for $i \neq j$, so only the kth term survives when we take the Euclidean inner product of both sides of the above with $\mathbf{e}_k$:

$$\langle \mathbf{v}, \mathbf{e}_k \rangle_{\mathbb{R}^n} = \langle c^\top \mathbf{e}, \mathbf{e}_k \rangle_{\mathbb{R}^n} = \left[\sum_{i=1}^{n} c_i \mathbf{e}_i^\top \right] \mathbf{e}_k = \sum_{i=1}^{n} c_i \mathbf{e}_i^\top \mathbf{e}_k = c_k \|\mathbf{e}_k\|_{\mathbb{R}^n}^2.$$

Dividing the scalar $\|\mathbf{e}_k\|^2 > 0$ through to both sides, this gives

$$c_k = \frac{\langle \mathbf{v}, \mathbf{e}_k \rangle}{\|\mathbf{e}_k\|^2},$$

where the inner product and norm are Euclidean.

Suppose, then, that a general square-integrable function $f \in L^2[-\pi, \pi]$ can be decomposed in the analogous way as linear combinations of members in $\mathcal{T}$. If this were possible, we'd have

$$f(x) = \frac{a_0}{2} \cdot 1 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)).$$

(The factor of $\frac{1}{2}$ in front of the constant-term 1's coefficient a_0 is a matter of convention; its purpose will become clear momentarily.)

What we'd then imagine is that the coefficients a_n and b_n are the **components** of f along the **basis directions** in $\mathcal{T}$. Just as we could in $\mathbb{R}^n$, we're able to extract the coefficients from the above by taking L^2 inner products with the appropriate member ϕ of $\mathcal{T}$:

$$\langle f, \phi \rangle = \int_{-\pi}^{\pi} \left[\frac{a_0}{2} \cdot 1 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \right] \phi(x)\, dx$$

$$
\begin{aligned}
&= \int_{-\pi}^{\pi} \frac{a_0}{2}\phi(x)\,dx + \int_{-\pi}^{\pi} \left[\sum_{n=1}^{\infty} a_n \cos(nx)\phi(x)\right] dx + \int_{-\pi}^{\pi} \left[\sum_{n=1}^{\infty} b_n \sin(nx)\phi(x)\right] dx \\
&= \frac{a_0}{2}\int_{-\pi}^{\pi} \phi(x)\,dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos(nx)\phi(x)\,dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin(nx)\phi(x)\,dx.
\end{aligned}
$$

(Here, we're allowed to interchange the order of summation and integration because the sines and cosines are all bounded functions — cf. the Bounded Convergence Theorem applied to an appropriately chosen constant $M > 0$.)

Suppose, for instance, that $\phi(x) = 1 \in \mathcal{T}$; then by definition,

$$\langle f, 1\rangle_{L^2} = \int_{-\pi}^{\pi} f(x)\,dx.$$

If we were indeed able to decompose f in terms of members of $\mathcal{T}$, we would have

$$\langle f, 1\rangle_{L^2} = \frac{a_0}{2}\int_{-\pi}^{\pi} 1\,dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi} \cos(nx)\,dx + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi} \sin(nx)\,dx.$$

We know that each of the $\int_{-\pi}^{\pi} \sin(nx)\,dx = 0$ and $\int_{-\pi}^{\pi} \cos(nx)\,dx = 0$, so this becomes

$$\langle f, 1\rangle_{L^2} = \frac{a_0}{2} \cdot 2\pi = a_0\pi.$$

Combining this supposed expression with the one we obtained from the definition, we obtain an expression for the coefficient a_0:

$$a_0 = \frac{1}{\pi}\langle f, 1\rangle_{L^2} = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx.$$

Now suppose that $\phi(x) = \cos(mx)$ for some $m \geq 1$. Again, by definition,

$$\langle f, \cos(mx)\rangle_{L^2} = \int_{-\pi}^{\pi} f(x)\cos(mx)\,dx.$$

Supposing the decomposition in terms of members of $\mathcal{T}$ exists, we'd have

$$\langle f, \cos(mx)\rangle_{L^2} = \frac{a_0}{2}\underbrace{\int_{-\pi}^{\pi} \cos(mx)\,dx}_{=0 \text{ by (I2)}} + \sum_{n=1}^{\infty} a_n \underbrace{\int_{-\pi}^{\pi} \cos(nx)\cos(mx)\,dx}_{=0 \text{ if } n \neq m \text{ by (I4)}} + \sum_{n=1}^{\infty} b_n \underbrace{\int_{-\pi}^{\pi} \sin(nx)\cos(mx)\,dx}_{=0 \text{ by (I5)}}.$$

The first term vanishes, the $b_n \sin(nx)$ terms vanish, and every term in the $a_n \cos(nx)$ sum vanishes *except* for $n = m$, which contributes

$$\langle f, \cos(mx)\rangle_{L^2} = a_m \int_{-\pi}^{\pi} \cos^2(mx)\,dx = a_m\pi.$$

Combining the two expressions for $\langle f, \cos(mx)\rangle_{L^2}$ and dividing through by π, we obtain

$$a_m = \frac{1}{\pi}\langle f, \cos(mx)\rangle_{L^2} = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(mx)\,dx, \quad m \geq 1.$$

Finally, suppose that $\psi(x) = \sin(mx)$ for some $m \geq 1$. Then

$$\langle f, \sin(mx) \rangle_{L^2} = \int_{-\pi}^{\pi} f(x) \sin(mx)\, dx,$$

and if the decomposition held,

$$\langle f, \sin(mx) \rangle_{L^2} = \frac{a_0}{2} \underbrace{int_{-\pi}^{\pi} \sin(mx)\, dx}_{=0 \text{ by (I1)}} + \sum_{n=1}^{\infty} a_n \underbrace{\int_{-\pi}^{\pi} \cos(nx) \sin(mx)\, dx}_{=0 \text{ by (I5)}} + \sum_{n=1}^{\infty} b_n \underbrace{\int_{-\pi}^{\pi} \cos(nx) \cos(mx)\, dx}_{=0 \text{ if } n \neq m \text{ by (I3)}}.$$

Only the $b_n \sin(nx)$ term for $n = m$ survives, contributing $b_n \pi$, so that we have

$$b_m = \frac{1}{\pi} \langle f, \sin(mx) \rangle_{L^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(mx)\, dx, \quad m \geq 1.$$

We now summarize these findings.

Definition 2.1.11 (Fourier Coefficients and Fourier Series)

Let $f \in L^2[-\pi, \pi]$ be a square-integrable function, and suppose f admits a trigonometric series representation

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)).$$

We say that the expansion on the right is the **Fourier series** of $f(x)$, where the **Fourier coefficients** $(a_0, a_1, b_1, a_2, b_2, \dots)$ of f can be determined by solving the integrals

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\, dx, \quad n = 0, 1, 2, \dots,$$
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx, \quad n = 1, 2, 3, \dots.$$

2.1.2 The Behavior of Fourier Series Representations

Fourier Series: Convergence Analysis

Before we get to the (important) question of *when* such a decomposition is possible, let us take a look at a somewhat revealing series of examples. To start, suppose that $f\colon [-\pi, \pi] \to \mathbb{R}$ is the **square wave** function

$$f(x) = \begin{cases} -1 & -\pi < x < 0, \\ 0 & x = 0, \\ 1 & 0 < x < \pi. \end{cases}$$

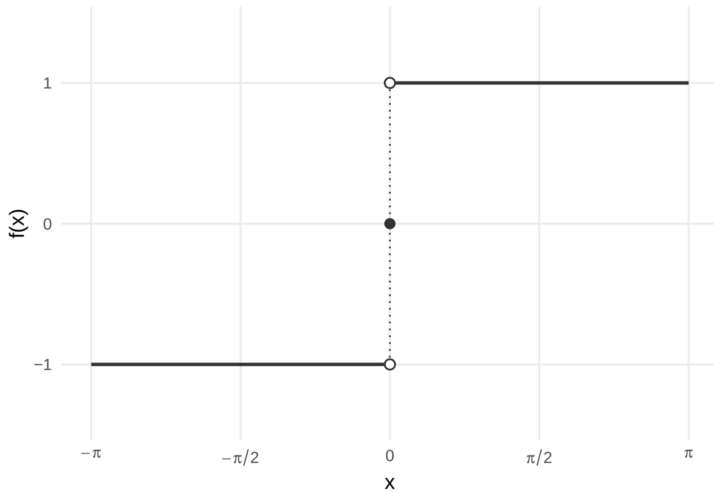


Figure: Plot of the square wave function $f(x)$.

Remark.

Note the value of $f(0) = 0$ here is set somewhat arbitrarily; it won't affect the evaluation of the integrals below, which is the primary concern. This is also why we've neglected to define $f(-\pi)$ and $f(\pi)$ — in general, we find that function evaluations at such singleton points $x \in \mathbb{R}$ don't contribute to integrals along continuous intervals; it is only when we change the function's value *along a continuum* that we observe a varying contribution to such an integral. As such, we may vary a function's evaluation at such **sets of measure zero** — e.g., singletons $\{x\} \subset \mathbb{R}$ for $x \in \mathbb{R}$ a fixed real number — without changing the resultant integral evaluations. This means that, for instance,

$$g(x) = \begin{cases} -1 & -\pi < x \leq 0, \\ 1 & 0 < x \leq \pi, \end{cases},$$

)
or

$$h(x) = \begin{cases} -1 & -\pi \leq x < 0, \\ 1 & 0 \leq x < \pi, \end{cases}$$

are "essentially the same function" as $f(x)$ if all we're concerned about is how they behave under integration. We'll explore this idea in more depth when we formally treat the Lebesgue integral.

So, given our square-wave function $f(x)$, we seek a trigonometric decomposition satisfying

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) = \begin{cases} -1 & -\pi < x < 0, \\ 0 & x = 0, \\ 1 & 0 < x < \pi. \end{cases}$$

Ultimately, we'll seek to convince ourselves that $f(x)$ is a quintessential candidate for representation by a Fourier series as presented in our definition above.. Whether such a decomposition exists is far from obvious — after all, we're claiming that a function with *sharp corners* and *jump* *discontinuities* can be built from combinations of *smooth*, *continuous* sines and cosines. Before we continue, then, let's consider why it may be reasonable to suppose it *could* exist for our given function $f(x)$.

First, once again letting $\mathcal{T}$ denote the set of basis functions of the form $\sin(nx)$ or $\cos(nx)$ for $n \in \mathbb{N}$ (and 1), what we observe is that *every* member $\phi(x)$ of $\mathcal{T}$ satisfies

$$\phi(x) = \phi(x + 2\pi k), \quad k \in \mathbb{Z};$$

in particular, each member of $\mathcal{T}$ is 2π-periodic (even if this isn't the "smallest period" we could assign to the function). That is, while $\sin(nx)$ and $\cos(nx)$ have **fundamental period** $p_n = 2\pi/n$, we clearly have $2\pi = n \cdot (2\pi/n) = np_n$, so that a function with fundamental period $p_n = 2\pi/n$ completes n full cycles on an interval of length 2π. We say that the members of $\mathcal{T}$ are *all* 2π-periodic — observe also that 2π is the *least common multiple* of the fundamental periods $\{2\pi/n\}_{n=1}^{\infty}$ of members of $\mathcal{T}$. What this means is that any linear combination of members of $\mathcal{T}$ will inherit this 2π-periodicity.

The reason we bring this up is that, if we're trying to decide whether or not it's worth pursuing a decomposition of our candidate square-wave function $f(x)$ in terms of trigonometric functions in $\mathcal{T}$, it would seem sensible to require that f itself be 2π-periodic — otherwise, we wouldn't have a member in $\mathcal{T}$ available to capture its non-periodic characteristics. Luckily, this isn't a problem for the square-wave f after one careful development: even though we've technically only defined f on the interval $[-\pi, \pi]$, this is already sufficient to ensure an **extension** $f^\star$ of f is available which is 2π-periodic. What we'll then derive is the Fourier series for the truly 2π-periodic $f^\star$ with the understanding that the Fourier series of the original function f is that of $f^\star$ restricted to a single period on $[-\pi, \pi]$.

In particular, if we simply define this extension to be

$$f^\star(x + 2\pi k) = f(x), \quad k \in \mathbb{Z}, \quad x \in [-\pi, \pi],$$

then $f^\star\colon\ \mathbb{R} \to \mathbb{R}$ is *automatically* a 2π-periodic function, seemingly regardless of the original behavior of $f\colon\ [-\pi, \pi] \to \mathbb{R}$. This will serve as the basis of many further extensions of Fourier theory to come, and it remains an implicit intuition baked into *many* Fourier-style computations. To be clear: a function "really doesn't need to be periodic," much less periodic on $[-\pi, \pi]$, to be "reasonable for representation by Fourier series" — we want to *manipulate* functions of interest into Fourier-suitable forms, perform the analysis there, then "wind everything back" to obtain the approximation of interest.

The function f is 2π-periodic after extension — the reason we postulated this as a sensible requirement of a candidate for decomposition by members of $\mathcal{T}$ is that those functions produce linear combinations which are 2π-periodic. Still, it would seem unclear whether this is *all* we need to proceed with confidence in our pursuit of a decomposition

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx). \tag{F1}$$

The answer to this question is, of course, nuanced. In the formal definition, we require that $f \in L^2[-\pi, \pi]$ be square-integrable, in the sense that

$$\int_{-\pi}^{\pi} |f(x)|^2 \, dx < \infty.$$

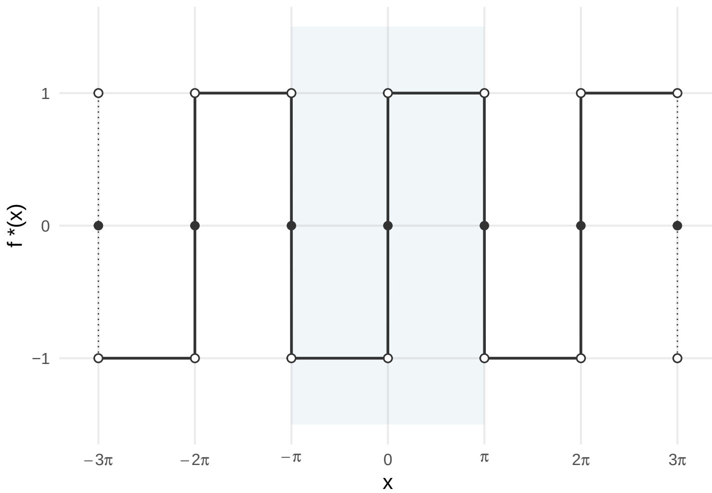


Figure: Plot of a few periods of the extension $f^\star(x)$.

This was in addition to our supposition that a trigonometric series *admitted* a trigonometric series of members in $\mathcal{T}$. One might have wondered why we require the square-integrability condition at all if it doesn't even guarantee the expansion's existence. The reason is that there are really *two separate questions* hiding inside the phrase "$f^\star$ admits a Fourier series":

(i) Can we *compute* the Fourier coefficients a_n and b_n in the Fourier series (F1)?

(ii) Given that (i) holds and we can compute the Fourier coefficients for $f^\star$, *how good* is the approximation on the right of (F1) to the true function's evaluation $f^\star(x)$ on the left for a given $x \in [-\pi, \pi]$? That is: *does* the series on the right converge to the actual value of the function on the left pointwise for $x \in [-\pi, \pi]$ and, if it does, *how many* Fourier coefficients do we need to compute to ensure its relative accuracy within some predetermined threshold, say $\epsilon > 0$?

The first question is the easier of the two. The Fourier coefficient formulas

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \tag{F2}$$

only require that the integrals on the right exist. But sines and cosines are clearly bounded — $|\cos(nx)| \le 1$ and $|\sin(nx)| \le 1$ for any n and any x — so

$$|a_n| = \frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|\cdot|\cos(nx)|\,dx \le \frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|\,dx < \infty$$

whenever $f \in L^1[-\pi, \pi]$ — that is, whenever f is **absolutely integrable**. This is actually a weaker condition than that of square-integrability, $f \in L^2[-\pi, \pi]$ — in particular, any square-integrable function is absolutely integrable, but the converse fails: there are many absolutely integrable functions which fail to be square-integrable (we can compactly write this as $L^2[-\pi, \pi] \subset L^1[-\pi, \pi]$ — the square-integrable functions on $[-\pi, \pi]$ are a subset of the absolutely integrable functions on $[-\pi, \pi]$.)

So, again, why would we require square-integrability in our definition? This is where we gesture toward question (ii) posed above. One who has taken a course in probability is likely familiar with the heavy-tailed Cauchy distribution — in particular, the kernel of this function is the familiar

$$g(x) = \frac{1}{1 + x^2}, \quad x \in \mathbb{R}.$$

Then $g(x) \in L^1[-\infty,\infty]$ is absolutely integrable on the whole real line, since the integrand is strictly positive,

$$\int_{-\infty}^{\infty} |g(x)|\,dx = \int_{-\infty}^{\infty} \left|\frac{1}{1+x^2}\right| dx = \int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \arctan(x)\Big|_{-\infty}^{\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi < \infty,$$

where, in the middle, we've identified $\frac{1}{(1+x^2)}$ as the antiderivative of $\arctan(x)$ from integration theory. (We could also have observed that $g(x)$ was an even function and obtained the same result: $2[\arctan(x)]_0^{\infty} = 2(\pi/2 - 0) = \pi$.)

Yet, as is perhaps its most identifiable property, the function $g \notin L^2[-\infty,\infty]$ — that is, g is absolutely integrable, but

$$\int_{-\infty}^{\infty} |g(x)|^2\,dx \text{ diverges to } +\infty, \text{ so } g \text{ is not square-integrable on } (-\infty,\infty).$$

We can construct similar examples on $[-\pi,\pi]$: one can verify that for the function

$$h(x) = |x|^{-1/2}, \quad x \in [-\pi,\pi],$$

we have that $h \in L^1[-\pi,\pi]$ is absolutely integrable on $[-\pi,\pi]$, but $h \notin L^2[-\pi,\pi]$ fails to be square-integrable on $[-\pi,\pi]$. (We leave this as an exercise — intuitively, however, this is because the function h blows up at the origin, but gently enough that $\int_{-\pi}^{\pi} |x|^{-1/2}\,dx$ converges, while $\int_{-\pi}^{\pi} |x|^{-1}\,dx$ does not.)

For such a function h, we can still compute every Fourier coefficient a_n and b_n using the fomulas in (F2) — we have

$$a_0 = \int_{-\pi}^{\pi} 1 \cdot h(x)\,dx = \int_{-\pi}^{\pi} h(x)\,dx \le \int_{-\pi}^{\pi} |h(x)|\,dx < \infty,$$

and that

$$a_n = \int_{-\pi}^{\pi} \cos(nx)h(x)\,dx \le \int_{-\pi}^{\pi} h(x)\,dx < \infty, \quad b_n = \int_{-\infty}^{\infty} \sin(nx)h(x)\,dx \le \int_{-\pi}^{\pi} h(x)\,dx < \infty,$$

so it is clear we would "obtain answers." So, what goes wrong? Well, recall that the interest is not just in calculating the real numbers a_n, b_n, but in applying these coefficients to obtain an approximation to a function: namely (cf. (F1)),

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} a_n \cos(nx) + \sum_{n=1}^{N} b_n \sin(nx).$$

Ideally, what we'd *like* to be able to say is that: given $\epsilon > 0$, we can find $N \in \mathbb{N}$ such that if we compute at least N of the coefficients $a_0, a_1, b_1, \dots, a_N, b_N$, $S_N(x)$ approximates $h(x)$ "reasonably well" on $[-\pi,\pi]$ — say,

$$h(x) \text{ is "close" to } S_N(x) \text{ for } x \in [-\pi,\pi].$$

To make headway, we need to rigorously characterize what we mean by "close."

There are (at least) two natural notions of closeness available to us. The first is **pointwise**: we say S_n approximates h pointwise on $[-\pi,\pi]$ if

$$\text{For each fixed } x \in [-\pi,\pi]\colon\ |h(x) - S_N(x)| \to 0 \quad \text{as} \quad N \to \infty. \tag{C1}$$

If $S_N(x)$ and $h(x)$ are real numbers for each fixed x, the natural choice for measuring this convergence is the Euclidean metric.

The second is **convergence in** L^2: we say that S_N approximates h in the L^2 sense if the total squared error, integrated over the whole interval $[-\pi, \pi]$, vanishes:

$$\|h - S_N\|_{L^2} := \int_{-\pi}^{\pi} |h(x) - S_N(x)|^2 \, dx \to 0 \quad \text{as} \quad N \to \infty. \tag{C2}$$

One might be tempted to believe that the pointwise convergence (C1) is stronger than the L^2 convergence (C2), if only because the latter "considers every point in $[-\pi, \pi]$ simultaneously when assessing the closeness of S_N to h." However, this is a misnomer; really, the L^2 norm is an *aggregate* measure — it tells us that the total area between h and S_N, squared and integrated, is small. But realize that integration *does not care* about behavior at individual points. There can even be a *countably infinite* number of points where S_N disagrees with h and still observe $S_n \to h$ in the L^2 sense — the point is that integration doesn't care about behavior on *sets of measure zero*. On the other hand, it is obvious that pointwise convergence (C1) is constructed specifically in order for this to occur.

Remark.

A third notion, that of **uniform convergence**, requires that the pointwise convergence (C1) hold in such a way that the "smallness of $|h(x) - S_N(x)|^2$ is controlled simultaneously over *all* $x \in [-\pi, \pi]$ at once."

Realize that in (C1), what we're really asserting is that given $\epsilon > 0$ *and* a particular point $x \in [-\pi, \pi]$, there exists an index in the tail of the sequence of partial sums S_N, say $K \in \mathbb{N}$, such that for all $N \geq K(\epsilon, x)$ further in the tail,

$$|h(x) - S_N(x)| < \epsilon.$$

The important point is that we choose the index in the tail K *only after* we're given the particular point $x \in [-\pi, \pi]$ for which we wish to assess convergence. In particular, it may be that "some points in $[-\pi, \pi]$ require K larger than others to observe ϵ-closeness under $|\cdot|$."

But what if the approximation S_N is "good enough" that we can choose the index in the tail K given *only* the desired error threshold $\epsilon > 0$ — i.e., suppressing the dependence on the point $x \in [-\pi, \pi]$ for which we desire $S_N(x) \to h(x)$ and instead seek that this convergence "hold simultaneously over all $x \in [-\pi, \pi]$ for all $N \geq K(\epsilon)$ independent of x"? In this case, we say that

$$\|h - S_N\|_{L^\infty} := \sup_{x \in [-\pi, \pi]} |h(x) - S_N(x)| \to 0 \quad \text{as} \quad N \to \infty. \tag{C3}$$

Whenever (C3) holds, we say that S_N converges **uniformly** to h. The only difference compared to (C1) is that we take the supremum over $x \in [-\pi, \pi]$ — realize that this intuitively ensures we can "choose N once and for all for every $x \in [-\pi, \pi]$ given only the threshold $\epsilon > 0$," though, since if the supremum is less than ϵ for all $N \geq K(\epsilon)$, then of course all other evaluations over points in the interval will be bounded by that supremum.

Now, for our function $h(x) = |x|^{-1/2}$, the L^2 notion of closeness (C2) is not even available: the expression $\|h - S_N\|_{L^2}$ requires computing

$$\int_{-\pi}^{\pi} |h(x) - S_N(x)|^2 \, dx,$$

but since $h \notin L^2[-\pi, \pi]$ to begin with, the integral will diverge. The function h lives entirely outside the Hilbert space $L^2[-\pi, \pi]$.

This means that the projection machinery we developed in the previous section — interpreting Fourier coefficients as *orthogonal projections* of functions f onto members of $\mathcal{T}$ and reading off coefficients by computing their inner products — loses its geometric foundation whenever those inner products aren't well-defined. To be clear, we can "still compute the formulas" for those coefficients; it is just that we cannot interpret the resultant series S_N as a projection of f onto the first several terms of $\mathcal{T}$ — in particular, if $f \notin L^2$, then we cannot guarantee *any* of the notions of convergence (C1)-(C3) of S_N to f hold.

Remark.

One might wonder: if L^2 membership of a function f guarantees convergence of S_N to f in the L^2 norm, and if $h \in L^1[-\pi, \pi]$ but $h \notin L^2[-\pi, \pi]$, would it be possible to simply replace the norm and analogously assert convergence of S_N to h under the L^1 norm instead of L^2? That is, we might imagine that if an absolutely integrable function $h \in L^1[-\pi, \pi]$ so that

$$\int_{-\pi}^{\pi} |h(x)|\, dx < \infty,$$

then, perhaps,

$$h \in L^1[-\pi, \pi] \overset{??}{\implies} S_n \to h \text{ under the } L^1 \text{ norm} : \int_{-\pi}^{\pi} |h - S_N|\, dx \to 0 \quad \text{as} \quad N \to \infty.$$

Remarkably, the answer here is *no*: there exist absolutely integrable functions $h \in L^1$ whose Fourier partial sums S_N diverge even in the L^1 norm. Really, if this *weren't* the case, we'd likely use L^1 spaces as the foundation for Fourier analysis — the space contains "more functions" than L^2, and if we *could* assert that Fourier approximations S_N behaved well for all members of L^1, we'd have no reason not to accept it as the canonical setting. Yet, the fact that this does occur is really a consequence of the structure of L^2 as a Hilbert space — namely, it is the *only* L^p space equipped with an inner product. We need inner products for concepts like "orthogonal projections" to be defined at all — it is the inner product which defines the familiar "angle," and it is the angle which gives rise to our notion of orthogonality in the first place. When $f \in L^2$, we're able to understand S_N as an orthogonal projection, and therefore it inherits many suitable geometric properties from the supposition that the target approximant f is in L^2. We'll more rigorously characterize these notions when we get to the Dirichlet conditions — the point we seek to emphasize is that L^2 is not merely a convenient strengthening of L^1, but rather the natural space in which the convergence question starts to make sense.

So the square-integrability condition in our definition is not there to make the Fourier coefficients a_n and b_n exist — L^1 membership suffices for tht. Rather, we require L^2 membership to ensure that the coefficients *tell us something useful* once we have them. Specifically, we have the following.

Theorem 2.1.12 (L^2 Convergence of Fourier Series)

If $f \in L^2[-\pi, \pi]$, then the Fourier partial sums

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} a_n \cos(nx) + \sum_{n=1}^{N} b_n \sin(nx)$$

converge to f in L^2 norm:

$$f \in L^2[-\pi, \pi] \implies \|f - S_N\|_{L^2}^2 = \int_{-\pi}^{\pi} |f(x) - S_N(x)|^2 \, dx \to 0 \quad \text{as} \quad N \to \infty.$$

This L^2 convergence is a powerful guarantee, but notice what it does *not* guarantee: namely, pointwise convergence in the sense of (C1) (nor does it promise uniform convergence (C3), which is stronger still). Indeed, pointwise convergence requires a few additional regularity assumptions on f beyond square-integrability — namely, assumptions on the tameness of f's discontinuities and the smoothness of its pieces. Instead, what we imagine L^2 convergence ensuring is that S_N will agree with f on regions of $\mathbb{R}$ of positive measure — i.e., intervals $I \subset \mathbb{R}$ of nonzero length $\text{len}(I) > 0$. Importantly, this set of intervals fails to include singleton points; we're not ensured anything on such measure zero sets.

Nevertheless, let us now return to our initial example of the square-wave,

$$f(x) = \begin{cases} -1 & -\pi < x < 0, \\ 0 & x = 0, \\ 1 & 0 < x < \pi. \end{cases}$$

We claim that f is sufficiently nice to compute its Fourier series coefficients and for the series to converge, so that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx).$$

— indeed, we have that

$$\|f\|_{L^1} = \int_{-\pi}^{\pi} |f(x)| \, dx = \int_{-\pi}^{0} |f(x)| \, dx + \int_{0}^{\pi} |f(x)| \, dx = \int_{-\pi}^{0} |-1| \, dx + \int_{0}^{\pi} |1| \, dx = 2\pi < \infty,$$

so $f \in L^1[-\pi, \pi]$, so at a minimum the Fourier coefficients of f all exist; i.e., we can compute the integrals

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx.$$

Furthermore, since

$$\|f\|_{L^2}^2 = \int_{-\pi}^{\pi} |f(x)|^2 \, dx = \int_{-\pi}^{0} |-1|^2 \, dx + \int_{0}^{\pi} |1|^2 \, dx = 2\pi < \infty,$$

it follows that $f \in L^2[-\pi, \pi]$. This means that if we continue to plug coefficients in to the Nth partial sum of the series

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} a_n \cos(nx) + \sum_{n=1}^{N} b_n \sin(nx),$$

then by the theorem above,

$$\|f - S_N\|_{L^2} = \int_{-\pi}^{\pi} |f(x) - S_N(x)|^2 \, dx \to 0 \quad \text{as} \quad N \to \infty.$$

Let us compute the coefficients explicitly. First, observe that

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{\pi} \int_{-\pi}^{0} -1 \, dx + \frac{1}{\pi} \int_{0}^{\pi} 1 \, dx = 0.$$

We observe that the integral of f over a symmetric interval $[-\pi, \pi]$ about 0 vanishes — in particular, the function f is odd. Just as when we compute products of odd and even integers, we have that

$$\text{odd} \times \text{odd} = \text{even}, \quad \text{odd} \times \text{even} = \text{odd}, \quad \text{even} \times \text{even} = \text{even}.$$

In particular, since $f(x)$ is odd and $\cos(nx)$ is even for any n, the function $f(x)\cos(nx)$ will be odd for any n, so that

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx = 0, \quad \forall n \in \mathbb{N}.$$

Similarly, since $f(x)\sin(nx)$ is even for any $n \geq 1$,

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx = \frac{2}{\pi} \int_{0}^{\pi} f(x) \sin(nx) \, dx.$$

On the interval $(0, \pi)$, the square-wave $f(x) = 1$ identically, so

$$b_n = \frac{2}{\pi} \int_{0}^{\pi} 1 \cdot \sin(nx) \, dx = \frac{2}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi} = \frac{2}{\pi} \left[1 - \cos(n\pi) \right].$$

But consider that for each n, we have $\cos(\pi) = -1$, $\cos(2\pi) = 1$, $\cos(3\pi) = -1$, and in general $\cos(n\pi) = (-1)^n$, so that

$$b_n = \frac{2}{\pi}[1 - (-1)^n] = \begin{cases} \dfrac{4}{n\pi} & n \text{ odd}, \\ 0 & n \text{ even}. \end{cases}$$

Now, with $a_n = 0$ for $n = 0, 1, 2, \ldots$ and $b_n = 4/n\pi$ for the odd $n = 1, 3, 5, \ldots, (2k+1), \ldots$, assemble the odd sine contributions in the series:

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx) \\ &= 0 + \frac{4}{\pi} \left(\sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \cdots \right) \\ &= \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}. \end{aligned} \tag{F3}$$

This is something worth pausing on: what we're claiming at this point is that the square-wave function $f(x)$ is approximated by the Nth term of the series above, in the sense that if we define

$$S_N(x) = \frac{4}{\pi} \sum_{k=0}^{\lfloor N/2 \rfloor} \frac{\sin((2k+1)x)}{2k+1}$$

then

$$\|f - S_N\|_{L^2} = \int_{-\pi}^{\pi} |f(x) - S_N(x)|^2 \, dx \to 0 \quad \text{as} \quad N \to \infty.$$

Again, this is nothing more than a consequence of the fact that $f \in L^2[-\pi, \pi]$ is square-integrable.

But, again, the question still remains: if our interest is in the value $f(x)$ for some particular $x \in [-\pi, \pi]$, will $S_N(x)$ approximate its evaluation "well enough" to be useful? Let us test at a few candidates $x \in [-\pi, \pi]$. To start, take $x = \pi/2$, where we note that $f(\pi/2) = 1$ since $\pi/2 > 0$. What we want to verify is whether or not $f(\pi/2) = \lim_{N\to\infty} S_N(\pi/2)$ — i.e., whether

$$1 = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin\left((2k+1)\frac{\pi}{2}\right)}{2k+1}.$$

It is really once we see this that the issue becomes somewhat clear: it really isn't obvious how to handle the series on the right, but if everything else we've stated were to hold, the relation above would hold as a natural consequence. In particular, we can write

$$\sin\left((2k+1)\tfrac{\pi}{2}\right) = \sin\left(k\pi + \frac{\pi}{2}\right) = \cos(k\pi) = (-1)^k,$$

so that the series on the right becomes

$$1 = \frac{4}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

Dividing the constant factor of $4/\pi$ through to the other side, we obtain a curious result: if it were the case that $S_N(x) \to f(x)$ pointwise for $x \in [-\pi, \pi]$ as $N \to \infty$, then it would necessarily follow that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

In fact, this is the famous **Leibniz formula** for computing π — that it should hold is not obvious from inspection, but again, it would follow if pointwise convergence of S_N to f were to hold in $[-\pi, \pi]$. Interestingly, this is the most straightforward means to compute the value of the series on the right — in an introductory Calulus course, perhaps we would realize that the terms $s_k = (-1)^k/(2k+1)$ are *alternating* in sign with terms $|s_k| = 1/(2k+1) \to 0$ as $k \to \infty$. In such a case, we know that the series on the right converges by the **Alternating Series Test**. Yet, the test doesn't tell us what the series converges to — indeed, if we think back on such an introductory treatment, we discuss convergence tests in some detail, but finding the exact value of such a series is a question primarily asked when the series is **geometric** — i.e., a series of the form $S = \sum_{k=0}^{\infty} ar^k$ with $|r| < 1$, in which case we have the closed form expression $S = \frac{a}{1-r}$. As we will see, Fourier series provide the other most common tool for evaluating such series.

Let us now test at a point where f is discontinuous. At $x = 0$, the square wave jumps from -1 to 1, and we defined $f(0) = 0$. The series gives

$$S_N(0) = \frac{4}{\pi} \sum_{k=0}^{\lfloor N/2 \rfloor} \frac{\sin\left((2k+1)\cdot 0\right)}{2k+1} = \frac{4}{\pi} \sum_{k=0}^{\lfloor N/2 \rfloor} \frac{\sin(0)}{2k+1} = 0$$

for every $N \in \mathbb{N}$ — that is, the partial sums $S_N(x)$ are identically zero for every N at $x = 0$, so trivially, $S_N(0) = 0 \to 0 = f(0)$ as $N \to \infty$.

Notice that we defined $f(0) = 0$ almost in anticipation of this — indeed, this was a deliberate choice on our end, and we now reveal why: since the **left-hand limit** of f at $x = 0$ — i.e., the limit of f as x approaches 0 from the left, taking arbitrarily small but strictly negative values — is

$$f(0^-) := \lim_{x \to 0^-} f(x) = \lim_{x \uparrow 0} f(x) = -1.$$

Similarly, the **right-hand limit** of f as x approaches 0 from the right from arbitrary small but strictly positive values is

$$f(0^+) := \lim_{x \to 0^+} f(x) = \lim_{x \downarrow 0} f(x) = 1.$$

Notice now that $f(0) = 0$ is actually the **midpoint** of the jump: it is the average of the left-hand limit and the right-hand limit,

$$f(0) := \frac{f(0^-) + f(0^+)}{2} = \frac{-1 + 1}{2} = 0.$$

At $x = \pi$, the situation is similar. Strictly, we defined $f(x)$ for $x \in (-\pi, \pi)$, not including the endpoints $x = \pm\pi$. But we defined the periodic extension $f^\star(x)$ to take on values in all of the real line $x \in \mathbb{R}$, where

$$f^\star(x + 2\pi k) = f(x), \quad k \in \mathbb{Z}, \quad x \in (-\pi, \pi).$$

This means that, in particular,

$$f^\star(\pi^-) = \lim_{x \uparrow \pi} f^\star(x) = \lim_{x \uparrow \pi} f(x) = 1,$$

which follows directly since f is defined for $x \uparrow \pi$ from the left. On the other hand, from the right, we have

$$f^\star(\pi^+) = \lim_{x \downarrow \pi} f^\star(x) = \lim_{x \downarrow -\pi} f^\star(x + 2\pi) = \lim_{x \downarrow -\pi} f(x) = -1,$$

where we've used periodicity and the definition of $f^\star$ in the second and third equalities. In particular,

$$f(\pi) := \frac{f^\star(\pi^+) - f^\star(\pi^-)}{2} = \frac{-1 + 1}{2} = 0.$$

We can directly verify that, by *defining* $f(\pi) := 0$, we force agreeance with the limit of partial sums:

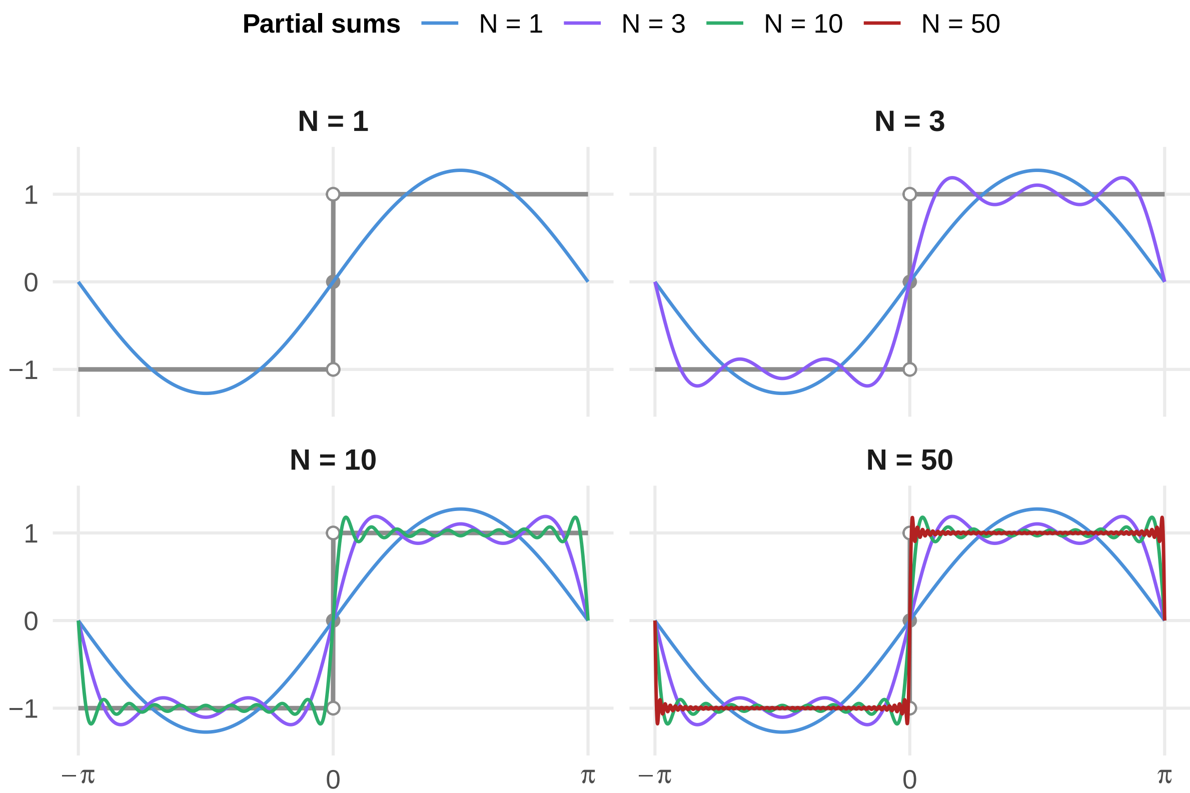
$$\frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin\left((2k+1) \cdot \pi\right)}{2k+1} = \frac{4}{\pi} \left(\sin(\pi) + \frac{\sin(3\pi)}{3} + \frac{\sin(5\pi)}{5} + \cdots \right) = 0,$$

for every $N \in \mathbb{N}$ since $\sin(\pm\pi) = 0$, meaning that

$$S_N(\pi) = 0 \to 0 = f(\pi) \quad \text{as} \quad N \to \infty.$$

We begin to see a pattern: at every point of discontinuity, the Fourier partial sums S_N converge not to $f(x)$ as we define it, but to the *average of its left- and right-hand limits* — in particular, realize that we set $f(0) = 0$ explicitly so that it was the average of $f(0^+)$ and $f(0^-)$, so that the value of $f(0)$ itself is entirely irrelevant. What we seek to emphasize is that, in practice, it is most commonly that we observe the behavior of the Fourier partial sums and define the value of f at peculiar points like $x = 0$ in accordance with that behavior — *not* that we're given a function whose value is "arbitrarily defined at a point" and asked to verify whether the Fourier series converges there.

Between the jumps, the evidence is equally compelling. At any point x in the interior of $(0, \pi)$ or $(\pi, 0)$, f is continuous (indeed, constant), so $f(x^-) = f(x^+) = f(x)$, and we would thus expect that $S_N(x) \to f(x)$ the average of the left- and right-limits at any such point. The Leibniz formula at $x = \pi/2$ is one verification of this.



At this point, we're close to verifying those functions f for which pointwise convergence of $S_N(x) \to f(x)$ occurs for each $x \in [-\pi, \pi]$. There is, however, one conspicuous feature in the plots that refuses to disappear: the persistent overshoot near each jump. In the plot, we see that these overshoots are visible even at $N = 50$. These points are troublesome not because they break pointwise convergence — indeed, we'll still have $S_N(x) \to f(x)$ as $N \to \infty$ for x inside these "pockets of overshoots" — but, rather, because they require us to collect relatively more coefficients of the series in order for the pointwise convergence of $S_N(x) \to f(x)$ to occur. This is not a numerical artifact, but a genuine phenomenon known as the **Gibbs effect**: as $N \to \infty$, the pocket around the overshoot grows *narrower*, sliding ever closer to the point of discontinuity, but its peak height converges to approximately 9% the jump size and never vanishes. Pointwise convergence is not threatened — for any fixed x near the jump, the pocket eventually slides past that point, and we'll begin to observe $S_N(x) \to f(x)$. But the convergence is *slower* for points closer to the discontinuity — in the language of (C1), the index in the tail $K(\epsilon, x)$ grows as x approaches the jump), and since there is always *some* point still caught in the pocket no matter how large N is, there is no way to choose a single $K(\epsilon)$ such that $N \geq K(\epsilon)$ implies $|S_N(x) - f(x)| < \epsilon$ for any $x \in [-\pi, \pi]$. In other words, it will *never* be that we can observe uniform convergence (C3) — the best we can do is pointwise convergence, (C1). The lack of uniform convergence is unfortunate from a theoretical perspective but, for most applications, pointwise convergence is more than sufficient and, as we will see, is capable of producing some astonishing results.

2.2 The Dirichlet Conditions

So, the picture that emerges from the square wave is the following: L^2 convergence (C2) follows whenever $f \in L^2[-\pi,\pi]$; pointwise convergence (C1) occurs everywhere when $f(x) = \frac{f(x^+)-f(x^-)}{2}$ for each $x \in [-\pi,\pi]$; yet, uniform convergence (C3) of S_N to f over all of $[-\pi,\pi]$ simultaneously is almost always impossible. The question now becomes: is this *always* the case for "reasonable" functions, like the square-wave $f(x)$?

The answer is given by the following classical results, which provide sufficient conditions under which pointwise convergence (C1) of $S_N(x) \to f(x)$ is guaranteed.

Theorem 2.2.1 (Dirichlet Hypothesis)

Let $f\colon \mathbb{R} \to \mathbb{R}$ be a 2π-periodic function and suppose that, on the interval $[-\pi,\pi]$:

(i) f is bounded: $|f(x)| \leq M$ for all $x \in [-\pi,\pi]$.

(ii) f has at most finitely many local extrema — or, more concretely, f is of **bounded variation**: for any partition $-\pi := x_0 < x_1 < x_2 < \cdots < x_{k-1} < x_k := \pi$, there exists a constant M such that

$$\sum_{j=0}^{k-1} |f(x_{j+1}) - f(x_j)| \leq M.$$

(iii) f has at most finitely many jump discontinuities $x_d \in [-\pi,\pi]$, and at every such point of discontinuity, the one-sided limits $f(x_d^-)$ and $f(x_d^+)$ both exist.

Then, the Fourier series of f converges for every $x \in [-\pi,\pi]$:

$$f(x) = \frac{f(x^-)+f(x^+)}{2} = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos(nx) + b_n\sin(nx)), \quad x \in [-\pi,\pi],$$

where the Fourier coefficients are given by

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad n = 1,2,\ldots$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx, \qquad n = 1,2,\ldots$$

At this point, we intend to rigorously prove the Dirichlet conditions — however, to do so will require a few pieces of machinery which we intend to develop in anticipation of their applicability.

Riemann-Lebesgue

To start, we'll also be utilizing the following lemma whose content, in our view, comprises some of the truly illuminating content of *why* pointwise convergence of the Fourier series is sensible to hypothesize in the first place.

Lemma 2.2.2 (Riemann-Lebesgue)

If $\phi \in L^1[a,b]$ is an abolutely integrable function on the interval $[a,b]$, then

$$\int_a^b \phi(t)\sin(\lambda t)\,dt \to 0 \quad \text{as} \quad \lambda \to \infty,$$

and

$$\int_a^b \phi(t)\cos(\lambda t)\,dt \to 0 \quad \text{as} \quad \lambda \to \infty.$$

Before proving the Riemann-Lebesgue lemma, let us take stock of its intuition: recall that, for any integer $n \in \mathbb{Z}$,

$$\int_{-\pi}^{\pi} \sin(nt)\,dt = \int_{-\pi}^{\pi} \cos(nt)\,dt = 0.$$

That is, a 2π-periodic sinusoidal function integrated over a complete period vanishes — the positive and negative half-cycles cancel exactly. The Riemann-Lebesgue Lemma says that this cancellation persists, in an approximate, limiting sense, even when the sinusoid is multiplied by an arbitrary integrable function $\phi \in L^1[a,b]$.

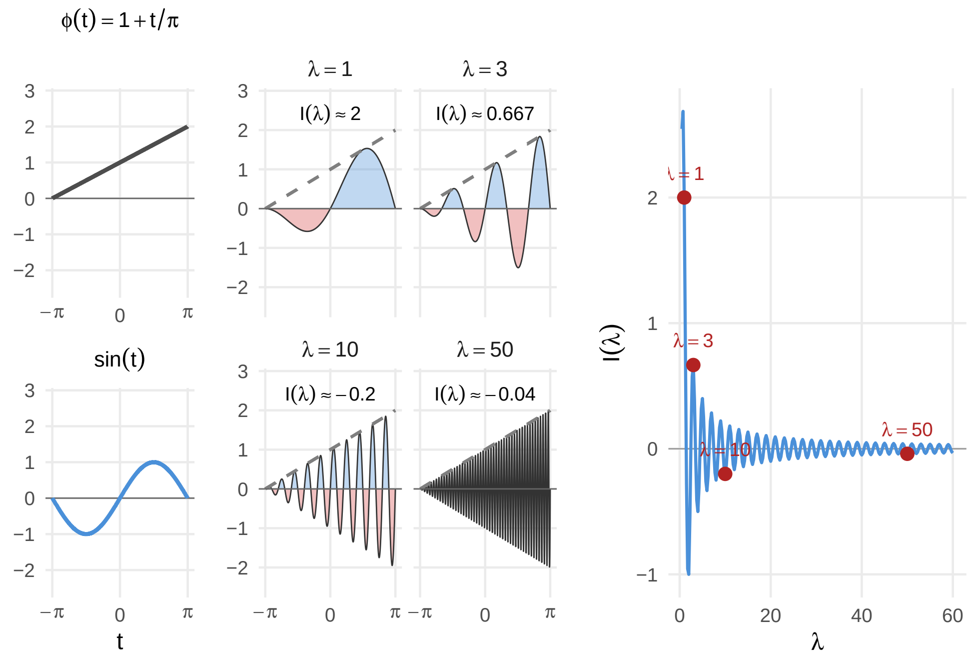


Figure (Illustrating Riemann-Lebesgue): Plot of $f(t \mid \lambda) = (1 + \frac{t}{\pi}) \cdot \sin(\lambda t)$ for $t \in [-\pi, \pi]$ and various $\lambda \in \mathbb{N}$. $I(\lambda)$ is the value of the integral over $[-\pi, \pi]$ at the given λ.

The plot makes the mechanism visible: imagining the integral as "slicing up the domain into small strips Δx, computing the values of f at each point, and summing the contributions," what essentially happens here is that we "multiply the function $\phi(t)$ by a number $\sin(\lambda t) \in [-1, 1]$, so that the contributions of each $\phi(t) \cdot \sin(\lambda t)$ are bounded within $[-\phi(t), \phi(t)]$ pointwise for each $t \in [-\pi, \pi]$. For small λ, the sinusoid $\sin(\lambda t)$ oscillates slowly relative to $\phi(t) = 1 + t/\lambda$, and the positive contributions (blue) and negative contributions (red) are unbalanced — the integral $I(\lambda)$ is nonzero. However, as we increase λ, the oscillations in $\sin(\lambda t)$ start to speed up; when they do, the unbalancedness of the positive and negative contributions begin to wash each other out. The half-cycles over which Δt is positive become so narrow that $\Delta\phi(t)$ is nearly constant across it, and because that near-constancy carries over to the next half-cycle of opposite sign, they nearly cancel. As $\lambda \to \infty$, these half-cycles become infinitesimally thin, and the cancellation becomes arbitrarily precise, so that $I(\lambda) = \int_{-\pi}^{\pi} \phi(t)\, \sin(\lambda t)\, dt \to 0$. The decay plot on the right shows this directly: $I(\lambda)$ oscillates with decreasing amplitude, asymptotically approaching $I(\lambda) \to 0$ as $\lambda \to \infty$.

That Riemann-Lebesgue holds for *some* functions shouldn't be terribly surprising — however, what *is* surprising is that it holds for such a general class as the absolutely integrable $f \in L^1[a, b]$. We shall explore the intuition for this after supplying a formal proof of the lemma.

Proof:

Since $\cos(\lambda x) = \sin(\lambda(x - \pi/2))$ for any $x \in \mathbb{R}$, it will be sufficient to prove the case for $\sin(\lambda x)$.

What we seek to formalize is the cancellation of competing half-cycles as we take finer and finer partitions of the interval $[-\pi, \pi]$. Writing

$$I(\lambda) := \int_a^b \phi(t) \sin(\lambda t)\, dt,$$

observe first that shifting t by half a period of $\sin(\lambda t)$ — that is, shifting $t \mapsto t + \pi/\lambda$ — flips the sign of the sinusoid:

$$t \mapsto t + \frac{\pi}{\lambda} \implies \sin\left(\lambda\left(t + \frac{\pi}{\lambda}\right)\right) = \sin(\lambda t + \pi) = -\sin(\lambda t).$$

Performing the substitution $u = t + \pi/\lambda$ in the integral (and extending ϕ by zero outside $[a, b]$ as necessary, so that $\phi(t)$ is defined appropriately for all t in the shifted interval,)

$$I(\lambda) = \int_a^b \phi(t) \sin(\lambda t)\, dt = \int_{a+\pi/\lambda}^{b+\pi/\lambda} \phi(u-\pi/\lambda) \sin\left(\lambda u - \pi\right)\, du = -\int_a^b \phi\left(u - \pi/\lambda\right) \sin(\lambda u)\, du.$$

Where in the last equality we've applied the sign-flip and shifted the limits back appropriately. Renaming u back to t, we obtain two equivalent expressions for $I(\lambda)$:

$$I(\lambda) = \int_a^b \phi(t) \sin(\lambda t)\, dt = -\int_a^b \phi\left(t - \frac{\pi}{\lambda}\right) \sin(\lambda t)\, dt.$$

Since both of these expressions equal $I(\lambda)$, their average will also equal $I(\lambda)$ — in particular,

$$I(\lambda) = \frac{1}{2}\int_a^b \left[\phi(t) - \phi\left(t - \frac{\pi}{\lambda}\right)\right] \sin(\lambda t)\, dt.$$

Taking absolute values and bounding $|\sin(\lambda t)| \leq 1$,

$$|I(\lambda)| = \frac{1}{2}\int_a^b \left|\phi(t) - \phi\left(t - \frac{\pi}{\lambda}\right)\right| \cdot |\sin(\lambda t)|\, dt \leq \frac{1}{2}\int_a^b \left|\phi(t) - \phi\left(t - \frac{\pi}{\lambda}\right)\right| dt.$$

Now, notice that as $\lambda \to \infty$, $t - \pi/\lambda \to t$, so that

$$\phi(t - \pi/\lambda) \to \phi(t)$$

pointwise at every point of continuity of ϕ. Since $\phi \in L^1[a, b]$, there are at most countably many points of discontinuity, so that ϕ is continuous almost everywhere on $[a, b]$ — in particular, $\phi(t - \pi/\lambda) \to \phi(t)$ pointwise a.e. $t \in [a, b]$. By the Bounded Convergence Theorem[a] (the integrand is bounded by $2|\phi(t)|$, which is a measurable function of the integrable $\phi(t)$, so it too is integrable), the integral on the right tends to zero:

$$|I(\lambda)| \leq \frac{1}{2}\int_a^b \left|\phi(t) - \phi\left(t - \frac{\pi}{\lambda}\right)\right| dt \to 0 \quad \text{as} \quad \lambda \to \infty.$$

This means that $I(\lambda) \to 0$ as $\lambda \to \infty$, completing the proof.

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In the context of Fourier analysis, this has an immediate and important consequence:

Theorem 2.2.3 (Decay of Fourier Coefficients)

If $f \in L^1[-\pi, \pi]$, then its Fourier coefficients satisfy

$$a_n \to 0 \quad \text{and} \quad b_n \to 0 \quad \text{as} \quad n \to \infty.$$

Proof:

This follows directly from Riemann-Lebesgue:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx \to 0$$

with $\phi = f \in L^2[-\pi, \pi] \subset L^1[-\pi, \pi]$ and $\lambda = n$. The analogous proof clearly holds for $\cos(nx)$ as well by applying the phase-shift of $\cos(x) = \sin(x - \pi/2)$.

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[a] We discuss this result in more detail in future sections; for now, it suffices to understand the intuition of *why* bounding the absolute value of an integral under investigation by the integral of another nonnegative function which we *know* to be integrable would suffice to show the integrability of the original function.

Convolutions and The Dirichlet Kernel

With the Riemann-Lebesgue Lemma in hand, we now turn to the second structural ingredient needed for the Dirichlet proof: **convolutions**.

Definition 2.2.4 (Convolution)

Given two 2π-periodic, integrable functions $f, g \in L^1[-\pi, \pi]$, their **convolution** is the function

$$(f * g)(x) := \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(x - t)\, dt.$$

At a glance, one would tend to wonder about the broader applicability of convolutions — it isn't immediately clear why such an operation would be so valuable as to warrant its own notation. Nevertheless, we can understand it best by considering a situation where it arises naturally. Suppose we're given the functions

$$f(x) = g(x) := \begin{cases} 1 & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Then f and g are simply indicator functions on $[0, 1]$ — i.e.,

$$f(x) = g(x) = \mathbf{1}_{[0,1]}(x), \quad x \in \mathbb{R}.$$

Suppose we're interested in finding the convolution of g by f — i.e., the function

$$(f * g)(x) = \int_{-\infty}^{\infty} f(t) g(x - t)\, dt.$$

Immediately, one must realize the peculiarly obvious fact that the convolution is a *function of x*. It is far too tempting to believe we can write

$$\int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(x-t)\,dt \overset{?}{=} \int_{0}^{1} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(x-t)\,dt \overset{??}{=} \int_{0}^{1} 1\,dt,$$

since we might incorrectly believe that "both indicators are over $[0,1]$, and so the integral should only turn on over $[0,1]$ where both are equal to 1."

But this is *wrong*, and seeing *why* it is wrong is the key to understanding the convolution. The issue is the term $\mathbf{1}_{[0,1]}(x-t)$ is not an indicator of the *variable of integration* t — it is an indicator of a *sliding window centered at* x as we integrate over t. This is, of course, in contrast with the term $\mathbf{1}_{[0,1]}(t)$, which *really is* the indicator of the variable of integration t. What we imagine, then, is that the convolution is computed by *sliding* g across f, and measuring their overlap at each position x. To see this, let us work through the integral more carefully.

Consider again the convolution

$$(f*g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\,dt = \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(x-t)\,dt.$$

Now, what really is true about the integrand $\mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(x-t)$ is that it is "only on" whenever $t \in [0,1]$ and $x-t \in [0,1]$, where we consider x to be fixed — it is as if we've written, for instance, $1-t \in [0,1]$, or $2-t \in [0,1]$.

In such a case where we have "two variables x and t, where x is fixed while t varies," it is advantageous to rewrite the domain explicitly in terms of the integrated variable t — e.g., here, to write

$$0 \le x-t \le 1 \implies x-1 \le t \le x.$$

The reason this is the "correct move" is that, now, we can write

$$(f*g)(x) = \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(x-t)\,dt = \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[x-1,x]}(t)\,dt.$$

For any fixed $x \in \mathbb{R}$, this integral is trivial — consider the case when $x=1$. Then the integrand is $\mathbf{1}_{[0,1]}(t)\,\mathbf{1}_{[0,1]}(t)$, so that the indicators overlap — i.e.,

$$\mathbf{1}_{[0,1]}(t)\,\mathbf{1}_{[1-1,1]}(t) = \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[0,1]}(t) = \mathbf{1}_{[0,1]}(t),$$

so that

$$(f*g)(1) = \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\,dt = \int_{0}^{1} 1\,dt = 1.$$

One can conduct similar arguments for the other cases. Nevertheless, realize that both f and g were nonzero strictly in $[0,1]$ — one might naturally assume that, in turn, the convolution $f*g$ would also be nonzero only for $x \in [0,1]$. This, however, is immediately shown to be false, since if $1 \le x \le 2$, we have that $0 \le x-1 \le 1$, so

$$\begin{aligned}(f*g)(x) &= \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[x-1,x]}(t)\,dt \\ &= \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\Big(\mathbf{1}_{[x-1,1]}(t) + \mathbf{1}_{(1,x]}(t)\Big)\,dt\end{aligned}$$

$$= \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(t)\mathbf{1}_{[x-1,1]}(t)\,dt + \int_{-\infty}^{\infty} \underbrace{\mathbf{1}_{[0,1]}(t)\mathbf{1}_{(1,x]}(t)}_{=0 \text{ for any } t\in\mathbb{R}}\,dt$$

$$= \int_{x-1}^{1} 1\,dt = 2 - x.$$

Similar arguments repeated over $x \in [0,1]$, $x < 0$, and $x > 2$ reveal

$$(f * g)(x) = \begin{cases} 0 & x \le 0, \\ x & 0 \le x \le 1, \\ 2 - x & 1 \le x \le 2, \\ 0 & x \ge 2. \end{cases}$$

Recalling our original rectangular functions

$$f(x) = g(x) = \begin{cases} 1 & 0 \le x \le 1, \\ 0 & \text{otherwise}, \end{cases}$$

what we've just shown is that their convolution $f * g$ is a *triangle* — a tent-shaped function increasing linearly from 0 to 1 on $[0,1]$ and decreasing back to 0 linearly on $[1,2]$.

The triangle didn't come from anywhere mysterious. The whole process essentially amounted to integrating the rectangle f remaining fixed on $[0,1]$ against a shifted version of g on $[x-1,x]$, and this result aligns with our physical intuition of "sliding rectangles and measuring their overlapping areas." Whenever we specified a shift x of g such that the two rectangles didn't overlap at all, their convolution is zero — there simply is no overlap of f with such a shifted version of g.

The rectangle example was chosen for its transparency, but we now turn our attention toward the convolution's applicability in the current setting — i.e., in analyzing the pointwise convergence of the Fourier series to its approximating function. As it turns out, convolution isn't just *related* to the convergence question — the entire convergence question *is* a convolution!

To see this, recall the Nth partial sum of the Fourier series of $f \in L^2[-\pi,\pi]$:

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N}(a_n \cos(nx) + b_n \sin(nx)),$$

where

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt, \qquad n = 1, 2, ...$$

Now, substitute the integrals directly into the formula for S_N: we have

$$S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,dt + \sum_{n=1}^{\infty}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt \cdot \cos(nx) + \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt \cdot \sin(nx)\right].$$

Since $\cos(nx)$ and $\sin(nx)$ aren't functions of the variable of integration t, we can pull each inside their respective integrals:

$$S_N(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,dt + \sum_{n=1}^{N}\left[\frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\cos(nx)\,dt + \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\sin(nx)\,dt\right]$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \left[\frac{1}{2} + \sum_{n=1}^{N} \Big(\cos(nt)\cos(nx) + \sin(nt)\sin(nx) \Big) \right] \, dt,$$

where the last line uses linearity of the integral to pull the factors of $1/\pi$ and $f(t)$ to the front.

But, now, recall the cosine addition/subtraction formulas:

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$

Looking to the form of the summand inside the integral, we observe it rsemblance to the right side of the cosine addition formula — indeed,

$$\cos(nt)\cos(nx) + \sin(nt)\sin(nx) = \cos\left(nt - nx\right) = \cos\Big(n(t-x)\Big) \equiv \cos\Big(n(x-t)\Big),$$

where the last equivalence follows since $\cos(\cdot)$ is even.

From this, we can write the Nth Fourier partial sum S_N as

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \left[\frac{1}{2} + \sum_{n=1}^{N} \cos(n(x-t)) \right] \, dt.$$

Look just at the bracketed expression above — in particular, write $u := x - t$, and let

$$D_N(u) := \frac{1}{2} + \sum_{n=1}^{N} \cos(nu).$$

Then (realizing the bounds of the integral are the same under this substitution), we can either write

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x-u) D_N(u) \, du = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(x-t) \, dt.$$

Recalling that we define the convolution

$$(f * g)(x) = \int_{-\pi}^{\pi} f(t) g(x-t) \, dt,$$

it should be readily apparent from the latter form of the expression above that S_N is nothing more than

$$S_N(x) = \frac{(f * D_N)(x)}{\pi} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(x-t) \, dt.$$

Indeed, the function D_N here is important enough to earn a definition:

Definition 2.2.5 (Dirichlet Kernel)

Given an integrable function $f \in L^1[-\pi, \pi]$, the Nth **Dirichlet Kernel** of f is

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nu).$$

And, in light of the preceding discussion, we have the following:

Theorem 2.2.6 (Integral Representation of Fourier Partial Sums)

Suppose the function f admits a Fourier expansion with Nth partial sum

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} (a_n \cos(nx) + b_n \sin(nx)).$$

Then, we equivalently have that

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(x-t)\, dt$$

Now, we're able to identify the question of whether $S_N(x) \to f(x)$ pointwise for $x \in [-\pi, \pi]$ by the question of whether the convolution of f with the Dirichlet kernel D_N recovers the original function f pointwise on $[-\pi, \pi]$ — i.e., by whether

$$(f * D_N)(x) \to f(x) \quad \text{pointwise for} \quad x \in [-\pi, \pi].$$

This is the real utility of convolutions; it may not be evident yet exactly *why* this will be useful, but it should be clear that we've consolidated the action of the Nth Fourier partial sum S_N into the actors f, D_N, and $f * D_N$. Before we utilize D_N further, let us derive a more tractable expression for it. Indeed, even though

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nu)$$

tells us how to evaluate $D_N(u)$, it requires us to compute N separate evaluations of $\cos(nu)$, $n = 1, \dots, N$. This is unfortunate; we would prefer a *closed-form* expression for D_N, if one were available. It isn't trivial to see, but by multiplying both sides of the expression above by $2\sin(u/2)$:

$$2\sin(u/2) D_N(u) = \sin(u/2) + \sum_{n=1}^{N} 2\sin(u/2)\cos(nu),$$

we can look to the terms inside the summand — i.e., those of the form

$$2\sin(u/2)\cos(nu), \quad n = 1, \dots, N.$$

Recalling the product-to-sum identity (P4):

$$\cos(\alpha)\sin(\beta) = \frac{\sin(\alpha+\beta) - \sin(\alpha-\beta)}{2} \implies \sin(\alpha+\beta) - \sin(\alpha-\beta) = 2\cos(\alpha)\sin(\beta),$$

we can substitute $\alpha = u/2$ and $\beta = nu$, in which case

$$2\sin(u/2)\cos(nu) = \sin\left((n+\tfrac{1}{2})u\right) - \sin\left((n-\tfrac{1}{2})u\right).$$

Now observe what happens when we sum these terms from $n = 1$ to N: we have

$$n = 1: \quad \sin\left(\tfrac{3}{2}u\right) - \sin\left(\tfrac{1}{2}u\right)$$

$$\begin{aligned}
n=2: &\quad \sin\left(\tfrac{5}{2}u\right)-\sin\left(\tfrac{3}{2}u\right)\\
n=3: &\quad \sin\left(\tfrac{7}{2}u\right)-\sin\left(\tfrac{5}{2}u\right)\\
\vdots &\qquad\qquad \vdots\\
n=N: &\quad \sin\left((N+\tfrac{1}{2}u)\right)-\sin\left((N-\tfrac{1}{2})u\right),
\end{aligned}$$

so that we obtain a **telescoping sum** wherein the alternating signs of the corresponding terms lead to most of the contributions cancelling; in particular, we have that

$$\begin{aligned}
\sum_{n=1}^{N} 2\sin(u/2)\cos(nu) &= \left[\sin\left(\tfrac{3}{2}u\right)-\sin\left(\tfrac{1}{2}u\right)\right]+\left[\sin\left(\tfrac{5}{2}u\right)-\sin\left(\tfrac{3}{2}u\right)\right]+\cdots+\left[\sin\left((N+\tfrac{1}{2})u\right)-\sin\left((N-\tfrac{1}{2})u\right)\right.\\
&= \sin\left((N+\tfrac{1}{2})u\right)-\sin\left(\tfrac{1}{2}u\right)+\underbrace{\left[\sin\left(\tfrac{3}{2}u\right)-\sin\left(\tfrac{3}{2}u\right)\right]}_{=0}+\underbrace{\left[\sin\left(\tfrac{5}{2}u\right)-\sin\left(\tfrac{5}{2}u\right)\right]}_{=0}+\cdots\\
&= \sin\left((N+\tfrac{1}{2})u\right)-\sin(u/2).
\end{aligned}$$

Substituting this back into

$$2\sin(u/2)D_N(u)=\sin()+\sum_{n=1}^{N}2\sin(u/2)\cos(nu),$$

we have that

$$2\sin(u/2)D_N(u)=\sin(u/2)+\sin\left((N+\tfrac{1}{2}u)\right)-\sin(u/2)=\sin\left((N+\tfrac{1}{2})u\right)$$

after cancelling the competing $\sin(u/2)$ terms.

Realize we now have everything we need to obtain our closed-form expression: dividing $2\sin(u/2)$ through to both sides (which is nonzero for $u\neq 0$ in $[-\pi,\pi]$):

$$D_N(u)=\frac{\sin\left((N+\frac{1}{2})u\right)}{2\sin(u/2)},\quad u\neq 0.$$

When $u=0$, L'Hopital's rule says that

$$\lim_{u\to 0}D_N(u)=\lim_{u\to 0}\frac{(N+\frac{1}{2})\cos\left((N+\frac{1}{2})u\right)}{\cos(u/2)}=\tfrac{1}{2}+N,$$

so we define

$$D_N(u):=\frac{1}{2}+\sum_{n=1}^{N}\cos(nu)=\begin{cases}\frac{\sin\left(\left(N+\frac{1}{2}\right)u\right)}{2\sin(u/2)} & u\in[-\pi,\pi]\quad\text{with}\quad u\neq 0,\\ N+\frac{1}{2} & u=0.\end{cases}$$

We summarize this, along with some other useful properties of the Dirichlet Kernel $D_N(u)$, in the following theorem.

Theorem 2.2.7 (Properties of the Dirichlet Kernel)

The Dirichlet kernel

$$D_N(u)=\frac{1}{2}+\sum_{n=1}^{N}\cos(nu),\quad N\in\mathbb{N}$$

satisfies the following properties:

(a) (Closed form) For $u \in [-\pi, \pi]$,

$$D_N(u) = \begin{cases} \frac{\sin\left(\left(N+\frac{1}{2}\right)u\right)}{2\sin(u/2)} & u \neq 0, \\ N + \frac{1}{2} & u = 0. \end{cases}$$

(b) (Evenness) The function D_N is even: i.e., $D_N(-u) = D_N(u)$ for all $u \in [-\pi, \pi]$.

(c) (Normalization) The integral of D_N over $[-\pi, \pi]$ exists, and in particular,

$$\int_{-\pi}^{\pi} D_N(t)\, dt = \pi.$$

As a consequence,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} D_N(t)\, dt = 1.$$

(d) (Symmetric half-integrals) As an immediate consequence of (b) and (c),

$$\int_{-\pi}^{\pi} D_N(t)\, dt = 2\int_{-\pi}^{0} D_N(t)\, dt = 2\int_{0}^{\pi} D_N(t)\, dt = \pi,$$

so that, in particular, we have

$$\frac{1}{\pi} \int_{-\pi}^{0} D_N(t)\, dt = \frac{1}{\pi} \int_{0}^{\pi} D_N(t)\, dt = \frac{1}{2}.$$

Proof:

We derived property (a) in the discussion above. For (b), observe that since $\cos(-u) = \cos(u)$ for every u, $\cos(-nu) = \cos(nu)$ for every $n \geq 1$ and every u. Thus,

$$D_N(-u) = \frac{1}{2} + \sum_{n=1}^{N} \cos(n(-u)) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nu) = D_N(u),$$

so that D_N is even on $[-\pi, \pi]$.

For (c), integrate the definition of D_N term by term:

$$\int_{-\pi}^{\pi} D_N(t)\, dt = \int_{-\pi}^{\pi} \frac{1}{2}\, dt + \frac{1}{\pi} \sum_{n=1}^{N} \int_{-\pi}^{\pi} \cos(nt)\, dt.$$

(The interchange in order of integration and summation is justified by the DCT since $|\cos(nt)| \leq 1$ and since the integral is over a bounded interval.) This means that

$$\int_{-\pi}^{\pi} \frac{1}{2}\, dt = \pi, \qquad \int_{-\pi}^{\pi} \cos(nt)\, dt = 0 \text{ by (I2) when } n \geq 1,$$

so

$$\int_{-\pi}^{\pi} D_N(t)\, dt = \pi \implies \frac{1}{\pi} \int_{-\pi}^{\pi} D_N(t)\, dt = 1.$$

For (d), since D_N is even by (b), substitute $u = -t$ in a half-integral over $[-\pi, 0]$ and apply evenness:

$$\int_{-\pi}^{0} D_N(t)\,dt = \int_{0}^{\pi} D_N(-u)\,du = \int_{0}^{\pi} D_N(u)\,du \implies \int_{-\pi}^{0} D_N(t)\,dt = \int_{0}^{\pi} D_N(t)\,dt$$

Since the two half-integrals are equal and their sum equals π by (c), it follows that

$$\frac{1}{\pi}\int_{-\pi}^{0} D_N(t)\,dt = \frac{1}{\pi}\int_{0}^{\pi} D_N(t)\,dt = \frac{1}{2}.$$

‖

It is worth pausing to appreciate the normalization property (c) of the Dirichlet kernel — in fact, it is the reason that Fourier series are such capable approximants over even strangely non-periodic functions. For instance, suppose that $f(x) = c$ a fixed constant for all $x \in [-\pi, \pi]$. In this case, it is immediate that f is square-integrable, since $\int_{-\pi}^{\pi} |f(t)|^2\,dt = 2\pi c^2 < \infty$ — in particular, $f \in L^2[-\pi, \pi]$, so should our current understanding hold, it should be the case that

$$\text{For all } x \in [-\pi,\pi]\colon \quad S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} (a_n \cos(nx) + b_n \sin(nx)) \overset{N\to\infty}{\longrightarrow} f(x) = c.$$

What we'll observe here visually is quite similar to what happened on one half of our motivating example, the square-wave. Indeed, let us now verify that $S_N(x) \to f(x)$ a constant for all $x \in [-\pi, \pi]$:

$$S_N(x) = (f * D_N)(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t) D_N(x-t)\,dt = \frac{c}{\pi}\int_{-\pi}^{\pi} D_N(x-t)\,dt.$$

Substitute $u = t - x$ so that $du = dt$ and the bounds shift from $-pi \mapsto x - \pi$ and $\pi \mapsto x + \pi$, apply the independence of starting point to the 2π-periodic D_N to shift these bounds back to $[-\pi, \pi]$, and apply the evenness property (b) of the Dirichlet kernel:

$$S_N(x) = \frac{c}{\pi}\int_{x-\pi}^{x+\pi} D_N(-u)\,du = \frac{c}{\pi}\int_{-\pi}^{\pi} D_N(-u)\,du = \frac{c}{\pi}\int_{-\pi}^{\pi} D_N(u)\,du.$$

Applying (c) of the properties, we have that $\int_{-\pi}^{\pi} D_N(u)\,du = \pi$, so

$$S_N(x) = \frac{c}{\pi}\cdot\pi = c = f(x)$$

explicitly, for every $N \in \mathbb{N}$. That is, here, we don't even need to take the limit. We can concretely verify that this coincides with what we observe directly: verify that

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt = \frac{c}{\pi}\cdot 2\pi = 2c,$$

that

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} c\cos(nt)\,dt = \frac{c}{\pi}\underbrace{\int_{-\pi}^{\pi}\cos(nt)\,dt}_{=0 \text{ by (I2)}} = 0,$$

and that

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} c\sin(nt)\,dt = \frac{c}{\pi}\underbrace{\int_{-\pi}^{\pi}\sin(nt)\,dt}_{=0 \text{ by (I1)}} = 0,$$

so, together, we have

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N}(a_n\cos(nx) + b_n\sin(nx)) = \frac{2c}{2} + 0 = c = f(x).$$

Proof of the Dirichlet Conditions

We are now in possession of the two structural ingredients previewed at the outset of this discussion:

(i) **Riemann-Lebesgue**: If $\phi \in L^1[a,b]$ is absolutely integrable, then

$$\int_a^b \phi(t)\sin(\lambda t)\,dt \to 0 \quad \text{and} \quad \int_a^b \phi(t)\cos(\lambda t)\,dt \to 0 \quad \text{as} \quad \lambda \to \infty.$$

(ii) **Convolutions**: The Nth partial sum S_N is a convolution of f with the Dirichlet kernel D_N:

$$S_N(x) = (f * D_N)(x),$$

where

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^{N}\cos(nu) = \begin{cases} \frac{\sin\left(\left(N+\frac{1}{2}\right)u\right)}{2\sin(u/2)} & u \neq 0, \\ N + \frac{1}{2} & u = 0, \end{cases}$$

is the Dirichlet kernel — an even, normalized function whose numerator oscillates with frequency $N + \frac{1}{2}$ and which integrates to π over every interval of length 2π.

Before stating and proving the theorem, we introduce one further condition that will play a central role in the argument. As we will see, the proof of pointwise convergence reduces to showing that a certain quotient $\phi(t) = [f(x-t) - f(x^-)]/(2\sin(t/2))$ is absolutely integrable on $[0,\pi]$. Away from $t = 0$, this is straightforward; the difficulty is near $t = 0$, where $\sin(t/2) \to 0$ and the denominator vanishes. Using the small-angle approximation $\sin(t/2) \approx t/2$, the integrability of ϕ near $t = 0$ is essentially equivalent to the finiteness of

$$\int_0^{\delta} \frac{|f(x-t) - f(x^-)|}{t}\,dt.$$

This motivates the following definition.

Definition 2.2.8 (Dini Condition)

A function f satisfies the **Dini condition from the left** at the point x if

$$\int_0^{\delta} \frac{|f(x-t) - f(x^-)|}{t}\,dt < \infty$$

for some $\delta > 0$. Similarly, f satisfies the **Dini condition from the right** at x if

$$\int_0^{\delta} \frac{|f(x+t) - f(x^+)|}{t}\,dt < \infty$$

for some $\delta > 0$.

We say that f satisfies the **Dini condition at** x if it satisfies both of these conditions.

Note that when f is continuous at x — so that $f(x^+) = f(x^-) = f(x)$ — the two one-sided limits can be combined via the triangle inequality, yielding the *single condition*

$$\int_0^\delta \frac{|f(x+t) + f(x-t) - 2f(x)|}{t}\,dt < \infty$$

for some $\delta > 0$. Again, this case is *implied* by the condition as stated in the definition, but ours is *strictly more general* — in particular, we can check a left- and right-Dini condition at jump discontinuities, which is a valuable tool for our current purposes.

The Dini condition is a precise formulation of the requirement that $f(x-t)$ approaches $f(x^-)$ "fast enough" relative to the vanishing of t for the resulting quotient to be integrable. It is strictly weaker than one-sided differentiability: if f is left-differentiable at x, then $[f(x-t) - f(x^-)]/t \to -f'(x^-)$, so the quotient is bounded near $t = 0$ and the integral is trivially finite. But the Dini condition allows the quotient to blow up, so long as it blows up slowly enough to remain integrable — for instance, the function $f(x-t) - f(x^-) = t^{1/2}$ gives a quotient $t^{-1/2}$, which diverges as $t \downarrow 0$ but is integrable. In particular, any piecewise smooth function — and indeed any function with one-sided derivatives at each point — satisfies the Dini condition everywhere.

We now state the theorem.

Theorem 2.2.9 (Pointwise Convergence of Fourier Series (Dini))

Let $f\colon \mathbb{R} \to \mathbb{R}$ be 2π-periodic. Suppose that on $[-\pi, \pi]$:

(i) f is bounded: $|f(x)| \leq M$ for all $x \in [-\pi, \pi]$.

(ii) f satisfies the Dini condition at x from both the left and the right.

Then

$$\lim_{N\to\infty} S_N(x) = \frac{f(x^-) + f(x^+)}{2}.$$

In particular:

(i) $S_N(x_0) \to f(x_0)$ at every point $x_0 \in [-\pi, \pi]$ at which f is continuous and satisfies the Dini condition.

(ii) $S_N(x_r) \to \lim_{x\to x_r} f(x)$ at any removable discontinuity x_r satisfying the Dini condition.

(iii) $S_N(x_j) \to \frac{f(x_j^-)+f(x_j^+)}{2}$ at any jump discontinuity x_j satisfying the Dini condition.

Proof:

Fix $x \in [-\pi, \pi]$. We want to show that

$$S_N(x) \to \frac{f(x^+) + f(x^-)}{2} \quad \text{as} \quad N \to \infty.$$

We showed above that the Nth Fourier partial sum can be written as the convolution

$$S_N(x) = (f * D_N)(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t) D_N(x-t)\,dt.$$

For the proof, it will be more convenient to have D_N evaluated at t directly. Substituting $u = x - t$ so that $t = x - u$, $du = -dt$, the bounds shift to $[x-\pi, x+\pi]$; since both f (after periodic extension) and D_N are 2π-periodic, the independence of starting point allows us to shift back to $[-\pi, \pi]$, giving

$$S_N(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x-t) D_N(t)\, dt.$$

Since the normalization property (c) gives $\frac{1}{\pi}\int_{-\pi}^{\pi} D_N(t)\, dt = 1$, we can write the target value as an integral against D_N. Since the symmetric half-integrals (d) give $\frac{1}{\pi}\int_0^{\pi} D_N(t)\, dt = \frac{1}{\pi}\int_{-\pi}^{0} D_N(t)\, dt = \frac{1}{2}$, we have

$$\frac{f(x^+) + f(x^-)}{2} = \frac{1}{\pi}\int_{-\pi}^{0} f(x^+) D_N(t)\, dt + \frac{1}{\pi}\int_0^{\pi} f(x^-) D_N(t)\, dt,$$

where we've assigned $f(x^+)$ to the integral over $(-\pi, 0)$ since, when $t < 0$, $x - t > x$ and we approach f from the right, and $f(x^-)$ to the integral over $(0, \pi)$ since, when $t > 0$, $x - t < x$ and we approach f from the left.

Splitting the integral for S_N at $t = 0$ and subtracting,

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = \frac{1}{\pi}\int_{-\pi}^{0} \big[f(x-t) - f(x^+)\big] D_N(t)\, dt + \frac{1}{\pi}\int_0^{\pi} \big[f(x-t) - f(x^-)\big] D_N(t)\, dt.$$

For the first integral, substitute $t \mapsto -t$ and use the evenness (b), $D_N(-t) = D_N(t)$:

$$\frac{1}{\pi}\int_{-\pi}^{0} \big[f(x-t) - f(x^+)\big] D_N(t)\, dt = \frac{1}{\pi}\int_0^{\pi} \big[f(x+t) - f(x^+)\big] D_N(t)\, dt.$$

It therefore suffices to show that both

$$I_N^+ := \frac{1}{\pi}\int_0^{\pi} \big[f(x+t) - f(x^+)\big] D_N(t)\, dt \quad \text{and} \quad I_N^- := \frac{1}{\pi}\int_0^{\pi} \big[f(x-t) - f(x^-)\big] D_N(t)\, dt$$

tend to zero as $N \to \infty$. The arguments are structurally identical, so we treat I_N^-.

Substituting the closed form (a) of the Dirichlet kernel $D_N(t)$, we can write the integral for I_N^- as

$$I_N^- = \frac{1}{\pi}\int_0^{\pi} \underbrace{\frac{f(x-t) - f(x^-)}{2\sin(t/2)}}_{=:\,\phi(t)} \sin\!\Big((N + \tfrac{1}{2})t\Big)\, dt = \frac{1}{\pi}\int_0^{\pi} \phi(t) \sin\!\Big((N + \tfrac{1}{2})t\Big)\, dt.$$

If $\phi \in L^1[0, \pi]$, then the Riemann-Lebesgue lemma gives $I_N^- \to 0$ as $N \to \infty$ immediately. The entire proof thus reduces to showing that ϕ is absolutely integrable.

On the interval $[\delta, \pi]$ for any fixed $\delta > 0$, there is no difficulty: $\sin(t/2) \geq \sin(\delta/2) > 0$ is bounded away from zero, and f is bounded by hypothesis, so

$$|\phi(t)| \leq \frac{|f(x-t)| + |f(x^-)|}{2\sin(\delta/2)} \leq \frac{M}{\sin(\delta/2)},$$

which is a constant. Thus $\phi \in L^1[\delta, \pi]$ for any $\delta > 0$.

Near $t = 0$, the denominator $\sin(t/2) \to 0$, and the bound above breaks down. Since $\sin(t/2) \sim t/2$ for small t, the integrability of ϕ near the origin is controlled by

$$\int_0^\delta |\phi(t)|\, dt \sim \int_0^\delta \frac{|f(x-t) - f(x^-)|}{t}\, dt,$$

which is precisely the Dini condition from the left at x. Since f satisfies the Dini condition at x by hypothesis, this integral is finite, and we conclude that $\phi \in L^1[0, \pi]$.

By the Riemann-Lebesgue lemma,

$$I_N^- = \frac{1}{\pi} \int_0^\pi \phi(t) \sin\Big((N + \tfrac{1}{2})t\Big)\, dt \to 0 \quad \text{as} \quad N \to \infty.$$

The identical argument with ϕ replaced by $\tilde{\phi}(t) = [f(x+t) - f(x^+)]/(2\sin(t/2))$ and the Dini condition from the right yields $I_N^+ \to 0$. Since

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = I_N^+ + I_N^-,$$

we conclude

$$S_N(x) \to \frac{f(x^+) + f(x^-)}{2} \quad \text{as} \quad N \to \infty.$$

‖

It is worth reviewing where each ingredient did its work. The convolution structure allowed us to write $S_N(x)$ as a single integral against D_N, converting a question about infinite series into a question about one integral. The normalization and evenness of D_N allowed us to subtract the target value cleanly, reducing the error to two symmetric integrals I_N^+ and I_N^-. Substituting the closed form of D_N revealed each as an integral of $\phi(t)$ against a high-frequency sinusoid, at which point the Riemann-Lebesgue lemma killed the integrals — provided $\phi \in L^1$. The Dini condition is precisely the hypothesis that makes this last step work: it ensures that ϕ is integrable near $t = 0$, where the denominator $\sin(t/2)$ vanishes.

One natural question is whether the Dini condition can be weakened further. After all, the classical Dirichlet conditions require only that f be bounded, of bounded variation, and have finitely many jump discontinuities — no explicit integrability condition on a difference quotient. Indeed, as it turns out, though the Dini condition was the natural tool for the proof we just gave, it is actually too strong for handling points of functions that "decay too slowly" as $f(x-t) - f(x^-) \to 0$ to counterbalance the denominator $t \downarrow 0$.

A Further Generalization: Breaking the Dini Condition

To see that the gap is genuine, consider the following example. Define $m\colon [0, 1/2] \to \mathbb{R}$ by

$$m(t) = \frac{1}{\log(1/t)}, \quad t \in (0, 1/2], \qquad m(0) = 0.$$

Then m is continuous, monotone increasing, bounded, and of bounded variation on $[0, 1/2]$. Its one-sided limit at 0 is $m(0^+) = \lim_{t \downarrow 0} 1/\log(1/t) = 0 = m(0)$, so there is no jump at the origin.

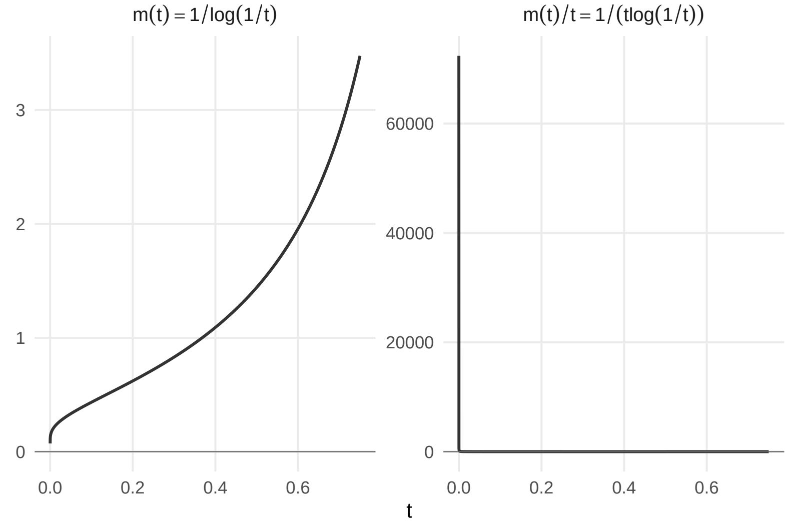

Figure: The function $m(t) = 1/\log(1/t)$ on $(0, 1/2]$. The function is monotone increasing, continuous, and of bounded variation, yet the difference quotient $m(t)/t = 1/(t\log(1/t))$ is not integrable near $t = 0$: the Dini condition fails.

In particular, m satisfies every one of the Dirichlet hypotheses. Yet the difference quotient

$$\frac{m(t) - m(0)}{t} = \frac{1}{t\log(1/t)}$$

is *not* integrable near $t = 0$: by the substitution $u = \log(1/t)$,

$$\int_0^{1/2} \frac{1}{t\log(1/t)}\,dt = \int_{\log 2}^{\infty} \frac{1}{u}\,du = +\infty.$$

What this would imply is that at $t = 0$, m fails to satisfy the Dini condition. According to our theorem as stated, this would seem to indicate that the Fourier series of m at $t = 0$ wouldn't converge — yet, if we look to the plots below, we observe a different phenomenon, one seeming to indicate the contrary:

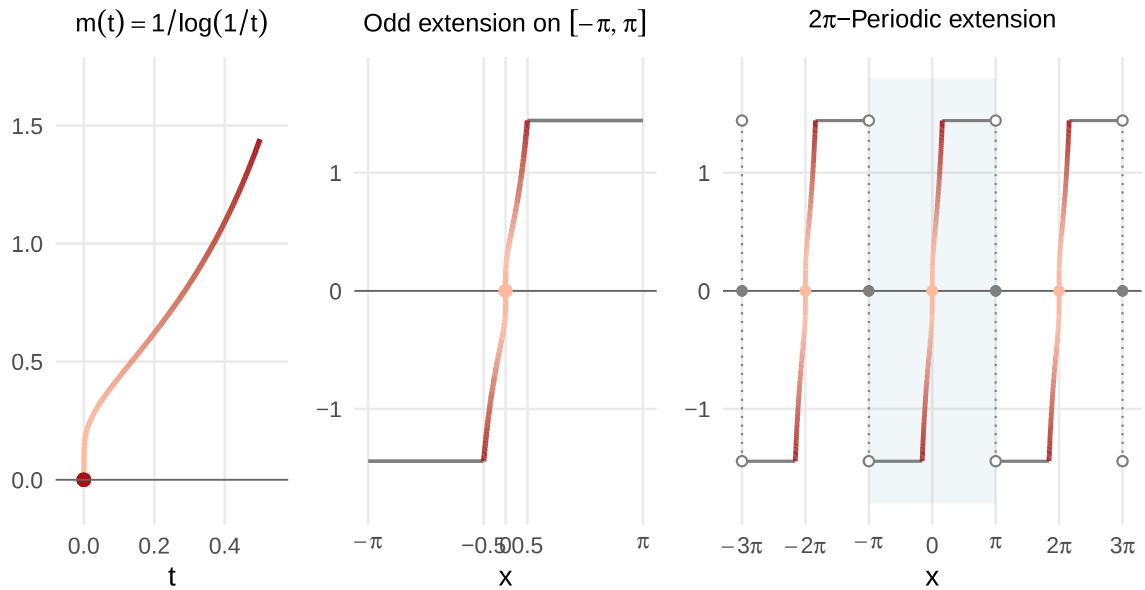

Figure: Construction of the 2π-periodic odd extension of $m(t) = 1/\log(1/t)$. Left: the original function on $(0, 1/2]$. Center: the odd extension $f(x)$ on $[-\pi, \pi]$, with the original piece and its odd reflection shown in color. Right: a few periods of the periodic extension $f^\star(x)$, with the home period $[-\pi, \pi]$ shaded.

The Dini condition fails for m at $t = 0$, even though m is continuous, monotone, and of bounded variation. The function approaches its limit at the origin, but it approaches *too slowly* — the decay of the numerator cannot outpace the blow-up of $1/t$ in the difference quotient $\frac{f(x-t)-f(x^-)}{t}$ as $t \downarrow 0$. *Nevertheless*, we observe directly in the plots that at $t = 0$, it appears as though $S_N(0) \approx m(0)$ once we reach around $N = 200$ — this certainly required us to compute more of the terms a_n, b_n in $S_N(x)$, yet we seem to have reached something of an impasse regardless. Indeed, if it *is* true that $S_N(0) \to m(0)$ as $N \to \infty$, then our statement of the Dirichlet conditions — in particular, the Dini condition — is simply too strong to cover the most ill-behaved points of some functions.

To find out whether or not this can be rectified, let us retrace our steps. In the proof we presented, we arrived at

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = I_N^- + I_N^+,$$

where we'd defined the integral

$$I_N^- = \frac{1}{\pi} \int_0^\pi [f(x-t) - f(x^-)] D_N(t)\, dt.$$

where

$$D_N(t) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nt).$$

We then proceeded to write the Dirichlet kernel D_N equivalently in closed-form as

$$D_N(t) = \frac{\sin\left((N + \frac{1}{2})t\right)}{2\sin(t/2)},$$

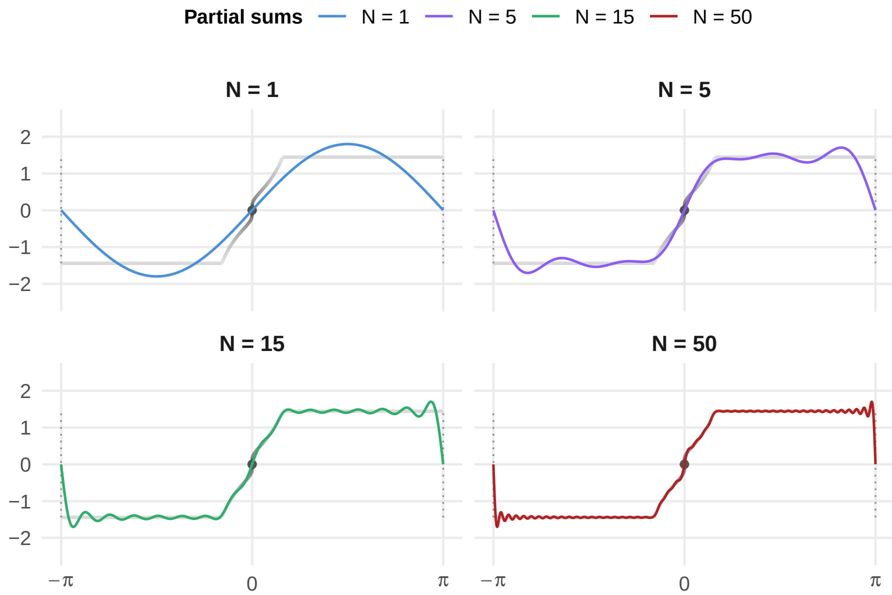

Figure: Fourier partial sums $S_N(x)$ for the odd extension of $m(t) = 1/\log(1/t)$ on $[-\pi, \pi]$. Gibbs overshoot is visible at $x = \pm\pi$ where the periodic extension introduces jump discontinuities.

which allowed us to rewrite the integrand

$$[f(x-t) - f(x^-)]D_N(t) = \underbrace{\frac{f(x-t) - f(x^-)}{2\sin(t/2)}}_{:=\phi(t)} \sin\left((N + \tfrac{1}{2})t\right).$$

It was the decision to "move the denominator $\sin(t/2)$ from D_N over to $f(x-t) - f(x^-)$," in turn defining

$$\phi(t) := \frac{f(x-t) - f(x^-)}{2\sin(t/2)}$$

that, at the time, seemed most appropriate for the proof. And, indeed, this setup works completely fine anytime $\sin(t/2)$ is bounded away from zero, or anytime the numerator $f(x-t) - f(x^-)$ decays "fast enough" as $t \downarrow 0$ to offset the rate at which $\sin(t/2) \approx t/2 \downarrow 0$ as $t \downarrow 0$ in the denominator. In either case, we're able to bound $\phi(t)$, so Riemann-Lebesgue allowed us to conclude that multiplying the bounded ϕ by a $\sin\left((N + \frac{1}{2})t\right)$ contribution (whose argument $N + \frac{1}{2} \to \infty$ as $N \to \infty$) leads to the original integral $I_N^- \to 0$ as $N \to \infty$. By a symmetric argument, $I_N^+ \to 0$ as $N \to \infty$, leading to pointwise convergence at any point satisfying the Dini condition.

But what happened at $t = 0$ for our counterexample $m(t) = 1/\log(1/t)$ is that the numerator decayed *too slowly* — in particular, if f is the 2π-extension of m, then

$$f(0^-) = \lim_{t \uparrow 0} f(t) = 0,$$

and

$$f(0-t) = f(-t) = -m(t).$$

Thus, near $t = 0$,

$$\phi(t) = \frac{f(0-t) - f(0^-)}{2\sin(t/2)} \approx \frac{-m(t)}{t} = -\frac{1}{t\log(1/t)},$$

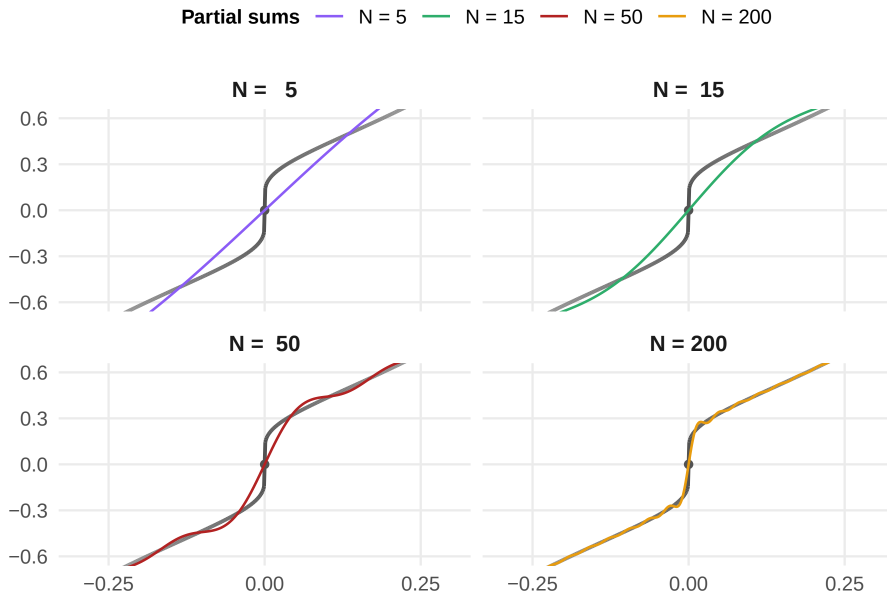

Figure: Zoomed view of Fourier partial sums near $x = 0$, where the Dini condition fails. The convergence is visibly slower than at points where f is smooth — the partial sums require more terms to track the function's logarithmically slow rise from the origin. Nevertheless, pointwise convergence occurs, as guaranteed by the Dirichlet-Jordan theorem.

and, as in the introduction to this ribbon, we substitute $u = \log(1/t)$, so $du = -1/t\,dt$ and conclude, for any $\delta \in (0, 1/2)$,

$$\int_0^\delta \frac{1}{t\log(1/t)}\,dt = \int_{\log(1/\delta)}^\infty \frac{1}{u}\,du = +\infty.$$

This means that the original function ϕ is not integrable near 0, so the Riemann-Lebesgue machinery breaks down — the intuition being that the multiplication by the $\sin\big((N+\frac{1}{2})t\big)$ term acts like weighting $\phi(t)$ by a number in $[-1, 1]$, so if ϕ is "too big" then we may not be sure that the sine-weights "dampen enough" for the worst points $t \mapsto \phi(t)$.

It is worth pointing out that, in the counterexample here, we've consolidated interest on an isolated "bad actor" at $t = 0$ — one would wonder whether we need to take such precautions over other points in the domain. Intuitively, we find it to be somewhat clear that the Dini condition will hold for *almost* every point whenever the Dirichlet conditions hold; that is, the set of points for which the Dini condition fails yet the Dirichlet series still converges pointwise is of *measure zero*. The real question, then, is not whether such points exist — they do, as we've just demonstrated — but whether our proof technique can be *adapted* to handle them. And the answer is yet, provided we make a different choice at a crucial step.

Going back to the integral I_N^-, let us forgo the explicit closed-form of D_N and instead work with it directly. In particular,

$$I_N^- = \frac{1}{\pi}\int_0^\pi \underbrace{[f(x-t) - f(x^-)]}_{:=g(t)} D_N(t)\,dt := \frac{1}{\pi}\int_0^\pi g(t)D_N(t)\,dt,$$

where we've defined $g(t) := f(x-t) - f(x^-)$ as a function of t. By forgoing the closed-form expression of D_N, we've precluded the denominator from entering the picture; we want to keep this structure intact, since it was that denominator that led to the singularity at $t = 0$ we're attempting to rectify.

Now, break the integral I_N^- above apart at an arbitrary $\delta > 0$ — we have that

$$I_N^- = \frac{1}{\pi}\int_0^\delta g(t)D_N(t)\,dt + \frac{1}{\pi}\int_\delta^\pi g(t)D_N(t)\,dt.$$

Here, we can freely apply the former proof technique to the second integral — on $[\delta, \pi]$, the denominator $\sin(t/2) \geq \sin(\delta/2) > 0$ is bounded away from zero, so we *can* "split the closed-form for D_N apart, define ϕ as before, realize its bounded, conclude $\phi \in L^1[\delta, \pi]$, so Riemann-Lebesgue applies to the integral $\phi(t)\sin\left((N+\frac{1}{2})t\right)$ since $\sin\left((N+\frac{1}{2})t\right)$ with $N + \frac{1}{2} \to \infty$ as $N \to \infty$."

The difficulty, then, is only in the first integral,

$$I_\delta := \frac{1}{\pi}\int_0^\delta g(t)D_N(t)\,dt, \qquad g(t) := f(x-t) - f(x^-).$$

To tackle the integral I_δ, we know we cannot emulate our previous approach and substitute the explicit form of $D_N(t)$ since it involves a denominator that goes to 0 as $t \downarrow 0$. Instead, we need a tool that allows us to work with the integrand $g(t)D_N(t)$ without ever introducing the denominator in the explicit form of D_N. What we need here is, actually, a generalization of the Riemann integral.

Riemann-Stieltjes Integral

First, we recall an important definition.

Definition 2.2.10 (Riemann Integral)

Let $f\colon [a,b] \to \mathbb{R}$ be bounded — i.e., $|f(t)| \leq M$ for all $t \in [a,b]$. Given a partition of $[a,b]$

$$\mathcal{P}_k := \{t_0, t_1, \dots, t_k\}, \qquad a = t_0 < t_1 < \cdots < t_k = b,$$

and a collection of sample points $t_i^\star \in [t_i, t_{i+1}]$ — say, $t_i^\star = t_i$, $t_i^\star = t_{i+1}$, $t_i^\star = \frac{t_i + t_{i+1}}{2}$, etc. — the **Riemann sum** of f with respect to $\mathcal{P}$ is

$$R(f, \mathcal{P}_k) := \sum_{i=0}^{k-1} f(t_i^\star)\Delta t_i, \qquad \Delta t_i := t_{i+1} - t_i.$$

The **Riemann integral** of f on $[a,b]$ is defined as the limit of these sums as we take finer and finer partitions of $[a,b]$ — in particular, if

$$\|\mathcal{P}_k\| := \max_i \Delta t_i \to 0 \quad \text{as} \quad k \to \infty$$

for

$$\lim_{\|\mathcal{P}_k\| \to 0} R(f, \mathcal{P}_k) \text{ exists,}$$

then we define the Riemann integral as this common limit of Riemann sums of f:

$$\int_a^b f(t)\,dt := \lim_{\|\mathcal{P}_k\| \to 0} R(f, \mathcal{P}_k).$$

The essential content of this definition is that we are "slicing up the domain $[a, b]$ into small strips of width Δt_i, evaluating f at a representative point $t_i^\star$ from within each strip, and summing the contributions $f(t_i^\star) \cdot \Delta t_i$." If there were something peculiar that happened when we chose, say, the left endpoint $t_i^\star = t_i$ as the representative instead of the right $t_i^\star = t_{i+1}$, then the Riemann integral wouldn't exist. An idiosyncrasy of this definition is that it is very sensitive to the choice of partition — realize that we must be able to make the limit of the Riemann sums agree on *any* such partition. This is, in fact, is the primary reason we wind up caring about the Lebesgue integral.

However, in the current setting, we're interested in something slightly different. Consider the integral $\int_a^b g(t)D_N(t)\,dt$ we've been working with. When we write it as a Riemann integral, we're treating $g(t) \cdot D_N(t)$ as a *single function* — a more honest notation might be $(g \cdot D_N)(t)$ — and integrating it against dt, the **differential** which, crucially, remains of *uniform width* as we evaluate over different strips of $[a, b]$. That is, when we imagine performing a Riemann integral, we're not imagining any sort of bias on behalf of the limiting partition.

On the other hand, this begs the question of what would happen if we *did* bias the partition. In particular, consider the following situation: suppose we have two functions f and g, both defined on $[a, b]$, and we're interested in evaluating

$$\int_a^b f(t)g'(t)\,dt.$$

Clearly we're supposing that g is differentiable on $[a, b]$ in the notation above — on the other hand, let us consider what happens in the Riemann sum: by definition, given any sequence of partitions $\mathcal{P}_k = \{t_i\}_{i=0}^k$ of $[a, b]$ $(t_0 = a < t_1 < \cdots < t_k = b)$such that $\|\mathcal{P}_k\| = \max_i \Delta t_i \to 0$ as $k \to \infty$, then the (Riemann) integral above is

$$\int_a^b f(t)g'(t)\,dt := \lim_{\Delta t_i \downarrow 0} \sum_{i=0}^{k-1} f(t_i^\star)g'(t_i^\star)\Delta t_i.$$

Now, let us recall an important theorem.

Theorem 2.2.11 (Mean Value Theorem (MVT))

Suppose f is continuous on $[a, b]$ and differentiable over (a, b). Then, there exists at least one point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

If f is linear, then the MVT holds for any $c \in [a, b]$. What is special, then, about the MVT is that it holds for *any* differentiable function. However, the visual proof just requires examination of the tangent line of such a differentiable function:

As for our purposes, we utilize the mean value theorem as follows: in the Riemann sum above, each term $f(t_i^\star)g'(t_i^\star)\Delta t_i$ has a factor $g'(t_i^\star)\Delta t_i$, which is the derivative of g evaluated at a point

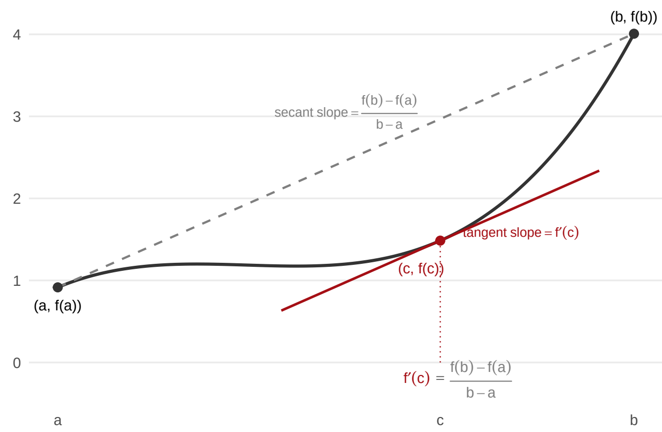

Figure: The Mean Value Theorem. The slope of the secant line (dashed) connecting $(a, f(a))$ and $(b, f(b))$ equals the slope of the tangent line (red) at some interior point $c \in (a, b)$.

$t_i^\star \in [t_i, t_{i+1}]$ from within the current increment $\Delta t_i = t_{i+1} - t_i$. Yet by the mean value theorem as stated, there exists some $c_i \in (t_i, t_{i+1})$ such that

$$g'(c_i) = \frac{g(t_{i+1}) - g(t_i)}{\Delta t_i};$$

therefore, for any increment in the Riemann sum Δt_i, $i = 0, 1, \dots, k$,

$$g'(c_i)\Delta t_i = g(t_{i+1}) - g(t_i) \quad \text{for some} \quad c_i \in (t_i, t_{i+1}).$$

Now, realize that $t_i^\star := c_i$ as defined above is a perfectly valid sample point satisfying our definition of the Riemann integral. Under this choice, we obtain

$$\sum_{i=0}^{k-1} f(c_i)g'(c_i)\Delta t_i = \sum_{i=0}^{k-1} f(c_i) \underbrace{[g(t_{i+1}) - g(t_i)]}_{:=\,\Delta g_i} := \sum_{i=0}^{k-1} f(c_i)\Delta g_i.$$

Supposing "all of this checks out," then the limit of the left-side above is still just the Riemann integral of $f(t)g(t)\,dt$; and, thus, equal to the limit of the Riemann sum on the right. What we might be inclined to write, then, is

$$\lim_{\|\Delta g_i\| \downarrow 0} \sum_{i=0}^{k-1} f(c_i)\Delta g_i := \int_a^b f(t)\,dg(t)$$

so that, in particular, we have

$$\int_a^b f(t)g'(t)\,dt = \int_a^b f(t)\,dg(t).$$

Remark.

In the **Riemann-Stieltjes sum** $\sum f(t_i^\star)\,\Delta g_i$, the increments $\Delta g_i = g(t_{i+1}) - g(t_i)$ are *inherently g-weighted*: as the partition $\mathcal{P}_k$ refines (i.e., $\|\mathcal{P}_k\| = \max_i \Delta t_i \to 0$), the uniform increments Δt_i

shrink at roughly the same rate across all strips, but the increments Δg_i need not. Where g is steep, Δg_i remains large relative to Δt_i; where g is flat, Δg_i is negligible by comparison. Of course, both will "still go to zero," but at the scale of infinitesimals, it is these smallest of nudges which can significantly alter a tool's applicability, as we'll soon verify firsthand.

Now, what is quite interesting about the formulation above is that, if we step back to the limit definition, writing

$$\lim_{\|\Delta g_i\| \downarrow 0} \sum_{i=0}^{k-1} f(c_i)\Delta g_i, \qquad \Delta g_i := g(t_{i+1}) - g(t_i),$$

we realize that nothing in this expression actually *requires* g to be differentiable — indeed, we *only* ever use the values g takes on at partition points $a = t_0 < t_1 < \cdots < t_k = b$ in the computation above (in particular, we don't require that $\Delta g_i / \Delta t_i$ to a finite limit for the limit on the left to make sense, and this is precisely what it means for g to be differentiable.) We only invoked differentiability to *arrive* at this expression (via the MVT), but the expression itself is a more primitive object: it asks only that g be *defined* at each of the t_i in our partition, and nothing more. This suggests that perhaps the integral $\int_a^b f(t)\,dg(t)$ ought to make sense in far greater generality than the case when g is differentiable, reducing it to $\int_a^b f(t)g'(t)\,dt$.

One further subtlety to note is that, in the derivation above, the sample points c_i were not chosen arbitrarily — they were the specific points furnished by the mean value theorem satisfying $g'(c_i)\Delta t_i = \Delta g_i$. We showed that the Riemann sum, evaluated at these particular c_i, takes the form $\sum f(c_i)\,\Delta g_i$. But if we are to take this expression seriously as a *definition* of a new integral — one that does not presuppose differentiability of g — then we must demand more: the limit

$$\lim_{\|\mathcal{P}_k\| \to 0} \sum_{i=0}^{k-1} f(t_i^\star)\,\Delta g_i$$

must converge to the same value regardless of the partition $\mathcal{P}_k$ and the representatives $t_i^\star$ chosen from within each subinterval. This is not something we've inherited from the derivation: it is an additional requirement that we impose in order for $\int f\,dg$ to be well-defined.

Now, we're ready to present a formal definition.

Definition 2.2.12 (Riemann-Stieltjes Integral)

Let $f, \alpha\colon\ [a,b] \to \mathbb{R}$ be bounded. Given a partition $\mathcal{P}_k = \{t_0, t_1, \dots, t_k\}$ of $[a,b]$ and sample points $t_i^\star \in [t_i, t_{i+1}]$, the **Riemann-Stieltjes sum** of f with respect to the **intergator** α is

$$S(f, \alpha, \mathcal{P}_k) := \sum_{i=0}^{k-1} f(t_i^\star)\Delta\alpha_i, \quad \Delta\alpha_i := \alpha(t_{i+1}) - \alpha(t_i).$$

If the limit

$$\lim_{\|\mathcal{P}_k\| \to 0} S(f, \alpha, \mathcal{P}_k)$$

exists and is the same for every choice of sample points $t_i^\star \in [t_i, t_{i+1}]$, then we say f is **Riemann-**

Stieltjes integrable with respect to the **integrator** α on $[a, b]$, and we write

$$\int_a^b f(t)\, d\alpha(t).$$

Remark.

Occasionally, we suppress the t's from the notation, and write the Riemann-Stieltjes integral of f w.r.t. α over $[a, b]$ simply as $\int_a^b f\, d\alpha$. This should not cause confusion, since both the function f and integrator α are defined as univariate functions on $[a, b]$.

When $\alpha(t) = t$, we have $\Delta\alpha_i = \Delta t_i$ and we recover the ordinary Riemann integral. When α is differentiable, we may write

$$d\alpha(t) = \alpha'(t)\, dt,$$

in which case we recover the familiar

$$\int_a^b f(t)\, d\alpha(t) = \int_a^b f(t)\alpha'(t)\, dt.$$

But, crucially, we must realize that the Riemann-Stieltjes integral is *strictly more general* than the Riemann integral — that is, though we've arrived at the framing through these particular examples, the definition as presented covers a variety of functions that simply would not make sense — namely, those for which the integrator α is non-differentiable. Indeed, we can have α be a non-differentiable, non-continuous, a step-function — all that matters is that the increments $\Delta\alpha_i$ are well-defined, and that the limit is independent of sample point.

Naturally, one asks: under what conditions does the Riemann-Stieltjes integral exist? The following classical result presents a sufficient condition for Riemann-Stieltjes integrability, wherein we recollect a familiar sub-condition — one that shall signal a return home.

Theorem 2.2.13 (Existence of the Riemann-Stieltjes Integral)

If f is continuous on $[a, b]$ and α is of **bounded variation** on $[a, b]$, then the Riemann-Stieltjes integral

$$\int_a^b f(t)\, d\alpha(t)$$

exists.

We defer the proof of this result, as it is primarily technical (the key ingredients being the uniform continuity of f on a compact interval and the control on the increments $\Delta\alpha_i$ provided by bounded variation). Instead, let us now attempt to motivate why *bounded variation* enters the story here, and what it provides that will be useful to us here.

Definition 2.2.14 (Bounded Variation)

Let $\alpha\colon [a,b] \to \mathbb{R}$. Define the **total variation** of α over $[a,b]$ to be

$$V_a^b(\alpha) := \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)|,$$

where the supremum is taken over all partitions $\mathcal{P}$ of $[a,b]$. The total variation thus measures the "total amount of movement" of α on $[a,b]$. If this total movement is finite — i.e., if

$$V_a^b(\alpha) < \infty,$$

then we say that α is of **bounded variation** on $[a,b]$.

Let us compute the total variation for a few functions of interest. First, suppose that α is **absolutely continuous** (really, you can just imagine this says **continuously differentiable**) on $[a,b]$. In this case, we have the following.
Fair enough, that's on me. Here's the clean version:

Definition 2.2.15 (Bounded Variation)

Let $\alpha\colon [a,b] \to \mathbb{R}$. Define the **total variation** of α over $[a,b]$ to be

$$V_a^b(\alpha) := \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)|,$$

where the supremum is taken over all partitions $\mathcal{P}$ of $[a,b]$. The total variation thus measures the "total amount of movement" of α on $[a,b]$. If this total movement is finite — i.e., if

$$V_a^b(\alpha) < \infty,$$

then we say that α is of **bounded variation** on $[a,b]$.

Let us compute the total variation for a few functions of interest. First, suppose that α is **continuously differentiable** on $[a,b]$. In this case, we have the following.

Theorem 2.2.16 (TV: Continuously Differentiable Functions)

Suppose that $\alpha\colon [a,b] \to \mathbb{R}$ is continuously differentiable. Then

$$V_a^b(\alpha) = \int_a^b |\alpha'(t)|\, dt.$$

Proof:

We prove both directions of the analogous inequality. Since $\alpha \in C^1[a,b]$, the fundamental theorem of calculus gives

$$\alpha(x) = \alpha(a) + \int_a^x \alpha'(t)\,dt, \qquad \forall x \in [a,b].$$

For any partition $\mathcal{P} = \{a = t_0, t_1, \dots, t_k = b\}$ of $[a,b]$, it follows that

$$|\alpha(t_{i+1}) - \alpha(t_i)| = \left|\int_{t_i}^{t_{i+1}} \alpha'(t)\,dt\right| \le \int_{t_i}^{t_{i+1}} |\alpha'(t)|\,dt,$$

where the inequality is an application of the triangle inequality. Summing over $\mathcal{P}$,

$$\sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| \le \sum_{i=0}^{k-1} \int_{t_i}^{t_{i+1}} |\alpha'(t)|\,dt = \int_a^b |\alpha'(t)|\,dt.$$

Since this holds for every partition $\mathcal{P}$, it holds for the supremum over $\mathcal{P}$, yielding

$$V_a^b(\alpha) \le \int_a^b |\alpha'(t)|\,dt.$$

For the reverse inequality, first observe that since α' is continuous on $[a,b]$ (by hypothesis), so too is the composition $|\alpha'|$ — in particular, $|\alpha'|$ is Riemann integrable. One way of formalizing the notion that $|\alpha'|$ is Riemann integrable — which, again, says that as we take finer and finer partitions $\mathcal{P}$ of the Riemann sum $\sum_{\mathcal{P}} |\alpha'(t_i)|\Delta t_i$, these sums converge to a common limit, which we call $\int_a^b |\alpha'(t)|\,dt$ — says that

- Given $\epsilon > 0$,
- There exists $\delta = \delta(\epsilon) > 0$ such that:
- For any partition $\mathcal{P} = \{a = t_0, t_1, \dots, t_k = b\}$ of $[a,b]$ with mesh $\|\mathcal{P}\| = \sup_i \Delta t_i < \delta$,
- And any evaluation points $\xi_i \in [t_i, t_{i+1}]$:

We have

$$\left| \underbrace{\sum_{i=0}^{k-1} |\alpha'(\xi_i)|\Delta t_i}_{\text{Riemann sum where } \|\mathcal{P}\| < \delta} - \int_a^b |\alpha'(t)|\,dt \right| < \epsilon.$$

Fix such a partition. Since α' is continuous on each closed subinterval $[t_i, t_{i+1}]$, the mean value theorem guarantees the existence of points $c_i \in (t_i, t_{i+1})$ with

$$\alpha(t_{i+1}) - \alpha(t_i) = \alpha'(c_i)\Delta t_i.$$

Taking absolute values and summing over the partition,

$$\sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| = \sum_{i=0}^{k-1} |\alpha'(c_i)|\Delta t_i.$$

But the points $c_i \in [t_i, t_{i+1}]$ are evaluation points (cf. ξ_i in the above). In particular, since $\|\mathcal{P}\| < \delta$, we have

$$-\epsilon < \sum_{i=0}^{k-1} |\alpha'(c_i)|\Delta t_i - \int_a^b |\alpha'(t)|\,dt < \epsilon;$$

working only with the left inequality and adding the integral,

$$\sum_{i=0}^{k-1} |\alpha'(c_i)|\Delta t_i > \int_a^b |\alpha'(t)|\,dt - \epsilon.$$

By the MVT, the left side is $\sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)|$, so

$$V_a^b(\alpha) = \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| \geq \underbrace{\sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)|}_{\text{Partition with } \|\mathcal{P}\|<\delta} > \int_a^b |\alpha'(t)|\,dt - \epsilon.$$

Since $\epsilon > 0$ was arbitrary,

$$V_a^b(\alpha) \geq \int_a^b |\alpha'(t)|\,dt.$$

Combining ($\leq$) and ($\geq$) gives the result.

$\|$

Next, we *no longer* suppose α is continuously differentiable on $[a, b]$, nor that α is even continuous. Rather, we only suppose that **monotone increasing** on $[a, b]$ — that is,

$$s \leq t \in [a, b] \implies \alpha(s) \leq \alpha(t) \in \mathbb{R}.$$

Then for any partition $\mathcal{P} = \{t_0, \dots, t_k\}$ (where recall we have $a = t_0 < t_1 < \cdots < t_k = b$), each absolute value inside the sum of $V_a^b(\alpha)$ is nonnegative — in particular, the resulting sum telescopes, so we obtain

$$\sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| = \sum_{i=0}^{k-1} \Big[\alpha(t_{i+1}) - \alpha(t_i)\Big] = \alpha(t_k) - \alpha(t_0) \equiv \alpha(b) - \alpha(a),$$

which is independent of the partition $\mathcal{P}$. In particular, for *any* monotone increasing function α, the total variation is

$$V_a^b(\alpha) = \alpha(b) - \alpha(a).$$

A symmetric argument for **monotone decreasing** functions gives

$$V_a^b(\alpha) = \alpha(a) - \alpha(b);$$

thus, combining the cases, we can assert that for any (possibly discontinuous) monotone function $\alpha\colon\ [a, b] \to \mathbb{R}$, we have

$$V_a^b(\alpha) = |\alpha(b) - \alpha(a)|.$$

We can extend the intuition from the example above to apply to **piecewise monotone functions** with finitely many turns. In particular, suppose that $\alpha\colon\ [a, b] \to \mathbb{R}$ is monotone on each of the subintervals $[a, c_1], [c_1, c_2], \dots, [c_p, b]$ for some finite collection of turning points $a < c_1 < \cdots < c_p < b$. Then the total variation is additive over subintervals — in particular, it follows by definition that

$$V_a^b(\alpha) = V_a^{c_1}(\alpha) + V_{c_1}^{c_2}(\alpha) + \cdots + V_{c_p}^b(\alpha),$$

and since α is monotonic on each piece, we have that $V_{c_j}^{c_{j+1}}(\alpha) = |\alpha(c_{j+1}) - \alpha(c_j)|$ — i.e.,

$$V_a^b(\alpha) = |\alpha(c_1) - \alpha(a)| + |\alpha(c_2) - \alpha(c_1)| + \cdots + |\alpha(b) - \alpha(c_p)|.$$

That is, the total variation of α over all of $[a, b]$ simply sums the change in α over each of its monotonic segments. The idea is that we're "tracking the total magnitude of change α accumulates over the whole of $[a, b]$" — hence the term "total variation."

Remark.

We emphasize that

Monotone $\neq$ Linear!!!

This is obvious, but it is easy to allow oneself to imagine a "baseline" for a monotonic function *as* a linear function. This is fine for establishing an initial intuition for the setting, but the *entire point* of the Stieltjes integral and associated bounded variation condition is that we *don't* require the monotonic integrator to be linear. Rather, try to imagine α as some sort of piecewise polynomial, or even a discontinuous step-function — this is more faithful to the broader applicability of this new material.

In particular, the class of piecewise monotone functions with "finitely many turning points" is *precisely* the class of functions with "finitely many local extrema" (feel comfortable with this equivalence before proceeding — both characterize the "total number of times a function changes from monotone decreasing to increasing," and vice-versa.) Now, if we recall the beginning of the chapter, we stated our initial Dirichlet Hypothesis in terms of the latter condition — i.e., we indicated this as a reasonable condition for ensuring a pointwise convergent Fourier series. And now, we can begin to see why — the condition that f have "finitely many local extrema" is exactly the condition that f be piecewise monotone with finitely many terms — which is *exactly* a condition for f be of bounded variation, as we've just shown.

Before we continue, we state some properties about the total variation $V_a^b(\alpha)$ which will be essential in the following material.

Theorem 2.2.17 (Total Variation Properties (I))

Let $\alpha \colon [a, b] \to \mathbb{R}$ be of bounded variation — i.e., we have

$$V_a^b(\alpha) = \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| < \infty.$$

Then:

(a) **(Additivity)** For any $c \in (a, b)$,

$$V_a^b(\alpha) = V_a^c(\alpha) + V_c^b(\alpha).$$

(b) **(Vanishing)** If α is right-continuous at a, then within a shrinking interval $[a, a+\delta]$, the total variation goes to zero:

$$\alpha \text{ right-continuous at } t = a \implies V_a^{a+\delta}(\alpha) \to 0 \quad \text{as} \quad \delta \downarrow 0.$$

Proof:

The first property is immediate from the definition: if we're given a partition $\mathcal{P}$ of $[a,b]$, then we can *refine* the partition as $\mathcal{P}^\star$ including a new point $c \in (a,b)$ without decreasing the sum (Exercise: prove this using the triangle inequality applied to a given increment $|\alpha(t_{i+1}) - \alpha(t_i)|$ and a new point $\alpha(c)$, $c \in [t_i, t_{i+1}]$.) In particular, letting $c \in (a,b)$ be arbitrary, we may assume without loss of generality that every partition $\mathcal{P}$ of $[a,b]$ contains c since, if it didn't, we could form $\mathcal{P}^\star$ as described and obtain a sum

$$\sum_{j=0}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| \leq \sum_{j=0}^{i-1} |\alpha(t_{j+1}) - \alpha(t_j)| + |\alpha(c) - \alpha(t_i)| + |\alpha(t_{i+1}) - \alpha(c)| + \sum_{j=i+1}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)|.$$

The left side is the sum over the partition $\mathcal{P}$, and the right is over $\mathcal{P}^\star$ — since the bounded variation concerns itself with the *supremum* over such partitions, the assertion follows naturally. Thus, assuming henceforth that $c \in \mathcal{P}$ for any partition $\mathcal{P}$, what we have is that

$$V_a^b(\alpha) := \sup_{\mathcal{P}} \sum_{\mathcal{P}} |\Delta\alpha_j| = \sup_{\mathcal{P}} \sum_{j=0}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| = \sup_{\mathcal{P}} \left[\sum_{j=0}^{i-1} |\alpha(t_{j+1}) - \alpha(t_j)| + \sum_{j=i-1}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| \right]$$

directly for any index i. Supposing, then, that $t_i = c$, what we would have is that the points $a := t_0 < t_1 < \cdots < t_i := c$ partition $[a,c]$, and $c := t_i < t_{i+1} < \cdots < t_k := b$ partition $[c,b]$. Furthermore, since $c \in (a,b)$ was arbitrary and it always winds up in the partition $\mathcal{P}$ for which the supremum is attained, what we intuitively imagine is that we're "allowed to pick the partitions $\mathcal{P}_L := \{t_0, t_1, \dots, t_i\}$ and $\mathcal{P}_R := \{t_i, t_{i+1}, \dots, t_k\}$ *independently*" — anytime we perform the latter procedure first, obtaining $\mathcal{P}_L$ and $\mathcal{P}_R$, we can adjoint the two at $t_i := c$ and obtain a partition $\mathcal{P} := \mathcal{P}_L \cup \mathcal{P}_R$ which is a valid partition of $[a,b]$ containing c, and vice-versa (clearly, from the preceding discussion).

Formally, we say that there is a **bijection** between partitions of $[a,b]$ containing c and pairs of partitions $(\mathcal{P}_L, \mathcal{P}_R)$ of $[a,c]$ and $[c,b]$ respectively, and that under this bijection, we have

$$\sup_{\mathcal{P}} \sum_{\mathcal{P}} |\Delta\alpha_j| = \sup_{(\mathcal{P}_L, \mathcal{P}_R)} \left[\sum_{j=0}^{i-1} |\Delta\alpha_j| + \sum_{j=i-1}^{k-1} |\Delta\alpha_j| \right] \overset{!!!}{=} \sup_{\mathcal{P}_L} \sum_{\mathcal{P}_L} |\Delta\alpha_j| + \sup_{\mathcal{P}_R} \sum_{\mathcal{P}_R} |\Delta\alpha_j|.$$

But these are just the expressions for the total variations in the relevant intervals $\mathcal{P}$, $\mathcal{P}_L$, and $\mathcal{P}_R$ are partitioning. In particular, we find that for any $c \in (a,b)$,

$$V_a^b(\alpha) = V_a^c(\alpha) + V_c^b(\alpha)$$

(In general, supremums do *not* split in the way demonstrated by the lattermost equality above — that is, though the *first* equality is quite obvious, it is the second — to find the "supremum over all partitions $\mathcal{P}$ of $[a,b]$ on the left" — which is not obvious to be the same as "taking the supremum over partitions $\mathcal{P}_L$ of $[a,c]$ independently of the supremum over partitions $\mathcal{P}_R$ of $[c,b]$, and adding the two together." We might imagine that the competing procedures would lead to an *inequality* — what we've accomplished in this derivation is asserting that it is actually an *equality*.")

For the second property, note by additivity,

$$V_a^b(\alpha) = V_a^{a+\delta}(\alpha) + V_{a+\delta}^b(\alpha) \implies V_a^{a+\delta}(\alpha) = V_a^b(\alpha) - V_{a+\delta}^b(\alpha).$$

As $\delta \downarrow 0$, the function $\delta \mapsto V_{a+\delta}^b(\alpha)$ is nondecreasing in δ and bounded above by the total variation over the full interval, i.e. $V_{a+\delta}^b(\alpha) \leq V_a^b(\alpha)$. Thus the bounded, nondecreasing $V_{a+\delta}^b(\alpha)$ converges to a limit $L \leq V_a^b(\alpha)$ — we claim that

$$L := \lim_{\delta\downarrow 0} V_{a+\delta}^b(\alpha) = V_a^b(\alpha).$$

To see this, take any partition $\mathcal{P} = \{a = t_0, t_1, \dots, t_k = b\}$ of $[a, b]$. For δ small enough such that $a + \delta < t_1$, form the partition $\mathcal{P}' = \{t_0' = a + \delta, t_1, \dots, t_k\}$ of $[a+\delta, b]$. Then, the total variation sums over $\mathcal{P}$ and $\mathcal{P}'$ differ only in their first terms, where $|\alpha(t_1) - \alpha(a)|$ in $\mathcal{P}$ is replaced by $|\alpha(t_1) - \alpha(a+\delta)|$ in $\mathcal{P}'$.

By the reverse triangle inequality ($\Big||x| - |y|\Big| \leq |x + y|$),

$$\Big||\alpha(t_1) - \alpha(a)| - |\alpha(t_1) - \alpha(a+\delta)|\Big| \leq \big|\alpha(a+\delta) - \alpha(a)\big|;$$

since the term on the left is strictly nonnegative,

$$|\alpha(t_1) - \alpha(a)| - |\alpha(t_1) - \alpha(a+\delta)| \leq \big|\alpha(a+\delta) - \alpha(a)\big|,$$

and rearranging this, we obtain

$$|\alpha(t_1) - \alpha(a+\delta)| \geq |\alpha(t_1) - \alpha(a)| - |\alpha(a+\delta) - \alpha(a)|.$$

In particular, this implies that the total variation sum over $\mathcal{P}' = \{a + \delta = t_0', t_1, \dots, t_k\}$, where we write $t_0 = a$ and $t_0' = a + \delta$ respectively, satisfies

$$\begin{aligned}
\sum_{\mathcal{P}'} |\Delta\alpha_j| &= |\alpha(t_1) - \alpha(t_0')| && + \sum_{j=1}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| \\
&\geq \Big(|\alpha(t_1) - \alpha(t_0)| - |\alpha(t_0') - \alpha(t_0)|\Big) && + \sum_{j=1}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| \\
&= \underbrace{|\alpha(t_1) - \alpha(t_0)| + \sum_{j=1}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)|}_{\sum_{j=0}^{k-1} |\alpha(t_{j+1}) - \alpha(t_j)| := \sum_{\mathcal{P}} |\Delta\alpha_j|} && - |\alpha(t_0') - \alpha(t_0)| \\
&= \sum_{\mathcal{P}} |\Delta\alpha_j| \quad - \quad |\alpha(t_0') - \alpha(t_0)|.
\end{aligned}$$

That is, for any partition $\mathcal{P}$ of $[a, b]$ and $\mathcal{P}'$ of $[a + \delta, b]$ for some $\delta > 0$, we have

$$\sum_{\mathcal{P}'} |\Delta\alpha_j| \geq \sum_{\mathcal{P}} |\Delta\alpha_j| - |\alpha(a+\delta) - \alpha(a)|.$$

Since $\mathcal{P}'$ is a particular partition of $[a+\delta, b]$, the supremum over all partitions of $[a+\delta, b]$ is at least as large:

$$\sup_{\mathcal{P}'} \sum_{\mathcal{P}'} |\Delta\alpha_j| := V_{a+\delta}^b(\alpha) \geq \sum_{\mathcal{P}} |\Delta\alpha_j| - |\alpha(a+\delta) - \alpha(a)|.$$

Since $\mathcal{P}$ is an arbitrary partition of $[a, b]$, the supremum over $\mathcal{P}$ will still satisfy this bound:

$$V_{a+\delta}^b(\alpha) \geq V_a^b(\alpha) - |\alpha(a+\delta) - \alpha(a)|,$$

or

$$V_a^b(\alpha) \leq V_{a+\delta}^b(\alpha) + |\alpha(a+\delta) - \alpha(a)|.$$

On the other hand, it is obvious that

$$V_{a+\delta}^b(\alpha) \leq V_a^b(\alpha) \quad \text{for any} \quad \delta > 0,$$

since every partition of $[a+\delta, b]$ partitions a subinterval of $[a, b]$, and additivity gives

$$V_a^b(\alpha) = V_a^{a+\delta}(\alpha) + V_{a+\delta}^b(\alpha) \geq V_{a+\delta}^b(\alpha).$$

Together, we have that

$$V_{a+\delta}^b(\alpha) \leq V_a^b(\alpha) \leq V_{a+\delta}^b(\alpha) + |\alpha(a+\delta) - \alpha(a)|.$$

Now is where we apply the assumption that α is *right-continuous* at a: if this is the case, then

$$|\alpha(a+\delta) - \alpha(a)| \downarrow 0 \quad \text{as} \quad \delta \downarrow 0;$$

thus, by the squeeze-theorem,

$$V_{a+\delta}^b(\alpha) \to V_a^b(\alpha) \quad \text{as} \quad \delta \downarrow 0.$$

Therefore, by additivity,

$$\lim_{\delta \downarrow 0} V_a^{a+\delta}(\alpha) = \lim_{\delta \downarrow 0} [V_a^b(\alpha) - V_{a+\delta}^b(\alpha)] = V_a^b(\alpha) - V_a^b(\alpha) = 0,$$

so that

$$V_a^{a+\delta}(\alpha) \to 0 \quad \text{as} \quad \delta \downarrow 0.$$

for any α right-continuous at $a \in [a, b]$.

$\|$

Next, we state an elegant result we'll use to derive some properties which will simplify the conditions we need to impose on f in the Dirichlet conditions.

Theorem 2.2.18 (Jordan Decomposition)

Every function $\alpha\colon [a, b] \to \mathbb{R}$ of bounded variation $V_a^b(\alpha) < \infty$ can be written as the difference of two monotone increasing functions,

$$\alpha(t) = V(t) - W(t),$$

where

$$t \leq s \implies V(t) \leq V(s) \quad \text{and} \quad W(t) \leq W(s).$$

Proof:
Start by defining

$$V(t) := V_a^t(\alpha), \quad t \in [a,b].$$

Then, by additivity, we have that V is monotone increasing: $t \geq s$ implies $V_a^t(\alpha) = V_a^s(\alpha) + V_s^t(\alpha)$ where $V_s^t(\alpha) \geq 0$, meaning $V_a^t(\alpha) \geq V_a^s(\alpha)$.

Now, write

$$\alpha(t) = V(t) - V(t) + \alpha(t) = V(t) - \underbrace{[V(t) - \alpha(t)]}_{:= W(t)}.$$

The function W is monotone increasing: for $s \leq t$,

$$W(t) - W(s) = [V(t) - V(s)] - [\alpha(t) - \alpha(s)] = V_s^t(\alpha) - [\alpha(t) - \alpha(s)].$$

We can consider $\alpha(t) - \alpha(s)$ to be the variation under the partition $\mathcal{P} = \{s, t\}$ of $[s, t]$, so that if $\mathcal{P}_s^t$ is an arbitrary partition of $[s,t]$, we have

$$\alpha(t) - \alpha(s) = |\alpha(t) - \alpha(s)| \leq \sup_{\mathcal{P}_s^t} \sum_{\mathcal{P}_s^t} |\Delta\alpha_j| = V_s^t(\alpha).$$

In particular, subtracting the second term over to the last yields the inequality

$$W(t) - W(s) := V_s^t(\alpha) - |\alpha(t) - \alpha(s)| \geq 0 \implies W(t) \geq W(s) \text{ whenever } t \geq s.$$

Thus, $\alpha(t) = V(t) - W(t)$ where $V(t) = V_a^t(\alpha)$ and $W(t) = V_a^t(\alpha) - \alpha(t)$ are each monotone increasing in $t \in [a,b]$, completing the proof.

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Using the Jordan decomposition, we have the following.

Theorem 2.2.19 (Total Variation Properties (II))

Let $\alpha \colon [a,b] \to \mathbb{R}$ be of bounded variation — i.e., we have

$$V_a^b(\alpha) = \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |\alpha(t_{i+1}) - \alpha(t_i)| < \infty.$$

Then:

(a) **(Boundedness)** α is bounded on $[a,b]$:

$$|\alpha(t)| \leq |\alpha(a)| + V_a^b(\alpha), \qquad \forall t \in [a,b].$$

(b) **(One-sided limits)** The one sided limits

$$\alpha(x^-) = \lim_{\delta \downarrow 0} \alpha(x - \delta) \qquad \text{and} \qquad \alpha(x^+) = \lim_{\delta \downarrow 0} \alpha(x + \delta)$$

exist for any $x \in [a,b]$.

(c) **(Countably Many Discontinuities)** α has at most countably many discontinuities on $[a,b]$.

Proof:

For any $t \in [a,b]$, consider the two-point partition $\mathcal{P} = \{a,t\}$ of $[a,t]$. Then

$$|\alpha(t) - \alpha(a)| \leq \sup_{\mathcal{P}} \sum_{\mathcal{P}} |\Delta\alpha_j| = V_a^t(\alpha) \leq V_a^b(\alpha),$$

where the last line follows by additivity ($V_a^t(\alpha) + V_t^b(\alpha) = V_a^b(\alpha)$ and $V_t^b(\alpha) \geq 0$.) By the triangle inequality,

$$|\alpha(t)| = |\alpha(t) - \alpha(a) + \alpha(a)| \leq |\alpha(a)| + |\alpha(t) - \alpha(a)| \leq |\alpha(a)| + V_a^t(\alpha).$$

For (d), by the Jordan decomposition, $\alpha = V - W$ where V and W are both monotone increasing on $[a,b]$. It suffices to show that any monotone increasing function has one-sided limits at every point, since the difference of two functions with one-sided limits has one-sided limits.

As such, let $h\colon [a,b] \to \mathbb{R}$ be monotone increasing, and fix $x \in (a,b]$. Consider the set

$$H_x := \{h(t)\colon\ a \leq t < x\}.$$

Since $x > a$, the set contains $h(a)$ and is thus nonempty. Furthermore, since h is monotone increasing, it must be that

$$\sup H_x = \sup_{a \leq t < x} h(t) \leq h(x).$$

Thus the set $H_x \subset \mathbb{R}$ is nonempty and bounded above; by the completeness of $\mathbb{R}$, it follows that the supremum exists. In particular, we have that

$$L := \sup_{t<x} h(t) \text{ exists in } \mathbb{R}.$$

We claim this limit equals the left-sided limit

$$L = h(x^-) = \lim_{t \downarrow 0} h(x-t).$$

Indeed, for any $\epsilon > 0$, by definition of the supremum $L = \sup_{t<x} h(t)$ there exists $t_0 < x$ such that $L - h(t_0) < \epsilon$, so $L - \epsilon < h(t_0)$. But then for any $t \in (t_0, x)$, monotonicity gives $h(t_0) \leq h(t)$, so

$$L - \epsilon < h(t_0) \leq h(t) \leq L,$$

or

$$L - \epsilon < h(t) \leq L \implies |h(t) - L| < \epsilon, \quad \forall t \in (t_0, x).$$

This is precisely the statement that

$$h(x^-) = \lim_{t \uparrow x} h(t) = L,$$

or that the monotone increasing function h has a left-sided limit. Other cases follow by the same style of argument; applying the result to each piece of the integrator $\alpha = V - W$ and recalling that the sum of functions with one-sided limits has a one-sided limit gives us (d).

Finally, for (e), we again use the Jordan decomposition and show that, in general, a monotone increasing function $h\colon [a,b] \to \mathbb{R}$ has at most countably many discontinuities;

the claim for $\alpha = V - W$ follows after applying the result to both parts separately and recalling that the sum of the continuous parts of V and $-W$ will remain continuous; in particular, the countable discontinuities of each V and W will have a countable union characterizing the discontinuities of α.

Applying the result from (d), we know that the function h has left- and right-sided limits at an arbitrary point of discontinuity $x \in [a, b]$ — first, we must realize that because h is monotone increasing, it simply *cannot* be the case that $h(x^+) = h(x^-)$ while $h(x) \neq h(x^+) = h(x^-)$. In particular, at any point $x \in [a, b]$, monotonicity forces

$$h(x^-) \leq h(x) \leq h(x^+),$$

so that either $h(x)$ equals each of the left and the right (hence $h(x^-) = h(x^+)$), or we have $h(x^-) < h(x)$ and/or $h(x) < h(x^+)$. In either of these latter cases, we have

$$h(x^-) < h(x^+).$$

Again, there are only two possible cases: either $h(x^-) = h(x) = h(x^+)$ — i.e., x is a point of continuity of h — or $h(x^-) < h(x^+)$ — i.e., x is a *jump* discontinuity of h. There is *no* situation giving rise to a removable discontinuity when h is monotone.

Thus, at any point of discontinuity $x \in [a, b]$ of h, we must have $h(x^-) < h(x^+)$, producing a nonempty, open interval

$$I_x := \Big(h(x^-), h(x^+)\Big) \subset \mathbb{R}.$$

From here, we realize that I_x is a nonempty interval of the real line for any point of discontinuity $x \in [a, b]$ of h. We know that the rationals $\mathbb{Q}$ are *dense* in the reals $\mathbb{R}$ — in particular, we can choose a rational $q_x \in I_x$ for any point of discontinuity. Moreover, these intervals are pairwise disjoint: if $x \leq y$ are two points of discontinuity, then $h(x^+) \leq h(y^-)$ by monotonicity, so that $I_x = (h(x^-), h(x^+))$ and $I_y = (h(y^-), h(y^+))$ don't overlap. In particular, the rationals q_x assigned to distinct discontinuities $x \in [a, b]$ of h are themselves distinct.

What we have constructed, then, is a map from the set of discontinuities of h into the rational numbers $\mathbb{Q}$: to each discontinuity x, assign the rational $q_x \in I_x$. Since the intervals I_x are pairwise disjoint, this map is *injective* — distinct discontinuities produce distinct rationals. But any set that injects into a countable set is itself countable; since the set of discontinuities $x \in [a, b]$ of h produces such an injective map into $\mathbb{Q}$, it follows that h has at most countably many discontinuities.

Since h was arbitrary, it follows that a monotone increasing function has at most countably many discontinuities. Since $\alpha = V - W$ is the difference of two monotone increasing functions with at most countably many discontinuities, it too will have at most countably many discontinuities, completing the proof. We note, however, that in applying this difference $V - W$ of monotone increasing functions, we can force removable discontinuities — i.e., just because a monotone increasing function only has jump discontinuities does *not* mean that sums or other compositions of monotone functions will retain this property — in particular, the BV integrator α is certainly allowed to have discontinuities of the *first-kind* — i.e., those where the one-sided limits exist at every point (including both removable and jump discontinuities as subcases.)

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Now, the reason we care about total variation in the current setting is that it provides a remarkably efficient way to conclude Riemann-Stieltjes integrability without ever needing integrate explicitly. In particular, we have the following.

Theorem 2.2.20 (Riemann-Stieltjes Estimate)

Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous and $\alpha\colon [a,b] \to \mathbb{R}$ is of bounded variation, so that the Riemann-Stieltjes integral

$$\int_a^b f\, d\alpha(t) \quad \text{exists.}$$

Then, we have the bound

$$\left|\int_a^b f\, d\alpha(t)\right| \leq \sup_{t\in[a,b]} |f(t)| \cdot V_a^b(\alpha).$$

Proof:
For any partition $\mathcal{P}_k$ and sample points $t_i^\star \in [t_i, t_{i+1}]$,

$$|S(f,\alpha,\mathcal{P}_k)| = \left|\sum_{i=0}^{k-1} f(t_i^\star)\, \Delta\alpha_i\right| \leq \sum_{i=0}^{k-1} |f(t_i^\star)|\, |\Delta\alpha_i| \leq \sup_{t\in[a,b]} |f(t)| \sum_{i=0}^{k-1} |\Delta\alpha_i| \leq \sup_{t\in[a,b]} |f(t)| \cdot V_a^b(\alpha),$$

where the first inequality is the triangle inequality, the second uses $|f(t_i^\star)| \leq \sup_{t\in[a,b]} |f(t)|$ for each i (since $t_i^\star \in [a,b]$), and the third uses $\sum |\Delta\alpha_i| \leq V_a^b(\alpha)$ by definition of the total variation.

Since the bound is independent of the partition and sample points, it persists in the limit $\|\mathcal{P}_k\| \to 0$, yielding

$$\left|\int_a^b f\, d\alpha(t)\right| \leq \sup_{t\in[a,b]} |f(t)| \cdot V_a^b(\alpha).$$

$\|$

The bound

$$\left|\int_a^b f\, d\alpha(t)\right| \leq \sup_{t\in[a,b]} |f(t)| \cdot V_a^b(\alpha)$$

is the Riemann-Stieltjes analogue of the familiar Riemann estimate

$$\left|\int_a^b f(t)\, dt\right| \leq \sup_{t\in[a,b]} |f(t)| \cdot (b-a).$$

(Verify that we recover this estimate when we take $\alpha(t) = t$ in the Riemann-Stieltjes definition. In particular, the role of $b-a$ — a *monotonic* function under the identity integrator $\alpha(t) = t$ — is replaced by the total variation, $V_a^b(\alpha)$.)

The second result we'll need is another extension of a familiar technique.

Theorem 2.2.21 (Integration by Parts (Riemann-Stieltjes))

If
$$\int_a^b f\,d\alpha \quad \text{exists,}$$
then
$$\int_a^b \alpha\,df \quad \text{also exists,}$$
and we have
$$\int_a^b f(t)\,d\alpha(t) = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha(t)\,df(t).$$

We often remember a shorthand for the formula suppressing the t's:
$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du.$$

In fact, this *is* the familiar integration by parts formula from an introductory calculus course: writing $f \equiv u$ and $\alpha \equiv v$ to get the familiar notation, we have
$$\int_a^b u(t)\,dv(t) = [u(t)v(t)]_a^b - \int_a^b v(t)\,du(t) = u(b)v(b) - u(a)v(a) - \int_a^b v(t)\,du(t).$$
Whenever u and v are differentiable, this says that
$$\int_a^b u(t)v'(t)\,dt = [u(t)v(t)]_a^b - \int_a^b v(t)u'(t)\,du,$$
which should be familiar.

Nevertheless, realize that the Riemann-Stieltjes formulation is strictly more general; it requires neither u nor v to be differentiable, only that one of the two integrals $\int f d\alpha$ or $\int \alpha df$ exist. In particular, the Riemann-Stieltjes formula allows us to integrate by parts against functions which are merely of bounded variation.

Dirichlet's Test

What we'll find in the midst of the proof we're about to give is a need to diagnose a particular series as convergent — one which seemingly eludes the methods we outlined in the last chapter. Indeed, we're ready to state a missing link from the chain of tests for diagnosing a series' convergence:

Theorem 2.2.22 (Dirichlet's Test)

If $(a_n)_{n\in\mathbb{N}}$ is a monotonic sequence of real numbers with
$$a_n \downarrow 0 \quad \text{as} \quad n \to \infty,$$

and if $(b_n)_{n\in\mathbb{N}}$ is a sequence of real (or complex) numbers with bounded partial sums:

$$|B_N| := \left|\sum_{n=1}^{N} b_n\right| \le C, \qquad \forall N \ge 1,$$

then the series

$$\sum_{n=1}^{\infty} a_n b_n \text{ converges.}$$

The proof of Dirichlet's test relies on *summation by parts* (also called Abel summation) — the discrete analogue of the integration by parts formula we developed in the preceding section.

Theorem 2.2.23 (Summation by Parts/Abel Summation)

Given sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$, we have that the the Nth partial sum of the product $\sum_{n=1}^{N} a_n b_n$ can be expressed as

$$\sum_{n=1}^{N} a_n b_n = a_N B_N - \sum_{n=1}^{N-1} (a_{n+1} - a_n) B_n, \qquad B_n := \sum_{k=1}^{n} b_k.$$

Compare with $\int u\,dv = [uv] - \int v\,du$ — on the left, the a_n play the role of u and the b_n play the role of dv; on the right, $a_N B_N$ plays the role of $[uv]$, the partial sums $B_n = \sum_{k=1}^{n} b_n$ play the role of v, and the increments $(a_{n+1} - a_n)$ play the role of du.

Proof:

To prove the summation by parts formula, the key is to recognize b_n as the analogous "discrete derivative" of its partial sums B_n. To see what we mean, start by defining $B_0 := 0$ (this will come back later), and $B_n := \sum_{k=1}^{n} b_k$ for $n \ge 1$ as before. Then, what we have is

$$b_n = B_n - B_{n-1}.$$

All this is saying is that "to recover b_n, we can subtract off the $(n-1)$th partial sum B_{n-1} from B_n." But what the formula encourages us to imagine is not b_n as an "element of the known sequence $(b_n)_{n\in\mathbb{N}}$," but rather as the *increment* accumulated by the sequence's partial sums between steps $n-1$ and n. In the continuous setting, we might write the analogous setup as something like

$$dB(t) = B'(t)\,dt = b(t)\,dt.$$

In the discrete setting, we replace the infinitesimal increment dt with a constant step size 1, leading to

$$\Delta B_n := B_n - B_{n-1} = b_n.$$

That is: b_n *is* the discrete differential dB, and $B_n = \sum_{k=1}^{n} b_k$ *is* the discrete antiderivative $B(n) = \int_0^n dB$.

Now, the rest of the proof is an exercise in algebra: with $b_n = B_n - B_{n-1}$, write

$$\sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N} a_n(B_n - B_{n-1}) = \sum_{n=1}^{N} a_n B_n - \sum_{n=1}^{N} a_n B_{n-1}.$$

Reindex the second sum: set $m := n-1$, so that as n runs from 1 to N, m runs from 0 to $N-1$, and (recalling that $B_0 := 0$),

$$\sum_{n=1}^{N} a_n B_{n-1} = \sum_{m=0}^{N-1} a_{m+1} B_m = a_1 \underbrace{B_0}_{=0} + \sum_{m=1}^{N-1} a_{m+1} B_m = \sum_{n=1}^{N-1} a_{n+1} B_n,$$

where we've relabelled the dummy index back from m back to n in the last step. In particular, we can substitute

$$\sum_{n=1}^{N} a_n B_{n-1} = \sum_{n=1}^{N-1} a_{n+1} B_n$$

in

$$\sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N} a_n B_n - \sum_{n=1}^{N} a_n B_{n-1} = \sum_{n=1}^{N} a_n B_n - \sum_{n=1}^{N-1} a_{n+1} B_n.$$

Peel off the $n = N$ term from the first sum:

$$\sum_{n=1}^{N} a_n b_n = a_N B_N + \sum_{n=1}^{N-1} a_n B_n - \sum_{n=1}^{N-1} a_{n+1} B_n = a_N B_N + \sum_{n=1}^{N-1} (a_n - a_{n+1}) B_n;$$

i.e., factoring out a negative from the sum,

$$\sum_{n=1}^{N} a_n b_n = a_N B_N - \sum_{n=1}^{N-1} (a_{n+1} - a_n) B_n,$$

as desired.

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Now, armed with the summation by parts formula, we're ready to give a proof of Dirichlet's test.

Proof:
We wish to show that $\sum_{n=1}^{\infty} a_n b_n$ converges. By the summation by parts formula,

$$\sum_{n=1}^{N} a_n b_n = a_N B_N - \sum_{n=1}^{N-1} (a_{n+1} - a_n) B_n, \qquad B_n = \sum_{k=1}^{n} b_k.$$

We show that both terms on the right converge as $N \to \infty$.
First, since $a_N \to 0$ and $|B_N| \le C$, we have

$$|a_N B_N| \le |a_N| \cdot C \to 0 \quad \text{as} \quad N \to \infty$$

Hence $a_N B_N \to 0$.

Now, we consider the sum

$$\sum_{n=1}^{N-1}(a_{n+1}-a_n)B_n.$$

We show it converges absolutely. Since $|B_n| \leq C$ for all n,

$$\sum_{n=1}^{N-1}|a_{n+1}-a_n||B_n| \leq C\sum_{n=1}^{N-1}|a_{n+1}-a_n|.$$

Since (a_n) is monotonically decreasing, $a_{n+1}-a_n \leq 0$ with $a_n \to 0$ as $n \to \infty$. In particular,

$$|a_{n+1}-a_n| = a_n - a_{n+1},$$

meaning the sum on the right above telescopes:

$$\sum_{n=1}^{N-1}|a_{n+1}-a_n| = \sum_{k=1}^{N-1}(a_n - a_{n+1}) = a_1 - a_N \to a_1 \quad \text{as} \quad N \to \infty.$$

Thus, in the limit,

$$\sum_{n=1}^{\infty}|a_{n+1}-a_n||B_n| \leq a_1 \cdot C < \infty,$$

so that $\sum(a_{n+1}-a_n)B_n$ converges absolutely. Since both terms on the right side of the summation by parts formula go to zero, we have that

$$\sum_{n=1}^{N}a_n b_n = a_N B_N - \sum_{n=1}^{N-1}(a_{n+1}-a_n)B_n \longrightarrow 0 \quad \text{as} \quad N \to \infty$$

whenever (a_n) is monotonically decreasing to zero and (b_n) has bounded partial sums $|B_n| \leq C$ for all n.

$\|$

Let us see some applications of Dirichlet's test. When $b_n = (-1)^n$, we have that

$$B_N = \left|\sum_{n=1}^{N}b_n\right| \leq 1 < \infty,$$

so that a sufficient condition for $\sum_{n=1}^{\infty}(-1)^n a_n$ to converge is for a_n to be monotonically decreasing to zero as $n \to \infty$. This is *exactly* the familiar Alternating Series test for convergence, and as we've just shown, it follows immediately as a special case of Dirichlet's test. However, it should be clear that Dirichlet's test applies much more broadly than the Alternating Series test — indeed, it is one of the most general and capable of our tools for diagnosing series convergence.

Let us apply this result to bound a particular integral of note — in particular, the **sine integral**

$$\mathrm{Si}(x) := \int_0^x \frac{\sin u}{u}\,du, \qquad x \geq 0.$$

We claim that $\mathrm{Si}(x)$ is *uniformly bounded* for all $x \geq 0$ — i.e., there exists a finite constant $M > 0$ such that $\mathrm{Si}(x) \leq M$. To see this, decompose the integral into intervals of length π: for arguments $x \mapsto \mathrm{Si}(x)$ that are in the Kth subinterval, $x \in [K\pi, (K+1)\pi]$, we can decompose

$$\mathrm{Si}(x) = \sum_{k=0}^{K-1} \underbrace{\int_{k\pi}^{(k+1)\pi} \frac{\sin u}{u}\, du}_{:= c_k} + \int_{K\pi}^{x} \frac{\sin u}{u}\, du.$$

Now, on each interval in the sum $[k\pi, (k+1)\pi]$, $k = 0, \ldots, K-1$, the function $\sin u$ completes a half-cycle in such a way that on the following interval $[(k+1)\pi, (k+2)\pi]$, the contributions c_k to the sum *alternate in sign*. Furthermore, their absolute values $|c_k|$ are monotonically decreasing: on each interval, $|\sin u|$ traces the same half-cycle while being divided by a progressively larger denominator $u \to \infty$:

$$|c_{k+1}| = \int_{(k+1)\pi}^{(k+2)\pi} \frac{|\sin u|}{u}\, du < \int_{k\pi}^{(k+1)\pi} \frac{|\sin u|}{u}\, du = |c_k|.$$

Moreover, $|c_k| \to 0$ as $k \to \infty$, since $|\sin u|/u \leq 1/u \to 0$ as $u \to \infty$.

Thus, $\sum_{k=0}^{\infty} c_k$ is an alternating series with terms decreasing in absolute value to zero — by the Alternating Series test (which we've just shown is a special case of Dirichlet's test), it converges. Furthermore (see below), we can bound the alternating series by its first term:

$$\left|\sum_{k=0}^{K-1} c_k\right| \leq |c_0| = \int_0^{\pi} \frac{\sin u}{u}\, du.$$

The leftover piece is similarly bounded:

$$\left|\int_{K\pi}^{x} \frac{\sin u}{u}\, du\right| \leq \left|\int_{K\pi}^{(K+1)\pi} \frac{\sin u}{u}\, du\right| = |c_K| \leq |c_0|.$$

Together, what we have is that

$$|\mathrm{Si}(x)| = \left|\sum_{k=0}^{K-1} \int_{k\pi}^{(k+1)\pi} \frac{\sin u}{u}\, du + \int_{K\pi}^{x} \frac{\sin u}{u}\, du\right| \leq 2\left|\int_0^{\pi} \frac{\sin u}{u}\, du\right| < 2\pi, \qquad \forall x \geq 0.$$

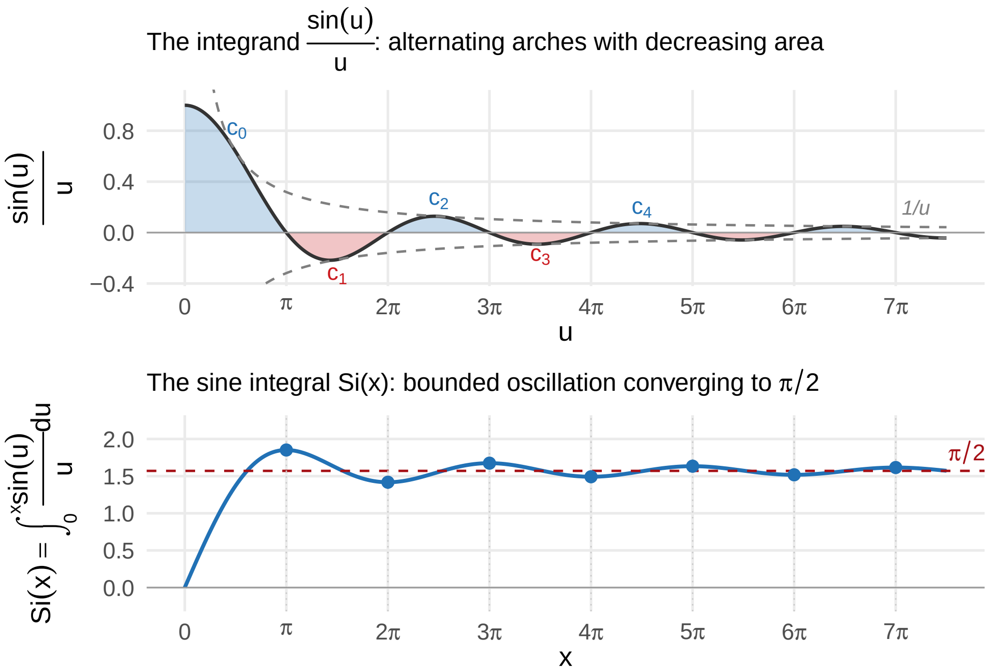
We now justify the alternating series bound from above:

Theorem 2.2.24 (Alternating Series Estimate)

If $\sum_{k=0}^{\infty} c_k$ is a convergent alternating series with $|c_k|$ decreasing, then the partial sums satisfy

$$\left|\sum_{k=0}^{K-1} c_k\right| \leq |c_0|$$

for all $K \geq 1$. That is, the partial sums of such an alternating series never exceed the first term in absolute value.



Top: the integrand $\sin(u)/u$ with alternating arches $c_k = \int_{k\pi}^{(k+1)\pi} \sin(u)/u\,du$ shaded; each arch has strictly smaller area than the last, and the signs alternate. Bottom: the sine integral $\mathrm{Si}(x) = \int_0^x \sin(u)/u\,du$, oscillating with decreasing amplitude and converging to $\pi/2$. The alternating series test guarantees the partial sums $\sum_{k=0}^{K-1} c_k$ remain bounded.

Proof: Suppose without loss of generality that $c_0 > 0$, so the terms alternate $+, -, +, -, \ldots$ with $|c_0| \geq |c_1| \geq |c_2| \geq \cdots$. Write $S_K = \sum_{k=0}^{K-1} c_k$. Grouping consecutive pairs in two ways:

$$S_K = c_0 + \underbrace{(c_1 + c_2)}_{\leq 0} + \underbrace{(c_3 + c_4)}_{\leq 0} + \cdots \leq c_0,$$

since each pair $(c_{2j-1} + c_{2j})$ is nonpositive (the negative term c_{2j-1} dominates in absolute value). On the other hand,

$$S_K = \underbrace{(c_0 + c_1)}_{\geq 0} + \underbrace{(c_2 + c_3)}_{\geq 0} + \cdots \geq 0,$$

since each pair $(c_{2j} + c_{2j+1})$ is nonnegative (the positive term c_{2j} dominates). Together, $0 \leq S_K \leq c_0 = |c_0|$ for all K.

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As a final example: we claim that, for any fixed $t \in (0, \pi]$, the series

$$\sum_{n=1}^{\infty} \frac{\sin(nt)}{n} \quad \text{converges.}$$

To see this, we apply Dirichlet's test by setting

$$a_n = \frac{1}{n} \quad \text{and} \quad b_n = \sin(nt).$$

The sequence $a_n = 1/n \to 0$ as $n \to \infty$ monotonically, so it remains to verify that the partial sums of b_n are bounded — i.e., that for some $C > 0$,

$$B_N(t) := \sum_{n=1}^{N} \sin(nt) \leq C.$$

To do this, we start from a somewhat surprising place: namely, by considering the series

$$\sum_{n=1}^{N} e^{int}.$$

This is a geometric series in n with common ratio $r = e^{it}$ and initial term e^{it} for $n = 1$, so by the usual geometric series formula $\sum_{n=1}^{N} ar^n = \frac{a(1-r^N)}{1-r}$,

$$\sum_{n=1}^{N} e^{int} = \frac{e^{it}(1 - e^{iNt})}{1 - e^{it}}.$$

We find this relevant due to *Euler's formula*, which gives

$$e^{int} = \cos(nt) + i\sin(nt).$$

We now observe that $\sin(nt)$ is explicitly involved in the former sum — in particular, what we have is that the imaginary part of the right is $\sin(nt)$, so

$$\mathrm{Im}(e^{int}) = \sin(nt).$$

From this (noting that $\mathrm{Im}(\cdot)$ is a linear operator, hence it distributes over sums), we can sum both sides of the above and obtain

$$\sum_{n=1}^{N} \sin(nt) = \sum_{n=1}^{N} \mathrm{Im}(e^{int}) = \mathrm{Im}\left(\sum_{n=1}^{N} e^{int}\right) = \mathrm{Im}\left(\frac{e^{it}(1 - e^{iNt})}{1 - e^{it}}\right).$$

To extract the imaginary part of the term on the right, identically multiply the numerator and denominator by $e^{it/2}$ to obtain

$$\frac{e^{it} - e^{i(N+1)t}}{1 - e^{it}} \cdot \frac{e^{-it/2}}{e^{-it/2}} = \frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{e^{-it/2} - e^{it/2}}.$$

The denominator here is the trick — as we can verify explicitly by manipulating Euler's formulas for e^{ix} and e^{-ix}, we have the identity

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i} \implies e^{ix} - e^{-ix} = 2i\sin(x).$$

applying this for $x = t/2$ gives

$$e^{it/2} - e^{-it/2} = 2i\sin(t/2).$$

Plugging this in, we have that

$$\frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{e^{-it/2} - e^{it/2}} = \frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{-2i\sin(t/2)}.$$

From the identity

$$\frac{1}{i} = -i,$$

write

$$\frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{-2i\sin(t/2)} = i \cdot \frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{2\sin(t/2)}.$$

Since, given a complex number $z = x + iy$, we have

$$\operatorname{Im}(i \cdot z) = \operatorname{Im}(ix - y) = x = \operatorname{Re}(z) \implies \operatorname{Im}(i \cdot z) = \operatorname{Re}(z),$$

we have

$$\operatorname{Im}\left(i \cdot \frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{2\sin(t/2)}\right) = \operatorname{Re}\left(\frac{e^{it/2} - e^{i\left(N+\frac{1}{2}\right)t}}{2\sin(t/2)}\right) = \frac{\operatorname{Re}\left(e^{it/2}\right) - \operatorname{Re}\left(e^{i\left(N+\frac{1}{2}\right)t}\right)}{2\sin(t/2)} = \frac{\cos(t/2) - \cos\left((N + \frac{1}{2}\right.}{2\sin(t/2)}$$

But the left side is equivalent to our original series — in particular, we've just verified that

$$\sum_{n=1}^{N} \sin(nt) = \frac{\cos(t/2) - \cos\left((N + \frac{1}{2})t\right)}{2\sin(t/2)}.$$

What we can say now is that

$$\left|\sum_{n=1}^{N} \sin(nt)\right| = \left|\frac{\cos(t/2) - \cos\left((N + \frac{1}{2})t\right)}{2\sin(t/2)}\right| \leq \frac{|\cos(t/2)| + \left|\cos\left((N + \frac{1}{2})t\right)\right|}{2|\sin(t/2)|} \leq \frac{2}{2|\sin(t/2)|} = \frac{1}{|\sin(t/2)|} < \infty$$

for any fixed $t \in (0, \pi]$.

What we've just shown, however, is that relative to our original series,

$$|B_N(t)| = \left|\sum_{n=1}^{N} \sin(nt)\right| \leq C;$$

i.e., the partial sums of $b_n(t) = \sin(nt)$ are bounded. Since $a_n = 1/n \downarrow 0$ as $n \to \infty$ monotonically and $b_n = \sin(nt)$ has bounded partial sums $B_N(t) \leq C$ for any $t \in (0, \pi]$, we conclude by Dirichlet's test that

$$\sum_{n=1}^{\infty} \frac{\sin(nt)}{n} \text{ converges for any fixed } t \in (0, \pi].$$

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The Dirichlet-Jordan Theorem

The general version of the Dirichlet conditions are given as follows.

Theorem 2.2.25 (Dirichlet-Jordan Conditions for Pointwise Convergence)

Let $f\colon \mathbb{R} \to \mathbb{R}$ be 2π-periodic, and suppose that f is of **bounded variation** on $[-\pi, \pi]$:

$$V_{-\pi}^{\pi}(f) := \sup_{\mathcal{P}} \sum_{i=0}^{k-1} |f(t_{i+1}) - f(t_i)| < \infty,$$

where the supremum is taken over partitions $\mathcal{P} = \{-\pi = t_0, t_1, \dots, t_k = \pi\}$ of $[-\pi, \pi]$.

Then for every $x \in [-\pi, \pi]$, the Fourier partial sums

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} (a_n \cos(nx) + b_n \sin(nx));$$

where we've defined the Fourier coefficients

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, dt, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt)\, dt, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt)\, dt;$$

converge to the midpoint of the left- and right-limits of f at x:

$$\lim_{N\to\infty} S_N(x) = \frac{f(x^+) + f(x^-)}{2}, \qquad x \in [-\pi, \pi].$$

Remark.

Here, we've greatly reduced the number of conditions we require on f — in particular, we've bundled the conditions that f be bounded, have finitely many extrema, and have finitely many jump discontinuities on $[a, b]$ into the single condition that f be of bounded variation on $[a, b]$. The reason is that bounded variation on a compact interval already implies each of these: boundedness follows from property (c) of the total variation properties, the existence of one-sided limits at every point — formerly a byproduct of all three classical conditions taken together — is guaranteed by property (d), and the set of discontinuities is at most countable by property (e). The classical Dirichlet conditions (finitely many extrema, finitely many discontinuities, bounded) are *sufficient* for bounded variation — finitely many extrema means piecewise monotone with finitely many pieces, which we showed gives finite total variation — but they are not *necessary*. A function can have infinitely many turning points and still be of bounded variation, provided the total excursion remains finite. The Jordan theorem's contribution is precisely this generalization: Dirichlet had the "finitely-many" version; Jordan realized that bounded variation was the real condition, which is why we've attributed him in the name of the revised theorem. However, see after the proof for a further note — counterintuitive as it may seem given our current pursuit, it *really isn't* the case that our good friend Dini has become obsolete!

You may wish to review the beginning of the proof we gave for the Dini case — we pick up, using the same notation, around halfway through.

\begin{proof}

As in the Dini proof, start by writing

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = I_N^- + I_N^+,$$

where

$$I_N^- = \frac{1}{\pi}\int_0^\pi g(t) D_N(t)\,dt, \quad g(t) := f(x-t) - f(x^-),$$

I_N^+ is defined symmetrically (with $f(x+t) - f(x^+)$ in place of g), and where

$$D_N(t) = \frac{1}{2} + \sum_{n=1}^N \cos(nt) = \frac{\sin\left((N+\frac{1}{2})t\right)}{2\sin(t/2)}$$

is the Dirichlet kernel. As before, it suffices to show $I_N^- \to 0$ as $N \to \infty$; the same for I_N^+ will follow immediately by an entirely analogous argument.

So, fixing $\delta \in (0, \pi)$, let us split the bounds of I_N^- at δ and write

$$I_N^- = \underbrace{\frac{1}{\pi}\int_0^\delta g(t) D_N(t)\,dt}_{:= I_\delta} + \underbrace{\frac{1}{\pi}\int_\delta^\pi g(t) D_N(t)\,dt}_{:= J_\delta}.$$

It is the integral I_δ which is of concern to us here; in particular, we now recall that the entire issue we're attempting to circumvent from our Dini-based proof is the denominator $\sin(t/2)$ in $D_N(t)$ going to zero as $t \downarrow 0$ in the integral. However, in J_δ, this isn't a concern — we have $\sin(t/2) \geq \sin(\delta/2) > 0$ for all $t \in [\delta, \pi]$, so that the denominator of the integrand in J_δ is bounded away from zero. What this means is that we may pursue a nearly identical argument to the one from before, substituting the classical conditions for our new bounded variation condition as appropriate.

In particular, again we define

$$\phi(t) := \frac{g(t)}{2\sin(t/2)}, \quad t \in [\delta, \pi],$$

where

$$\int_\delta^\pi |\phi(t)|\,dt \leq \frac{1}{2\sin(\delta/2)}\int_\delta^\pi |g(t)|\,dt.$$

Now, $g(t) = f(x-t) - f(x^-)$ where f is assumed to be BV on $[-\pi, \pi]$. Recall (cf. Total Variation Properties (c)) that a BV function f on $[-\pi, \pi]$ is automatically bounded on the same interval: $|f(t)| \leq M$ for some M and all $t \in [-\pi, \pi]$. Using this and the triangle inequality, we obtain

$$|g(t)| \leq |f(x-t)| + |f(x^-)| \leq 2M,$$

since $x - t$ and x are points in $[-\pi, \pi]$, whence satisfy the bound above. In particular, we have that

$$\int_\delta^\pi |\phi(t)|\,dt \leq \frac{1}{2\sin(\delta/2)}\int_\delta^\pi |g(t)|\,dt \leq \frac{2M}{2\sin(\delta/2)}\int_\delta^\pi dt = \frac{M(\pi-\delta)}{\sin(\delta/2)} < \infty,$$

so that $\phi \in L^1[\delta, \pi]$.

Thus, we have

$$J_\delta = \frac{1}{\pi}\int_\delta^\pi \phi(t)\sin\Big((N+\tfrac{1}{2})t\Big)\,dt$$

with $\phi \in L^1[\delta,\pi]$ and $N+\frac{1}{2}\to\infty$. By Riemann-Lebesgue, it follows that

$$J_\delta \to 0 \quad \text{as} \quad N\to\infty.$$

Now, we're ready to tackle I_δ. This is where the Dini argument breaks down, and where we'll be putting our new Riemann-Stieltjes machinery to real use. To start, define the antiderivative of the Dirichlet kernel D_N on $(0,t)$ as a function of t:

$$\sigma_N(t) := \int_0^t D_n(s)\,ds = \frac{t}{2} + \sum_{n=1}^N \frac{\sin(nt)}{n},$$

where we've obtained the expression by integrating the definition of D_n term by term. At this point, we observe that the sum on the right,

$$\sum_{n=1}^N \frac{\sin(nt)}{n},$$

is the same as the one we encountered at the end of the previous ribbon. In particular, there, we found that

$$\lim_{N\to\infty}\sum_{n=1}^N \frac{\sin(nt)}{n} = \sum_{n=1}^\infty \frac{\sin(nt)}{n} \text{ converges for any } t\in[0,\pi].$$

In particular, since the series converges and $t/2$ is a constant in N for each fixed $t\in[0,\pi]$, it follows that $\sigma_N(t)$ is bounded for each fixed t.

Moreover, σ_N is **uniformly bounded** in both N and t. To see this, note that $\sigma_N(t) = \int_0^t D_N(s)\,ds$ — substituting our closed form,

$$\sigma_N(t) = \int_0^t \frac{\sin\Big((N+\frac{1}{2})s\Big)}{\sin(s/2)}\,ds.$$

Write

$$\frac{1}{2\sin(s/2)} = \frac{1}{s} + \underbrace{\left(\frac{1}{2\sin(s/2)} - \frac{1}{s}\right)}_{:=\rho(s)},$$

so that

$$\sigma_N(t) = \int_0^t \frac{\sin\Big((N+\frac{1}{2})s\Big)}{s}\,ds + \int_0^t \rho(s)\sin\Big((N+\tfrac{1}{2})s\Big)\,ds.$$

Here, we observe that by making the substitution $u = \Big(N+\frac{1}{2}\Big)s$ in the first integral, we have the bound $t\mapsto \Big(N+\frac{1}{2}\Big)t$, so

$$\int_0^t \frac{\sin\Big((N+\frac{1}{2})s\Big)}{s}\,ds = \int_0^{(N+1/2)t}\frac{\sin u}{u}\,du = \mathrm{Si}\Big((N+\tfrac{1}{2})t\Big) \le 2\pi$$

uniformly in $t\in[0,\pi]$ and $N\ge 1$, owing to our bound on the Sine integral we derived in the previous ribbon.

What remains is to bound the "correction term,"

$$\int_0^t \rho(s)\sin\Big((N+\tfrac{1}{2})s\Big)\,ds, \qquad \rho(s) := \frac{1}{2\sin(s/2)} - \frac{1}{s}.$$

Again, we recall the original issue: if the denominator of $\rho(s) \to \infty$ as $s \downarrow 0$, then we'd be done; it is why we needed a "different proof" from the Dini approach in the first place. In essence, what we've been building to is the following: notice that on expanding

$$\sin(s/2) = \frac{s}{2} - \frac{s^3}{48} + \frac{s^5}{3840} - \cdots \approx \frac{s}{2},$$

with the approximation becoming more and more reasonable as $s \downarrow 0$, we have that

$$\rho(s) \approx \frac{1}{2(s/2)} - \frac{1}{s} \to 0 \quad \text{as} \quad s \downarrow 0.$$

Realize the significance of what we've accomplished here — by extending ρ such that

$$\rho(s) := \begin{cases} \dfrac{1}{2\sin(s/2)} - \dfrac{1}{s} & s \in (0,\pi], \\ 0 & s = 0, \end{cases}$$

we will have succeeded in our original goal: $\rho\colon [0,t] \to \mathbb{R}$ is now *continuous* on the compact interval $[0,\pi]$, and hence, is bounded: for some $R \geq 0$,

$$|\rho(s)| \leq R, \qquad s \in [0,\pi].$$

Since $\sin\Big((N+\tfrac{1}{2})s\Big) \leq 1$, we obtain

$$\left|\int_0^t \rho(s)\sin\Big((N+\tfrac{1}{2})s\Big)\,ds\right| \leq R\int_0^t ds = Rt \leq R\pi$$

for all $t \in [0,\pi]$ and $N \geq 1$.

Combining the two bounds in

$$\sigma_N(t) = \int_0^t \frac{\sin\Big((N+\tfrac{1}{2})s\Big)}{s}\,ds + \int_0^t \rho(s)\sin\Big((N+\tfrac{1}{2})s\Big)\,ds.$$

we have that

$$|\sigma_N(t)| \leq R\pi + 2\pi := C \quad \text{for all} \quad t \in [0,\pi], \quad N \geq 1.$$

Here is where we make the Stieltjes leap: observing that, by construction and the FTC,

$$\sigma_N'(t) = D_N(t),$$

what we have is

$$D_N(t)\,dt = \sigma_N'(t)\,dt := d\sigma_N(t),$$

where it is the lattermost notation where we've adopted the Stieltjes notation. In particular, we can rewrite the integral I_δ as

$$I_\delta = \frac{1}{\pi}\int_0^\delta g(t)D_N(t)\,dt = \frac{1}{\pi}\int_0^\delta g(t)\sigma_N'(t)\,dt = \frac{1}{\pi}\int_0^\delta g(t)\,d\sigma_N(t).$$

This latter expression says I_δ is the Riemann-Stieltjes integral of $g(t) = f(x-t) - f(x^-)$, which inherits bounded variation on $[0,\delta]$ from f's presumed BV on $[-\pi,\pi]$, against a *smooth* integrator $d\sigma_N(t)$, which (on a bounded interval) automatically implies bounded variation. In particular (cf. the Riemann-Stieltjes Existence Theorem), what we've just guaranteed is that for any $\delta > 0$, the integral

$$I_\delta = \frac{1}{\pi}\int_0^\delta g(t)\,d\sigma_N(t) \quad \text{exists.}$$

Now apply integration by parts in the Riemann-Stieltjes sense — with g as the integrand and σ_N as the integrator:

$$\int_0^\delta g(t)d\sigma_N(t) = \Big[g(t)\sigma_N(t)\Big]_0^\delta - \int_0^\delta \sigma_N(t)dg(t) = g(\delta)\sigma_N(\delta) - g(0^+)\sigma_N(0) - \int_0^\delta \sigma_N(t)\,dg(t).$$

It is this latter expression,

$$g(\delta)\sigma_N(\delta) - g(0^+)\sigma_N(0) - \int_0^\delta \sigma_N(t)\,dg(t),$$

which we will proceed in analyzing.

First, let's look at the boundary terms. We have that

$$g(0^+)\sigma_N(0) = g(0^+)\int_0^0 D_N(s)\,ds = 0,$$

which vanishes identically *regardless* of the value of $g(0^+)$ for any N. This is extremely important — in the Dini proof, the behavior of $g(t) = f(x-t) - f(x^-)$ near $t = 0$ was the entire source of difficulty, since it determined whether $\phi(t) = g(t)/(2\sin(t/2))$ was integrable on $[0,t]$. However, when we multiply this troublesome contribution by the vanishing $\sigma_N(0)$, it simply disappears. We've rid ourselves of the significance of the singularity *at the point* $t = 0$.

Second, since $g(\delta) = f(x-\delta) - f(x^-)$ depends on δ and not on N and since $|\sigma_N(t)| \leq C$ uniformly in t for all $N \geq 1$, we have that

$$|g(\delta)\sigma_N(\delta)| \leq |g(\delta)| \cdot C,$$

where

$$\lim_{\delta\downarrow 0} g(\delta) = \lim_{\delta\downarrow 0}[f(x-\delta) - f(x^-)] = f(x^-) - f(x^-) = 0,$$

so that

$$|g(\delta)\sigma_N(\delta)| \leq |g(\delta)| \cdot C \to 0 \quad \text{as} \quad \delta \downarrow 0.$$

Since this convergence depends only on δ and not on N, in particular, it holds *uniformly* in N.

Finally, we look at the integral

$$S_\delta := \int_0^\delta \sigma_N(t)\,dg(t).$$

We seek to verify two things: first, that this Riemann-Stieltjes integral exists; second, that it can be made small.

For existence, realize that $\sigma_N(t)$ is smooth on $[0,\delta]$, and that the integrator $g(t) = f(x-t) - f(x^-)$ is of bounded variation on $[0,\delta]$ by the assumption that f is of bounded variation on all of $[-\pi,\pi]$. By the Riemann-Stieltjes Existence Theorem (continuous integrand, BV integrator), the integral S_δ exists.

For the bound, apply the Riemann-Stieltjes Estimate (see the previous ribbon) directly: since $|\sigma_N(t)| \le C$ uniformly in N for every t,

$$|S_\delta| = \left|\int_0^\delta \sigma_N(t)\,dg(t)\right| \le \sup_{t\in[0,\pi]} |\sigma_N(t)| \cdot V_0^\delta(g) \le C \cdot V_0^\delta(g).$$

Now, recall that $g(0^+) = f(x^-) - f(x^-) = 0$, so defining $g(0) = 0$ makes $g(0^+) = g(0)$ so that g is right-continuous at $t = 0$. In particular, g is of bounded variation on $[0,\pi]$ and right-continuous at $t = 0$, the Vanishing Property (cf. Total Variations Property (b)) says that

$$V_0^\delta(g) \to 0 \quad \text{as} \quad \delta \downarrow 0.$$

But what this means is that we can control the size of $|S_\delta|$ just by taking $\delta > 0$ sufficiently small — and the constant C, nor the total variation $V_0^\delta(g)$, depend on $N \ge 1$, so that the bound $|S_\delta| \le C \cdot V_0^\delta(g)$ is, again, *uniform* in N.

Now, what we've just shown is that

$$I_\delta = \frac{1}{\pi}\int_0^\delta g(t)\,d\sigma_N(t) = \frac{1}{\pi}\left[g(\delta)\sigma_N(\delta) - g(0^+)\sigma_N(0) - \int_0^\delta \sigma_N(t)\,dg(t)\right],$$

where

$$|g(\delta)\sigma_N(\delta)| \le |g(\delta)| \cdot C \to 0 \quad \text{as} \quad \delta \downarrow 0 \quad \text{uniformly for} \quad N \ge 1,$$

tgat

$$|g(0^+)\sigma_N(0)| = 0 \quad \text{for all} \quad N \ge 1,$$

and that

$$\left|\int_0^\delta \sigma_N(t)\,dg(t)\right| \le C \cdot V_0^\delta(g) \quad \text{uniformly for} \quad N \ge 1.$$

We thus have

$$|I_\delta| = \frac{1}{\pi}\left|\int_0^\delta g(t)\,d\sigma_N(t)\right| \le \frac{1}{\pi}\left[|g(\delta)| \cdot C + 0 + C \cdot V_0^\delta(g)\right] = \frac{C}{\pi}[|g(\delta)| + V_0^\delta(g)].$$

So, let $\epsilon > 0$. Since $|g(\delta)| \to 0$ and $V_0^\delta(g) \to 0$ as $\delta \downarrow 0$, we can choose $\delta > 0$ small enough that

$$|I_\delta| < \frac{\epsilon}{2}, \quad \text{uniformly in } N.$$

With δ now *fixed*, the integral J_δ is handled as we described earlier: since

$$J_\delta = \frac{1}{\pi}\int_\delta^\pi \phi(t)\sin\left((N+\tfrac{1}{2})t\right)dt$$

with $\phi \in L^1[\delta,\pi]$ and $N + \frac{1}{2} \to \infty$, it follows by Riemann-Lebesgue that for any fixed $\delta > 0$

$$J_\delta \to 0 \quad \text{as} \quad N \to \infty.$$

In particular, there exists $N_0 \ge 1$ such that $|J_\delta| < \epsilon/2$ for all $N \ge N_0$.

Together, we have that for any $\delta > 0$ and all $N \geq N_0$,

$$|I_N^-| = |I_\delta + J_\delta| \leq |I_\delta| + |J_\delta| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, we conclude that

$$I_N^- \to 0 \quad \text{as} \quad N \to \infty.$$

By a symmetric argument (replacing $g(t) = f(x-t) - f(x^-)$ with $f(x+t) - f(x^+)$ throughout), we also have

$$I_N^+ \to 0 \quad \text{as} \quad N \to \infty.$$

Therefore,

$$S_N(x) - \frac{f(x^+) + f(x^-)}{2} = I_N^- + I_N^+ \to 0 \quad \text{as} \quad N \to \infty,$$

which is to say that, for any $x \in [-\pi, \pi]$, we have

$$\lim_{N\to\infty} S_N(x) = \frac{f(x^+) - f(x^-)}{2},$$

which completes the proof.

Remark.

We promised a word about Dini. In fact, though we've stated the theorem above as the "most general" version of the Dirichlet conditions ensuring pointwise convergence of S_N, what we really have now is a set of *complementary* tools — i.e., the Dini condition has not become superfluous in light of the Jordan-Dirichlet conditions. In particular, realize that what Dirichlet-Jordan gives is a means of ensuring pointwise convergence across the *entirety* of $[-\pi, \pi]$ by checking only a single *global* condition — namely, that f be of bounded variation on $[-\pi, \pi]$, $V_{-\pi}^{\pi}(f) < \infty$. If the condition holds, convergence is guaranteed everywhere, and one never needs to inspect the behavior of f at any individual point.

On the other hand, the Dini condition is *inherently local*: it asks whether f approaches its one-sided limit "fast enough" at a *specific* $x \in [-\pi, \pi]$, and if so, it guarantees convergence *at that point alone* — with no assumptions whatsoever on f's behavior elsewhere on the interval $[-\pi, \pi]$.

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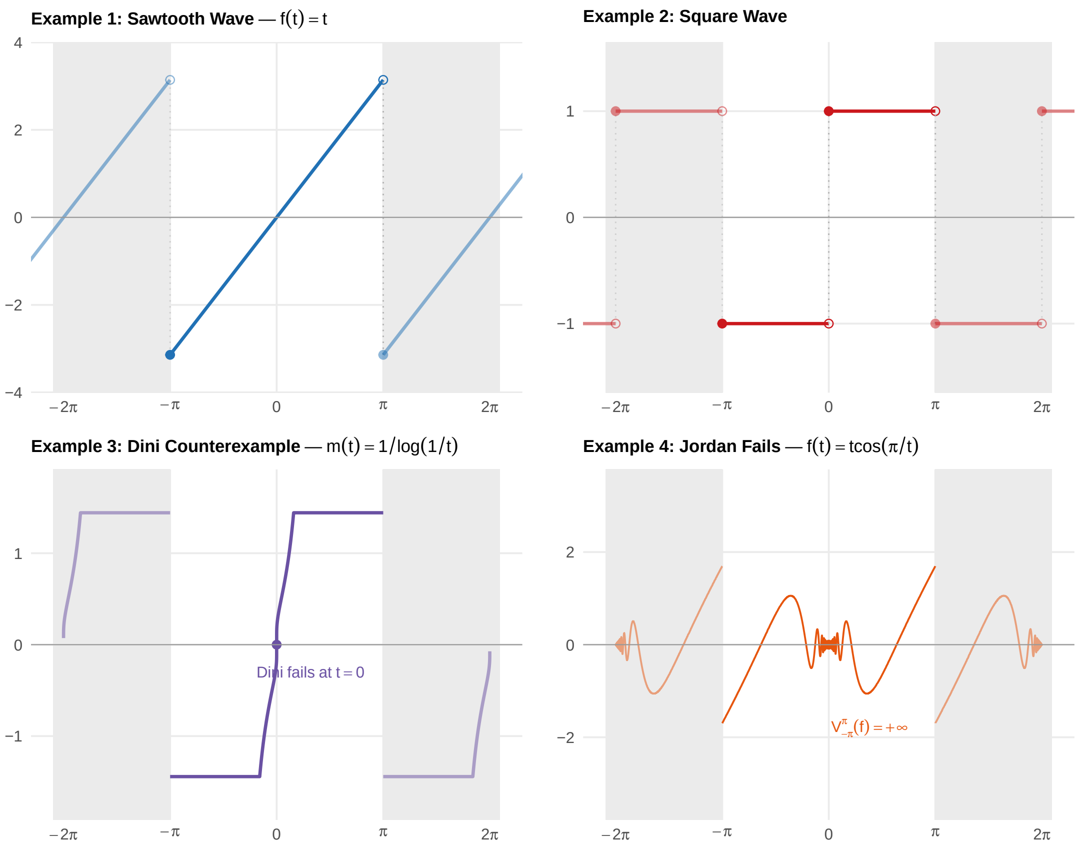
Examples

Let us see some examples in action.

Example 2.2.26 (The Fourier Sawtooth):
Let $f\colon [-\pi, \pi] \to \mathbb{R}$ be defined by

$$f(t) = t, \quad t \in (-\pi, \pi),$$



The four 2π-periodic extensions of functions considered in the examples below. Examples 1–3 are of bounded variation and fall under the Dirichlet-Jordan theorem; Example 4 has infinite total variation near the origin, so Jordan does not apply, but Dini gives convergence at every point away from $t = 0$.

where f is discontinuous at $\pm\pi$, but with well-defined left- and right-limits

$$f(\pi^-) = \lim_{t \uparrow \pi} f(t) = \pi, \quad f(-\pi^+) = \lim_{t \downarrow -\pi} f(t) = -\pi.$$

Extend f to be 2π-periodic:

$$f(x + 2\pi) = f(x), \quad x \in (-\pi, \pi),$$

so that f is the classical **sawtooth wave**. We wish to determine the Fourier series S_N of f and verify its convergence on $[-\pi, \pi]$ using the Dirichlet-Jordan Theorem.

To start, note that the function $f(t) = t$ is monotone increasing on $[-\pi, \pi]$, so

$$V_{-\pi}^{\pi}(f) = |f(\pi) - f(-\pi)| = |\pi - (-\pi)| = 2\pi < \infty,$$

so that f is of bounded variation on $[-\pi, \pi]$. As such, the Dirichlet-Jordan theorem applies: the Fourier partial sums

$$S_N(x) \to \frac{f(x^+) + f(x^-)}{2} \quad \text{as} \quad N \to \infty \quad \text{for every} \quad x \in [-\pi, \pi].$$

Let us now compute the Fourier series itself. Recall that

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N}(a_n \cos(nx) + b_n \sin(nx)),$$

where the Fourier coefficients are defined to be

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\,dt, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt.$$

We can compute the coefficients directly: we have that

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} t\,dt = \frac{1}{\pi}\left[\frac{t^2}{2}\right]_{-\pi}^{\pi} = \frac{1}{\pi}\left[\frac{\pi^2}{2} - \frac{(-\pi)^2}{2}\right] = 0;$$

alternatively, observe that $f(-t) = -t = -f(t)$ for all $t \in [-\pi, \pi]$, so that $f(t)$ is odd; it follows that the integral of f over any symmetric interval about 0 will vanish.

Similarly, observe that in

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t\cos(nt)\,dt,$$

the function t is odd, while the function $\cos(nt)$ is even for any $n \geq 1$. In particular, for every n, the integrand is odd: $t\cos(nt) = \text{odd} \times \text{even} = \text{odd}$. Thus, we have

$$a_n = 0, \qquad n \geq 1.$$

On the other hand, since $\sin(nt)$ is odd for any $n \geq 1$, the product $t\sin(nt) = \text{odd} \times \text{odd} = \text{even}$, so that

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t\sin(nt)\,dt = \frac{2}{\pi}\int_{0}^{\pi} t\sin(nt)\,dt.$$

Integrating by parts with $u = t$ and $dv = \sin(nt)$, we have that

$$\begin{aligned} u &= t, & dv &= \sin(nt), \\ du &= dt, & v &= -\frac{1}{n}\cos(nt) \end{aligned}$$

so

$$\int_0^{\pi} t\sin(nt)\,dt = \left[-\frac{t\cos(nt)}{n}\right]_0^{\pi} - \frac{1}{n}\int_0^{\pi}\cos(nt)\,dt = -\frac{\pi\cos(n\pi)}{n} + \frac{1}{n}\underbrace{\left[\frac{\sin(nt)}{n}\right]_0^{\pi}}_{=0} = -\frac{\pi(-1)^n}{n},$$

since $\cos(nx) = (-1)^n$, $n = 0, 1, 2, \ldots$. Thus, we have that

$$b_n = \frac{2}{\pi}\int_0^{\pi} t\sin(nt) = \frac{2}{\pi} \cdot -\frac{\pi(-1)^n}{n} = \frac{2(-1)^{n+1}}{n}, \qquad n \geq 1.$$

Thus, we have that $a_0 = 0$ and that for $n \geq 1$, $a_n = 0$ and $b_n = \frac{2(-1)^{n+1}}{n}$. Plugging these into S_N,

$$S_N(x) = \sum_{n=1}^{N} b_n \sin(nx) = \sum_{n=1}^{N} \frac{2(-1)^{n+1}}{n}\sin(nx) = 2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \cdots\right).$$

By the Dirichlet-Jordan theorem, at every point of continuity $x \in (-\pi, \pi)$, $S_N(x) \to f(x) = x$. At the jump discontinuity $x = \pm\pi$,

$$S_N(\pi) \to \frac{f(\pi^-) + f(-\pi^+)}{2} = \frac{\pi + (-\pi)}{2} = 0,$$

and similarly

$$S_N(-\pi) \to \frac{f(-\pi^+) + f(\pi^-)}{2} = \frac{-\pi + \pi}{2} = 0.$$

Interestingly, if we set $x = \pi/2 \in (-\pi, \pi)$, then from

$$f(x) = x = \lim_{N\to\infty} S_N(x),$$

we obtain

$$\frac{\pi}{2} = 2\left(\underbrace{\sin\tfrac{\pi}{2}}_{=1} - \underbrace{\frac{\sin\pi}{2}}_{=0} + \underbrace{\frac{\sin\frac{3\pi}{2}}{3}}_{=-1/3} - \underbrace{\frac{\sin 2\pi}{4}}_{=0} + \underbrace{\frac{\sin\frac{5\pi}{2}}{5}}_{=1/5} + \cdots\right) = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right);$$

rearranging, we can again obtain Leibniz's formula (which, recall, we earlier derived starting from the Fourier series for a different function — namely, the square-wave):

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

Now, consider the function which caused us so much trouble earlier:

$$m(t) = \begin{cases} \dfrac{1}{\log(1/t)} & t \in (0, 1/2], \\ 0 & t = 0. \end{cases}$$

What we found earlier was that, at the point $t = 0$, the function m failed to satisfy the Dini condition — we thus could not conclude pointwise convergence of $S_N(0)$ to $m(0)$ at the point $t = 0$.

Now, we've replaced the pointwise check at $t = 0$ with a global check: verify that since m is monotone increasing on $[0, 1/2]$, we have that

$$V_0^{1/2}(m) = m(1/2) - m(0) = \frac{1}{\log 2} < \infty,$$

so that m is of bounded variation on $[0, 1/2]$. In particular, then, after defining the 2π-periodic extension f of m on $[-\pi, \pi]$, we have that

$$V_{-\pi}^{\pi}(f) = V_{-\pi}^{-1/2}(f) + V_{-1/2}^{1/2}(f) + V_{1/2}^{\pi}(f)$$

by additivity, where

$$V_{-\pi}^{-1/2}(f) = V_{1/2}^{\pi}(f) = 0,$$

and where

$$V_{-1/2}^{1/2}(f) = 2V_0^{1/2}(m) = \frac{2}{\log 2},$$

so that, together,

$$V_{-\pi}^{\pi}(f) = \frac{2}{\log 2} < \infty.$$

In particular, f — the odd 2π-periodic extension of m — is of bounded variation on $[-\pi, \pi]$, so that the Dirichlet-Jordan theorem guarantees

$$S_N(0) \to \frac{f(0^+) + f(0^-)}{2} = \frac{0+0}{2} = 0 = f(0).$$

This is the convergence we *observed* visually earlier in the chapter, but which remained elusive to us armed with only the pointwise Dini condition, since

$$\int_0^\delta \frac{|f(0-t) - f(t^-)|}{t}\, dt = \int_0^\delta \frac{1}{t\log(1/t)}\, dt = +\infty.$$

The Jordan approach handles this by never dividing by t, never checking the rate at which $m(t) \to 0$ as $t \downarrow 0$, and thus never encountering the logarithmic singularity. It only asks whether m is of bounded variation, and since m is monotone increasing from $m(0) = 0$ continuously on the compact interval $[0, 1/2]$, it certainly is.

The final example we demonstrate is one where Jordan fails on the whole interval, while Dini survives. As we'll see, it is really that there is "one problematic point" which breaks bounded variation, and if we were to remove this point, we could actually verify bounded variation on the rest of the interval. Nevertheless, since BV is inherently a global check, it is nice to have something local like the Dini condition to tell us just how widespread the problems are, and whether or not we can rectify the issue by removing those points from the domain.

In particular, consider

$$g(t) = \begin{cases} t\cos(\pi/t) & t \in (0,1], \\ 0 & t = 0 \end{cases}.$$

Extend g to the 2π-periodic f, and observe that f is odd on $[-\pi, \pi]$. We showed earlier that g is of unbounded variation on $[0, 1]$:

$$V_0^1(g) = V_0^1(f) = +\infty.$$

This is because oscillations of $\cos(\pi/t)$ near $t = 0$ produce a "harmonic series' worth of excursion" about the origin — we imagine that we're "pacing slower as we approach $t \to 0$, but never slow enough that our total journey winds up finite since the excursions decay like $1/t$, whose integral diverges for intervals $(-\delta, \delta)$ about 0."

However, at any point $x_0 \in (0, 1]$ *away* from the origin, f is differentiable, with

$$f'(x_0) = \cos(\pi/x_0) + \frac{\pi}{x_0}\sin(\pi/x_0).$$

In particular, near x_0, we can bound (cf. Taylor)

$$|f(x_0 - t) - f(x_0)| \le t \cdot |f'(x_0)| + O(t^2),$$

so that

$$\int_0^\delta \frac{|f(x_0 - t) - f(x_0)|}{t}\, dt \le |f'(x_0)| \cdot \delta + O(\delta^2) < \infty$$

for fixed $\delta > 0$. In particular, the Dini condition holds at $x_0 \in (0, 1]$; since it was arbitrary, the original Dini proof gives

$$S_N(x_0) \to f(x_0) \quad \text{as} \quad N \to \infty.$$

Realize that Jordan cannot see the nuance here: if we were to compute the total variation $V_\delta^1(m)$ for some fixed $\delta > 0$, we would observe $V_\delta^1(m) < \infty$, so that we reach the same conclusion; nevertheless, it is often more practical in situations where the initial BV check fails to diagnose the behavior using a Dini argument as above. Indeed, Jordan's theorem can cover situations that Dini can't, and Dini can cover some situations that Jordan can't. Both are useful, though we often begin by checking the Jordan condition in practice as it automatically guarantees us convergence on all of $[-\pi, \pi]$ whenever it holds.

Techniques for Computing Fourier Series

Thus far, we've consolidated our attention primarily on the class of 2π-periodic functions $f\colon [-\pi, \pi] \to \mathbb{R}$. The reason for this is that the interval $[-\pi, \pi]$ is the *natural* domain of the trigonometric basis $\mathcal{T} = \{1, \cos(nt), \sin(nt)\}_{n\in\mathbb{N}}$ — on $[-\pi, \pi]$, the nth harmonics $\cos(nx)$ and $\sin(nx)$ complete exactly n full periodic cycles. In particular, we require an interval of length 2π because the functions $\sin(x)$ and $\cos(x)$ are each periodic with period exactly equal to 2π, and all other harmonics have periods $2\pi n$ equal to integer multiples of the largest period.

Nevertheless, it should be clear that there is nothing sacred about the interval $[-\pi, \pi]$. We've already discussed the process of *extending* the square-wave from $[-1, 1]$ to $[-\pi, \pi]$. Indeed, in general, the passage to an arbitrary symmetric interval $[-\ell, \ell]$ is nothing more than a linear change of variables.

Theorem 2.2.27 (Fourier Series on Symmetric Intervals)

Suppose that $f \in L^1[-\ell, \ell]$ for some $\ell > 0$. Then the Fourier series of f is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{\ell}\right) + b_n \sin\left(\frac{n\pi x}{\ell}\right)\right),$$

where

$$a_0 = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\,dt, \qquad a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\cos\left(\frac{n\pi t}{\ell}\right)dt, \qquad b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\sin\left(\frac{n\pi t}{\ell}\right)dt.$$

The function f is then extended to be 2ℓ-periodic by requiring that

$$f(x + 2\ell) = f(x), \quad \forall x \in \mathbb{R}.$$

Proof:

We seek to turn the problem into one we already know how to solve. As such, given $t \in [-\ell, \ell]$, define the rescaled variable

$$u = \frac{\pi t}{\ell},$$

so that

$$t = \frac{\ell u}{\pi}, \quad dt = \frac{\ell}{\pi} du.$$

Then, as we integrate over $t \in [-\ell, \ell]$, the new variable u ranges over $[-\pi, \pi]$ — in particular, we have that

$$a_0 = \frac{1}{\ell} \int_{-\ell}^{\ell} f(t)\, dt = \frac{1}{\ell} \int_{-\pi}^{\pi} f\left(\frac{\ell u}{\pi}\right) \cdot \frac{\ell}{\pi} du = \frac{1}{\pi} \int_{-\pi}^{\pi} g(u)\, du,$$

where we've defined

$$g(u) := f\left(\frac{\ell u}{\pi}\right), \quad u \in [-\pi, \pi].$$

Since $f \in L^1[-\ell, \ell]$ and $u \mapsto \ell u/\pi$ is a continuous function for $u \in [-\pi, \pi]$, the composition $u \mapsto \ell u/\pi \mapsto f(\ell u/\pi) = g(u)$ provides a continuous argument to the integrable f, whence g remains integrable:

$$f \in L^1[-\ell, \ell] \implies g \in L^1[-\pi, \pi].$$

Thus, we can obtain the standard Fourier series for g:

$$g(u) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nu) + b_n \sin(nu)),$$

with coefficients

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(u) \cos(nu)\, du, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} g(u) \sin(nu)\, du.$$

Now, simply undo the substitution: with $u = \pi t/\ell$, we have $du = (\pi/\ell) dt$ and that the limits at $u = \pm\pi$ correspond with $t = \pm\ell$:

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(u) \cos(nu)\, du \\ &= \frac{1}{\pi} \int_{-\ell}^{\ell} f(t) \cos\left(\frac{n\pi t}{\ell}\right) \frac{\pi}{\ell}\, dt \\ &= \frac{1}{\ell} \int_{-\ell}^{\ell} f(t) \cos\left(\frac{n\pi t}{\ell}\right) dt, \end{aligned}$$

and similarly

$$b_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(t) \sin\left(\frac{n\pi t}{\ell}\right) dt.$$

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Now, suppose the interval is not symmetric: that is, suppose $f\colon [a, b] \to \mathbb{R}$ for arbitrary $a < b$. The idea is the same: shift the interval so that it *becomes* symmetric, apply what we just derived, then shift back. In particular, we have the following.

Theorem 2.2.28 (Fourier Series on General Intervals)

Let $[a,b] \subset \mathbb{R}$ with $a < b$ and set

$$\ell := \frac{b-a}{2}, \qquad c := \frac{a+b}{2}.$$

Suppose $f \in L^1[a,b]$. Then the Fourier series of f on $[a,b]$ is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi(x-c)}{\ell}\right) + b_n \sin\left(\frac{n\pi(x-c)}{\ell}\right)\right),$$

where

$$a_0 = \frac{1}{\ell}\int_a^b f(t)\,dt, \quad a_n = \frac{1}{\ell}\int_a^b f(t)\cos\left(\frac{n\pi(t-c)}{\ell}\right)\,dt, \quad b_n = \frac{1}{\ell}\int_a^b f(t)\sin\left(\frac{n\pi(t-c)}{\ell}\right)\,dt.$$

The function f is then extended to be $(b-a)$-periodic by requiring

$$f(x+(b-a)) = f(x), \quad \forall x \in [a,b].$$

Proof:

The proof is nothing more than an application of the symmetric case after defining the symmetrized $\tilde{f}(t) := f(t+c)$ for $t \in [-\ell, \ell]$ (where $c = \frac{a+b}{2}$ and $\ell = \frac{b-a}{2}$ as above). Then $f \in L^1[a,b]$ implies $\tilde{f} \in L^1[-\ell,\ell]$, and we apply the previous theorem to obtain $\tilde{f}$'s Fourier series. By then re-substituting $t \leftarrow t-c$ and re-shifting the bounds from $[-\ell,\ell]$ to $[a,b]$, we recover the desired form.

$\|$

Remark.

When the interval is already symmetric — i.e., when $a = -\ell$ and $b = \ell$ — then $c = 0$ as we've defined it, and the shift $x - c = x$. In particular, we recover precisely the same formula from the symmetric case from the general case. By then setting $\ell = \pi$, we recover the familiar, canonical formulas on $[-\pi,\pi]$. In this way, we only need to remember the *general form* described above. Nevertheless, we tend to *work with* the symmetrized version when stating results in general.

The critical quantity, of course, is

$$\ell = \frac{b-a}{2};$$

Once ℓ is known, the normalization is $1/\ell$, the frequency of the nth harmonic is $\frac{\pi n}{\ell}$, and the period of the function's extension is 2ℓ. (Verify these agree with what we know from the case when $\ell = \pi$.)

When computing Fourier series in practice, the following facts tend to be useful.

Theorem 2.2.29 (Fourier Integration Techniques)

Let $f \in L^1[-\ell, \ell]$ be piecewise continuous. Suppose we wish to compute the Fourier coefficient integrals

$$a_0 = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\,dt, \quad a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\cos\left(\frac{n\pi t}{t}\right)\,dt, \quad b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\sin\left(\frac{n\pi t}{t}\right)\,dt.$$

The following collection of techniques covers the evaluation of these integrals for a large collection of functions encountered in practice.

(i) **(Endpoint Evaluations)** For any $n \in \mathbb{Z}$, we have that

$$\sin(n\pi) = 0, \qquad \cos(n\pi) = (-1)^n.$$

(ii) **(Half-integer Evaluations)** We have that

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & n\,\text{even}, \\ 1 & n = 1, 5, 9, 13, \ldots \\ -1 & n = 3, 7, 11, 15, \ldots \end{cases} \implies \left|\sin\left(\frac{n\pi}{2}\right)\right| = \begin{cases} 0 & n\,\text{even}, \\ 1 & n\,\text{odd}, \end{cases}$$

and that

$$\cos\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & n\,\text{odd}, \\ 1 & n = 0, 4, 8, 12, \ldots \\ -1 & n = 2, 6, 10, 14, \ldots \end{cases} \implies \left|\cos\left(\frac{n\pi}{2}\right)\right| = \begin{cases} 0 & n\,\text{odd}, \\ 1 & n\,\text{even}. \end{cases}$$

That is, the two half-integer evaluations $\sin(n\pi/2)$ and $\cos(n\pi/2)$ cycle between $\{0, 1, 0, -1\}$ and $\{1, 0, -1, 0\}$ respectively with period 4 in n.

(iii) **(Parity and Symmetry)** If f is even on $[-\ell, \ell]$, then $f(t)\sin\left(\frac{n\pi t}{\ell}\right)$ is odd for every $n \geq 1$. In particular,

$$f \text{ even on } [-\ell, \ell] \implies \forall n \geq 1\colon \quad b_n = 0 \quad \text{and} \quad a_n = \frac{2}{\ell}\int_0^{\ell} f(t)\cos\left(\frac{n\pi t}{\ell}\right)\,dt$$

Similarly, for an odd function $g \in L^1[-\ell, \ell]$, we have that

$$g \text{ odd on } [-\ell, \ell] \implies \forall n \geq 1\colon \quad a_n = 0 \quad \text{and} \quad b_n = \frac{2}{\ell}\int_0^{\ell} g(t)\sin\left(\frac{n\pi t}{\ell}\right)\,dt.$$

(iv) **(The Constant Coefficient)** In practical computations, we tend to treat the zeroth cosine coefficient

$$a_0 = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\,dt$$

separately from the other cosine coefficients a_n, $n \geq 1$. What we seek to distinguish here is that, though we often refer to a_0 as a case of the formula for a_n for $n = 0$, a common mistake is to "compute a closed-form expression for a_n" and then apply it at $n = 0$ to find a_0, when such closed form involves a *denominator* in n. This happens often enough that it is typically simpler to reserve the general formula for a_n, $n \geq 1$, and to compute a_0 simply by integrating f as above.

(v) **(Tabular/DI Method for Integration By Parts)** We often prefer to compute integrals requiring successive applications of integration by parts $\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$ by applying the tabular method; see the figure below for an illustration of this technique.

(vi) **(Product-to-Sum Formulas)** As derived earlier, we have the following formulas for any $\alpha, \beta \in \mathbb{R}$:

$$\cos\alpha\cos\beta = \frac{\cos(\alpha-\beta)+\cos(\alpha+\beta)}{2}, \tag{P1}$$

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta)-\cos(\alpha+\beta)}{2}, \tag{P2}$$

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}, \tag{P3}$$

$$\cos\alpha\sin\beta = \frac{\sin(\alpha+\beta)-\sin(\alpha-\beta)}{2}. \tag{P4}$$

(vii) **(Computing by Inspection)** Whenever the target function f is itself composed of finitely many trigonometric parts — say,

$$f(t) = c_0 + \sum_{k=1}^{M}\left(\alpha_k\cos\left(\frac{k\pi t}{\ell}\right) + \beta_k\sin\left(\frac{k\pi t}{\ell}\right)\right),$$

then the orthogonality relations immediately give

$$a_0 = 2c_0; \quad a_k = \alpha_k, \quad b_k = \beta_k, \qquad k = 1, \dots, M;$$

and that all other Fourier coefficients vanish:

$$a_{M+1} = a_{M_2} = \cdots = 0; \qquad b_{M+1} = b_{M+2} = \cdots = 0.$$

(viii) **(Even–Odd Reindexing.)** Recall from (i) that $\cos(n\pi) = (-1)^n$. Since nearly every Fourier coefficient integral is evaluated at $t = \pm\ell$ — producing $\cos(n\pi) = (-1)^n$ or $\sin(n\pi) = 0$ at the boundary — the factor $(-1)^n$ appears in the closed-form expression for a_n or b_n with overwhelming frequency. In particular, one commonly encounters the expressions

$$1-(-1)^n = \begin{cases} 0 & n \text{ even}, \\ 2 & n \text{ odd}; \end{cases} \qquad 1+(-1)^n = \begin{cases} 2 & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

In particular, it is often useful to know that *if* a_n contains $1-(-1)^n$ as a factor, then $a_n = 0$ for all even n — the series involves only odd harmonics. Substituting $n = 2k+1$ (odd):

$$\sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi t}{\ell}\right) = \sum_{k=0}^{\infty} a_{2k+1}\cos\left(\frac{(2k+1)\pi t}{\ell}\right).$$

Similarly, if a_n contains $1+(-1)^n$ as a factor, then $a_n = 0$ for all odd n and we substitute $n = 2k$ (even):

$$\sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi t}{\ell}\right) = \sum_{k=1}^{\infty} a_{2k}\cos\left(\frac{2k\pi t}{\ell}\right).$$

$$\int f(t)\,\cos\omega_n t\ dt$$

d		i
$f(t)$	$+$	$\cos\omega_n t$
$f'(t)$	$-$	$\omega_n^{-1}\sin\omega_n t=\omega_n^{-1}\cos\left(\omega_n t+\frac{\pi}{2}\right)$
$f''(t)$	$+$	$-\omega_n^{-2}\cos\omega_n t=\omega_n^{-2}\cos\left(\omega_n t+\pi\right)$
$f'''(t)$	$-$	$-\omega_n^{-3}\sin\omega_n t=\omega_n^{-3}\cos\left(\omega_n t+\frac{3\pi}{2}\right)$
$f^{(4)}(t)$		$\omega_n^{-4}\cos\omega_n t=\omega_n^{-4}\cos\left(\omega_n t+2\pi\right)$
$\vdots$		$\vdots$
$f^{(n)}(t)$		$\omega_n^{-n}\cos\left(\omega_n t-\frac{n\pi}{2}\right)$

$$\int f(t)\,\sin\omega_n t\ dt$$

d		i
$f(t)$	$+$	$\sin\omega_n t$
$f'(t)$	$-$	$-\omega_n^{-1}\cos\omega_n t=\omega_n^{-1}\sin\left(\omega_n t+\frac{\pi}{2}\right)$
$f''(t)$	$+$	$-\omega_n^{-2}\sin\omega_n t=\omega_n^{-2}\sin\left(\omega_n t+\pi\right)$
$f'''(t)$	$-$	$\omega_n^{-3}\cos\omega_n t=\omega_n^{-3}\sin\left(\omega_n t+\frac{3\pi}{2}\right)$
$f^{(4)}(t)$		$\omega_n^{-4}\sin\omega_n t=\omega_n^{-4}\sin\left(\omega_n t+2\pi\right)$
$\vdots$		$\vdots$
$f^{(n)}(t)$		$\omega_n^{-n}\sin\left(\omega_n t-\frac{n\pi}{2}\right)$

The tabular method (DI method) for $\int f(t)\cos\omega_n t\,dt$ and $\int f(t)\sin\omega_n t\,dt$, where $\omega_n := n\pi/\ell$. The d column lists successive derivatives of f and the i column lists successive antiderivatives of the trigonometric factor. Each colored diagonal pairs a d entry with the i entry one row below; the product, multiplied by the sign on the diagonal, gives one term in the antiderivative. Signs alternate $+,-,+,-,\ldots$; the kth diagonal (counting from $k=0$) carries sign $(-1)^k$. The grayed-out entry at the top of each i column has no d partner and does not contribute a term. The general nth i entry is the nth successive antiderivative of the trigonometric factor, written compactly using the phase-shift identity: $\omega_n^{-n}\cos(\omega_n t-n\pi/2)$ and $\omega_n^{-n}\sin(\omega_n t-n\pi/2)$ for the cosine and sine cases, respectively.

We will now work through several examples that exercise the techniques above for functions defined on the general interval $[-\ell,\ell]$ (again, recalling that we can translate a function f defined on general interval $[a,b]$ to an equivalent function defined on the symmetric interval $[-\ell,\ell]$ where $\ell=\frac{b-a}{2}$).

Example 2.2.30 (Example 1):

Let

$$f(t) := |t|, \quad t\in[-\ell,\ell].$$

Since for all $t\in[-\ell,\ell]$, we have

$$f(-t)=|-t|=|t|=f(t),$$

it follows that f is an even function on $[-\ell,\ell]$. By (iii), it immediately follows that the sine components of f's Fourier series vanish — i.e., we have

$$b_n = 0, \quad \forall n \in \mathbb{N}.$$

Notice that we did this without needing to compute $\int_{-\ell}^{\ell} |t| \sin(nt)\, dt$.

Next, in line with the philosophy described in (iv), we compute the constant coefficient deliberately (applying evenness of $f(t) = |t|$) as

$$a_0 = \frac{1}{\ell} \int_{-\ell}^{\ell} |t|\, dt = \frac{2}{\ell} \int_0^{\ell} t\, dt = \frac{2}{\ell} \cdot \frac{\ell^2}{2} = \ell.$$

For the remaining cosine coefficients, we again apply the even-function reduction from (iii):

$$a_n = \frac{2}{\ell} \int_0^{\ell} t \cos\left(\frac{n\pi t}{\ell}\right) dt, \quad n = 1, 2, \ldots$$

To compute the integral

$$\int_0^{\ell} t \cos\left(\frac{n\pi t}{\ell}\right) dt,$$

one applies the DI method (along with the) as

$$D\colon\ t \mapsto 1 \mapsto 0$$

$$I\colon\ \cos\left(\frac{n\pi t}{t}\right) \mapsto \frac{\ell}{n\pi} \sin\left(\frac{n\pi t}{\ell}\right) \mapsto -\frac{\ell^2}{n^2\pi^2} \cos\left(\frac{n\pi t}{\ell}\right)$$

which gives

$$\int_0^{\ell} t \cos\left(\frac{n\pi t}{\ell}\right) dt = \left[\frac{t\ell}{n\pi} \sin\left(\frac{n\pi t}{\ell}\right) - \frac{\ell^2}{n^2\pi^2} \cos\left(\frac{n\pi t}{\ell}\right)\right]_{t=0}^{t=\ell}$$

At $t = \ell$, we have

$$\sin\left(\frac{n\pi\ell}{\ell}\right) = \sin(n\pi) = 0, \quad \cos\left(\frac{n\pi\ell}{\ell}\right) = \cos(n\pi) = (-1)^n, \qquad \forall n \in \mathbb{N},$$

and at $t = 0$, that

$$\sin(0) = 0, \quad \cos(0) = 1.$$

In particular, we have that

$$\int_0^{\ell} t \cos\left(\frac{n\pi t}{\ell}\right) dt = \left[\frac{\ell^2}{n^2\pi^2}(-1)^n\right] - \left[\frac{\ell^2}{n^2\pi^2}\right] = -\frac{\ell^2}{n^2\pi^2}[1 - (-1)^n].$$

By (vii), the term $[1-(-1)^n]$ vanishes for all even n, and for odd $n = 2k+1$, we have $1-(-1)^{2k+1} = 1 + 1 = 2$, so that

$$a_{2k+1} = 2 \int_0^{\ell} t \cos\left(\frac{n\pi t}{\ell}\right) dt = 2 \cdot \left[-\frac{2\ell}{(2k+1)^2\pi^2}\right] = \frac{-4\ell}{(2k+1)^2\pi^2}, \quad \forall k \in \mathbb{N}.$$

Now, we have that

$$a_0 = \ell, \quad a_{2k+1} = \frac{-4\ell}{(2k+1)^2\pi^2}, \quad a_{2k} = 0, \quad \text{and} \quad b_k = 0; \quad k \in \mathbb{N};$$

we can thus assemble the series:

$$\begin{aligned}
f(x) = |x| &\sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{\ell}\right) + b_n \sin\left(\frac{n\pi x}{\ell}\right)\right) \\
&= \frac{\ell}{2} - \frac{4\ell}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{(2k+1)^2} \cos\left(\frac{(2k+1)\pi x}{\ell}\right) \\
&= \frac{\ell}{2} - \frac{4\ell}{\pi^2} \left(\cos\left(\frac{\pi x}{\ell}\right) + \frac{1}{9}\cos\left(\frac{3\pi x}{\ell}\right) + \frac{1}{25}\cos\left(\frac{5\pi x}{\ell}\right) + \cdots \right).
\end{aligned}$$

Now, we can apply this formula at $x = 0 \in [-\ell, \ell]$ for any $\ell \geq 0$: since $\cos(0) = 1$ and $f(0) = |0| = 0$, the formula gives us

$$0 = \frac{\ell}{2} - \frac{4\ell}{\pi^2}\left(1 + \frac{1}{9} + \frac{1}{25}\cdots\right),$$

which implies that

$$1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

Compare this with Leibniz's formula for π which arose from the square wave on $-\ell, \ell$:

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}.$$

It is curious indeed that they look so similar. And, indeed, we can take this comparison further: realize that, if we square Leibniz's formula, we obtain

$$\left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right]^2 = \left[\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}\right]^2;$$

for finite N, recall that

$$\left(\sum_k a_k\right)^2 = \sum_k a_k^2 + 2\sum_{j<k} a_j a_k.$$

In particular, this allows us to write

$$\left(\sum_{k=0}^{N} \frac{(-1)^k}{2k+1}\right)^2 = \sum_{k=0}^{N} \frac{1}{(2k+1)^2} + 2\sum_{k=0}^{N}\sum_{j<k} \frac{(-1)^{k+j}}{(2j+1)(2k+1)}.$$

But, now, the left side is

$$\left(\sum_{k=0}^{N} \frac{(-1)^k}{2k+1}\right)^2 \to \left(\frac{\pi}{4}\right)^2 = \frac{\pi^2}{16} \quad \text{as} \quad N \to \infty,$$

and we *also* know that the first term in the sum is just

$$\sum_{k=0}^{N} \frac{1}{(2k+1)^2} \to \frac{\pi^2}{8} \quad \text{as} \quad N \to \infty.$$

After subtracting the above through to both sides, we would seem to force

$$-2\sum_{k=0}^{\infty}\sum_{j<k}\frac{(-1)^{j+k}}{(2j+1)(2k+1)} = \sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} - \left(\sum_{k=0}^{\infty}\frac{(-1)^k}{2k+1}\right)^2 = \frac{\pi^2}{8} - \frac{\pi^2}{16} = \frac{\pi^2}{16}.$$

What we've discovered here is quite analogous to the familiar statistical identity,

$$\underbrace{\mathrm{Var}(X)}_{:=\mathbb{E}[(X-\mu_X)^2]} = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

which, here, analogously maps to

$$\underbrace{-2\sum_{k=0}^{\infty}\sum_{j<k}\frac{(-1)^{j+k}}{(2j+1)(2k+1)}}_{\sim\mathrm{Var}(X)} = \underbrace{\sum_{k=0}^{N}\frac{1}{(2k+1)^2}}_{\sim\mathbb{E}[X^2]} - \underbrace{\left(\sum_{k=0}^{N}\frac{(-1)^k}{2k+1}\right)^2}_{\sim(\mathbb{E}[X])^2}.$$

It is tempting to see the factor of 2 between $\pi^2/16$ and $\pi^2/8$ as multiplicative — as if squaring each term individually "doubles" the result of squaring the whole sum. But the relationship is additive, not multiplicative: we have

$$\underbrace{\frac{\pi^2}{16}}_{\text{cross terms: } -2\sum_{j<k} a_j a_k} = \underbrace{\frac{\pi^2}{8}}_{\text{sum of squares: } \sum a_k^2} - \underbrace{\frac{\pi^2}{16}}_{\text{square of sum: } (\sum a_k)^2}$$

so it is more or less a coincidence specific to this series that the cross terms happen to equal $\pi^2/16$ exactly, producing the illusion of a clean multiplicative effect where the intuition for the real result lies elsewhere. We will return to this example later, when we discuss *Parseval's Theorem*.

Differentiation and Computation

At this point, let us catalogue two results of note from this chapter. First, recall that we showed the *square wave*

$$f(x) = \begin{cases} -1 & -\pi < x < 0, \\ 0 & x = 0, \\ 1 & 0 < x < \pi, \end{cases}$$

had Fourier series

$$f(x) \sim \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\sin((2k+1)x)}{2k+1}.$$

Second, we just showed that the absolute value

$$g(x) = |x|, \quad x \in (-\pi, \pi),$$

has Fourier series (for $\ell = \pi$)

$$g(x) \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\cos((2k+1)x)}{(2k+1)^2}.$$

Visually, the similarities between the series expressions for f and g are quite evident — we're summing over odd sines and cosines, both are multiplied by a factor of $4/\pi$, both involve a factor of $1/(2k+1)$ in the denominator — but let us look at where they *differ*. The addition of the constant $\pi/2$ term in g is quite insignificant, as is the factor of -1 in front of the infinite series itself. On the other hand, what is *not* insignificant is the fact that the nonzero coefficients in the respective series expansions *decay at different rates* — namely, like

$$b_k(f) \sim O\left(\frac{1}{k}\right), \quad a_k(g) \sim O\left(\frac{1}{k^2}\right).$$

This observation, though seemingly mundane, reveals quite the interesting structural connection between Fourier series and *differentiation* of the original functions f and g. To see why, consider first what happens when we explicitly compute the coefficients $b_n(f)$ once again:

$$b_n(f) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-1)\sin(nt)\,dt + \int_{0}^{\pi}(1)\sin(nt)\,dt\right].$$

What we observe here is that, to find $b_n(f)$, we *never actually* need to integrate by parts — instead, f serves to split the integral, at which point we simply "integrate $\sin(nt)$ to get $-n^{-1}\cos(nt)$" as usual.

On the other hand, look at what happens when we compute $a_n(g)$:

$$a_n(g) = \frac{1}{\pi}\int_{-\pi}^{\pi} |t|\cos(nt)\,dt = \frac{2}{\pi}\int_{0}^{\pi} t\cos(nt)\,dt.$$

From here, we *do* need to integrate by parts, but *just once*: applying the DI method to

$$\text{D}: \ t \mapsto 1 \mapsto 0,$$
$$\text{I}: \ \cos(nt) \mapsto n^{-1}\sin(nt) \mapsto -n^{-2}\cos(nt)$$

we wind up applying integration by parts once to turn $\int t\cos(nt)\,dt$ into $[n^{-1}t\sin(nt)] - n^{-1}\int \sin(nt)\,dt$, and then we integrate $-n^{-1}\int\sin(nt)\,dt$, obtaining $-n^{-2}\cos(nt)$ as usual, to finish it out.

At this point, we've made a series of seemingly disconnected observations about the "number of times we can perform integration by parts on $\int f(t)\cos(nt)\,dt$ and/or $\int f(t)\sin(nt)\,dt$" — the gut punch here is that, actually, it is this underlying behavior which *endows* the Fourier coefficients with their rates of decay, and this behavior is directly to the *number of continuous derivatives* f possesses — or more precisely, that f's periodic extension possesses — on $\mathbb{R}$.

```
fourier_coefficients <- function(f, a, b, N){

  ell <- (b - a) / 2
  cc <- (a + b) / 2

  # Compute a_0, a_1, ..., a_N, b_1, ..., b_N for f on [-ell, ell]:
  # Nth trig argument:
  omega <- function(n) n*pi / ell

  # Integration helper:
```

```r
  safe_integrate <- function(g) {
  integrate(g, a, b, subdivisions = 1000L,
            rel.tol = 1e-10, stop.on.error = FALSE)$value
  }

  # Constant a_0:
  a.0 <- (1 / ell) * safe_integrate(f)

  # Cosine a_n:
  a.n <- sapply(1:N, function(n) {
    (1/ell) * safe_integrate(function(t) f(t) * cos(omega(n) * (t - cc)))
  })

  # Sine b_n:
  b.n <- sapply(1:N, function(n) {
    (1/ell) * safe_integrate(function(t) f(t) * sin(omega(n) * (t - cc)))
  })

  list(
    a.0 = a.0,
    a.n = a.n,
    b.n = b.n,
    lower = a,
    upper = b,
    ell = ell,
    cc = cc
  )
}

fourier_partial_sum <- function(coefs, N, x){
  # Evaluate S_N(x) = a.0/2 +
  # + sum_{n=1}^N [a.n cos(n pi (x - cc) / ell) + b.n sin(n pi (x - cc) / ell)

  ell <- coefs$ell
  cc <- coefs$cc

  # To vectorize, copy the scaled constant a_0/2 into a vector of length(x)
  S <- rep(coefs$a.0 / 2, length(x))

  # Loop:
  for (n in seq_len(min(N, length(coefs$a.n)))){
    S <- S +
      coefs$a.n[n] * cos(n * pi * (x - cc) / ell) +
      coefs$b.n[n] * sin(n * pi * (x - cc) / ell)
  }
  S
```

```r
}

compute_fourier <- function(f, a, b, N, x){
  coefs <- fourier_coefficients(f, a, b, N)
  sum <- fourier_partial_sum(coefs, N, x)

  list(
    coefs = coefs,
    S.n = sum
  )
}

(abs.fourier <- compute_fourier(abs, -pi, pi, 200, 0))
```

```
$coefs
$coefs$a.0
[1] 3.141593

$coefs$a.n
  [1] -1.273240e+00  2.220446e-16 -1.414711e-01 -2.775558e-17 -5.092958e-02
  [6]  1.369405e-16 -2.598448e-02 -4.926615e-16 -1.571901e-02 -5.300924e-17
 [11] -1.052264e-02 -1.744888e-16 -7.533962e-03 -2.208719e-17 -5.658842e-03
 [16]  6.262352e-16 -4.405673e-03  7.034769e-16 -3.526979e-03 -1.590277e-16
 [21] -2.887165e-03 -1.828819e-15 -2.406880e-03 -3.363326e-15 -2.037183e-03
 [26]  1.369405e-16 -1.746556e-03  4.770832e-16 -1.513959e-03 -6.885680e-16
 [31] -1.324911e-03  1.237508e-15 -1.169182e-03  1.889835e-16 -1.039379e-03
 [36]  3.205403e-16 -9.300508e-04  3.335165e-16 -8.371069e-04  1.139699e-15
 [41] -7.574298e-04  3.180555e-16 -6.886098e-04 -1.228048e-15 -6.287603e-04
 [46]  1.108225e-15 -5.763873e-04 -4.062248e-15 -5.302955e-04  5.389273e-16
 [51] -4.895192e-04  1.410267e-15 -4.532715e-04 -1.221421e-15 -4.209056e-04
 [56] -1.099942e-15 -3.918866e-04  4.130304e-16 -3.657683e-04  3.224729e-16
 [61] -3.421767e-04  6.063277e-16 -3.207961e-04  3.649858e-15 -3.013585e-04
 [66]  6.424265e-16 -2.836355e-04 -4.832003e-16 -2.674311e-04  5.797886e-16
 [71] -2.525768e-04 -5.092477e-16 -2.389265e-04 -8.901136e-16 -2.263537e-04
 [76]  7.404729e-16 -2.147478e-04 -2.734394e-15 -2.040121e-04 -1.581442e-15
 [81] -1.940618e-04  1.673104e-16 -1.848221e-04 -1.485363e-15 -1.762269e-04
 [86] -1.766975e-16 -1.682177e-04  6.138029e-15 -1.607423e-04  3.728317e-15
 [91] -1.537543e-04 -1.198368e-15 -1.472123e-04  3.667577e-15 -1.410792e-04
 [96] -1.599595e-16 -1.353215e-04  8.326869e-16 -1.299091e-04 -1.075646e-15
[101] -1.248152e-04 -1.766975e-16 -1.200150e-04  3.236049e-15 -1.154866e-04
[106] -9.950277e-16 -1.112097e-04  1.902259e-15 -1.071660e-04  2.916613e-15
[111] -1.033390e-04 -1.210378e-15 -9.971333e-05  1.605267e-15 -9.627520e-05
[116] -1.929868e-15 -9.301187e-05  1.146254e-15 -8.991170e-05 -8.484240e-16
[121] -8.696397e-05 -1.925942e-14 -8.415887e-05 -1.230478e-15 -8.148733e-05
[126] -2.999655e-15 -7.894101e-05  1.223734e-14 -7.651220e-05  9.985446e-15
[131] -7.419379e-05 -1.313245e-15 -7.197917e-05 -2.989396e-15 -6.986225e-05
[136] -1.529831e-15 -6.783737e-05  1.708750e-15 -6.589926e-05 -1.724733e-15
```

```
[141] -6.404303e-05  7.323103e-16 -6.226415e-05 -1.895943e-15 -6.055836e-05
[146] -4.825706e-15 -5.892172e-05 -1.195193e-15 -5.735055e-05  2.654364e-15
[151] -5.584139e-05  6.827287e-15 -5.439103e-05 -3.345755e-16 -5.299644e-05
[156] -2.541683e-15 -5.165482e-05  1.533313e-15 -5.036350e-05  2.319154e-15
[161] -4.912000e-05  2.271847e-15 -4.792200e-05 -1.821226e-15 -4.676729e-05
[166]  5.705820e-16 -4.565383e-05  5.006337e-15 -4.457966e-05  2.434554e-15
[171] -4.354295e-05  2.083926e-15 -4.254200e-05  8.853655e-15 -4.157517e-05
[176] -1.639974e-16 -4.064093e-05 -1.627258e-15 -3.973782e-05  1.400328e-15
[181] -3.886449e-05 -6.448485e-16 -3.801963e-05 -6.772828e-16 -3.720203e-05
[186] -4.413926e-15 -3.641052e-05 -4.508547e-16 -3.564401e-05 -3.562626e-15
[191] -3.490144e-05 -5.776791e-15 -3.418185e-05 -4.997452e-15 -3.348427e-05
[196] -6.355588e-16 -3.280784e-05  3.047847e-15 -3.215170e-05  2.810594e-16

$coefs$b.n
  [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[112] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[149] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
[186] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

$coefs$lower
[1] -3.141593

$coefs$upper
[1] 3.141593

$coefs$ell
[1] 3.141593

$coefs$cc
[1] 0


$S.n
[1] 0.003183072
```

2.3 Fourier: A Statistical Perspective

2.3.1 Cesàro Means and Expectation

Cesàro Means

In the previous examples, we saw ways in which the sequence of Fourier partial sums $S_n f$ of the Fourier series $\mathcal{S}f$ can behave quite poorly. Let us take a moment to try and diagnose the issue.

Again letting $f(x)$ be the square-wave on $[-\pi, \pi]$,

$$f(x) = \begin{cases} -1 & -\pi < x < 0, \\ 0 & x = 0, \\ 1 & 0 < x < \pi, \end{cases}$$

what we observed was that, as we near the jump point at $x = 0$ and enter a neighborhood, say, $N_\epsilon = (-\epsilon, \epsilon)$ about $x = 0$, the Fourier approximation starts to *undershoot* the left limit $f(0^-) = -1$ by "dipping down" in preparation for the jump discontinuity, converging pointwise to the midpoint of the limits $\frac{f(x^-)+f(x^+)}{2} = 0$ at $x = 0$, and then *overshooting* the right limit $f(0^+) = 1$ as a sort of "recoil" from how rapidly we forced the approximation to change.

What is so peculiar about this behavior — the **Gibbs effect** — is that it *never goes away.* In particular, given any $\epsilon > 0$, if $x_0 \in [-\ell, \ell]$ is a jump discontinuity of f with Fourier partial sums $S_N f$, we have that

$$\lim_{N\to\infty} \sup_{x\in N_\epsilon(x_0)} |S_N f(x) - f(x)| \approx 0.09 \times |f(x_0^+) - f(x_0^-)|.$$

That is, no matter how many terms N we include in the partial sum $S_N f$, the overshoot near the discontinuity x_0 persists at roughly 9% of the jump size, $|f(x_0^+) - f(x_0^-)|$. For the square-wave at $x_0 = 0$, the jump is

$$|f(0^+) - f(0^-)| = 2,$$

so that the partial sums $S_N f$ persistently overshoot the left- and right-limits $f(0^-) = -1$ and $f(0^+) = 1$ by about $0.09 \times 2 \times 100\% = 18\%$ no matter how many terms N we take in the Fourier approximation $S_N f$.

Remark.

The exact Gibbs constant is

$$\mathcal{G} := \frac{1}{\pi}\int_0^\pi \frac{\sin t}{t}\, dt - \frac{1}{2} \approx 0.0895;$$

we use 0.09 as a reasonable approximation, but in reality, the overshoot beyond the limit at a jump of height h converges to h times this constant $\mathcal{G}$.

We might find all of this rather confusing. To be clear, we're *not* saying that we don't observe pointwise convergence at $x = \pm 1$; indeed, this is ensured by the Dirichlet-Jordan theorem since the square-wave f is of bounded variation. Rather, the issue is that the convergence is *not uniform* near the jump — the overshoot doesn't *shrink* as N grows larger. But for any fixed $N \in \mathbb{N}$, it's always possible to choose $\epsilon > 0$ such that points inside the neighborhood still exhibit the overshoot; this is the problem, since it means we need to *choose the number N only after characterizing the point $x_0 \in [-\ell, \ell]$ we seek to analyze.* Compare this with the more convenient setting where, given $\epsilon > 0$, we're able to choose N based *purely off* ϵ such that convergence holds *for any point simultaneously.*

This, again, characterizes the difference between *pointwise* and *uniform* convergence. We've

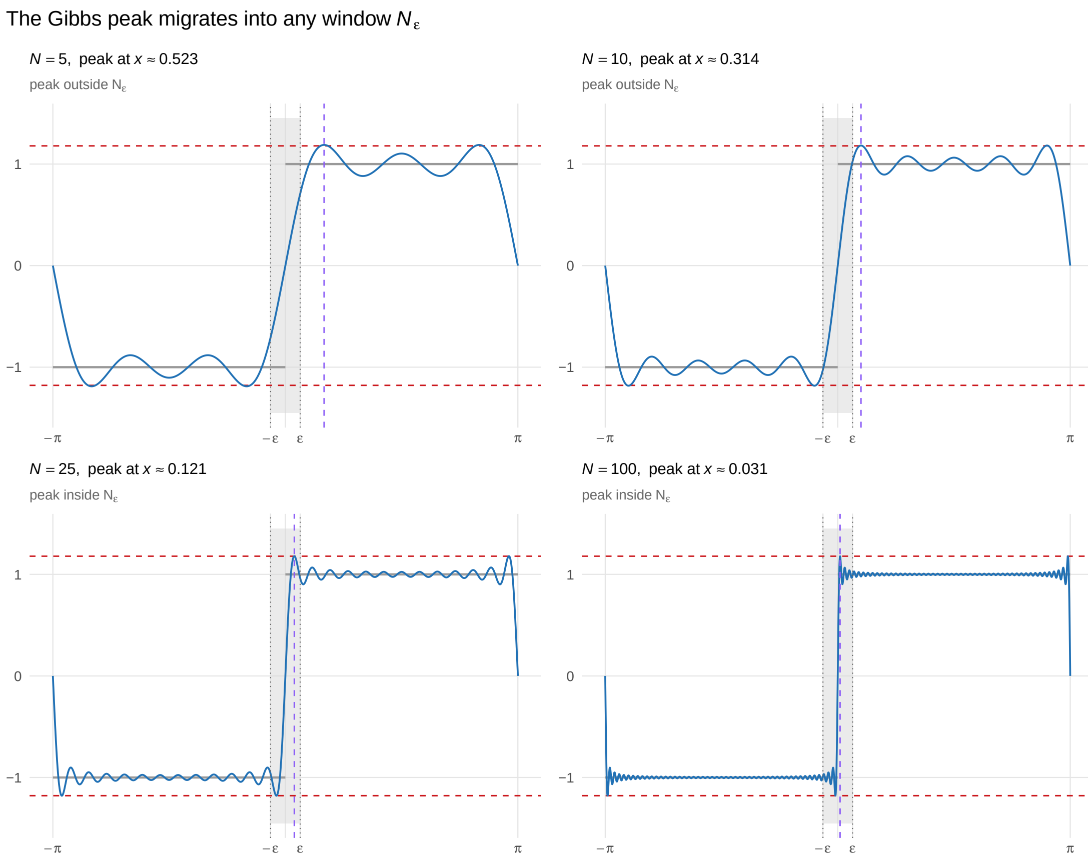

Illustration of the Gibbs Effect for the square-wave function.

discussed this previously, but now, we want to dig a bit deeper; namely, into whether or not an analogous approximating mechanism *does* display uniformity in its convergence, and whether we could do so without overhauling the existing Fourier machinery.

At this point, we're ready to reveal that, indeed, we *can do better* than the mere pointwise convergence currently in our possession by a rather simple modification to our current setup.

Definition 2.3.1 (Cesàro Mean)

The N**th Cesàro mean** of the Fourier series $\mathcal{S}f$ is the average of its first N Fourier partial sums:

$$\sigma_N f(x) := \frac{1}{N+1} \sum_{n=0}^{N} S_n f(x).$$

What we observe here is astonishingly simple in its conceit: we know that the sequence of Fourier partial sums $(S_N f)_{N\in\mathbb{N}}$ converges pointwise, but with the rather annoying feature of "letting sinusoidal noise accumulate" near jump discontinuities. What we intuitively imagine happening is that each new Fourier coefficient we add contributes a non-negligible amount of "compensatory

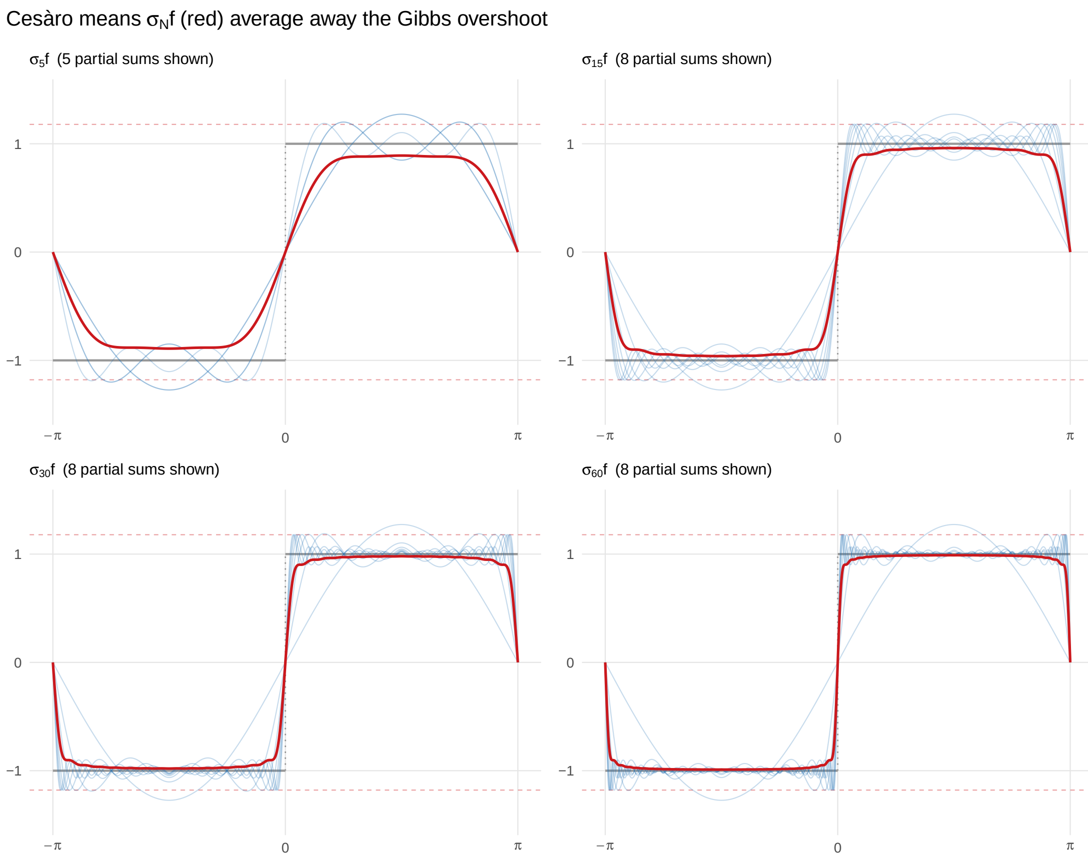

Illustration of Cesàro means for the square-wave function.

wiggliness" around jump discontinuities in order to facilitate the rapid change in function evaluation. We imagine that the coefficients further out in the tail of the sequence characterize sinusoids growing more and more peaked within a shrinking window, and that the "rate at which the window shrinks" fails to overcome the "steepness of the peaks."

But what we also often observe is something to the effect of "the Nth partial sum overshoots, the $(N+1)$th undershoots," and so on. Indeed, the noise is *oscillatory* in nature — what we might imagine is that the overshoots are analogous to *mean zero noise*. In statistics, we often use the Law of Large Numbers as justification for the fact that

$$\frac{1}{N}\sum_{n=1}^{N} X_n \to \mathbb{E}[X] \quad \text{as} \quad N \to \infty;$$

and, indeed, this is essentially how the Cesàro mean operates.

To make this analogy precise, we need to understand what σ_N looks like as an *integral operator*. In particular, recall (cf. the Fourier Partial Sum Integral Representation) that each Fourier partial sum $S_n f(x)$ admits a convolution representation against the Dirichlet kernel:

$$S_N f(x) = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t) D_N\left(\frac{\pi(x-t)}{\ell}\right)\, dt,$$

where D_n is the Dirichlet kernel

$$D_n(u) = \frac{1}{2} + \sum_{k=1}^{n} \cos(ku).$$

Substitute this into the definition of the Nth Cesàro mean $\sigma_N f$:

$$\sigma_N f(x) = \frac{1}{\ell(N+1)} \sum_{n=0}^{N} \left[\int_{-\ell}^{\ell} f(t) D_n \left(\frac{\pi(x-t)}{\ell} \right) dt \right] = \frac{1}{\ell} \int_{-\ell}^{\ell} f(t) \underbrace{\left[\frac{1}{N+1} \sum_{n=0}^{N} D_n \left(\frac{\pi(x-t)}{\ell} \right) \right]}_{:= F_N\left(\frac{\pi(x-t)}{\ell}\right)} dt.$$

The bracketed object F_N is a new kernel — the arithmetic mean of the first $N+1$ Dirichlet kernels $D_0, D_1, \dots, D_N$. This object is significant enough to earn a definition.

Definition 2.3.2 (Fejér Kernel)

The N**th Fejér kernel** is

$$F_N(u) := \frac{1}{N+1} \sum_{n=0}^{N} D_n(u),$$

where D_n is the nth Dirichlet kernel, $n = 0, \dots, N$.

With this definition, what we have is that the Nth Cesàro mean $\sigma_N f(x)$ admits an integral representation as f integrated against the Nth Fejér kernel F_N:

$$\sigma_N f(x) = \frac{1}{\ell} \int_{-\ell}^{\ell} f(t) F_N \left(\frac{\pi(x-t)}{\ell} \right) dt.$$

We will return to this representation shortly.

Below, we plot a few of the Fejér kernels centered at $u = 0$, overlaid with a sample of their comprising Dirichlet kernels; in particular, take note of what happens when we average these *signed* contributions D_N into the *average* F_N:

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> If we squint at the blue Dirichlet kernels (particularly in the bottom set of panels), they seem to be tracing out a symmetric funnel centered about the x-axis — in particular, what we observe is that the positive and negative fluctuations *perfectly interfere* with one another, so that what is left after averaging is precisely the nonnegative envelope given by the Fejér curve in red. This is curious — there really is no reason to believe a priori that the signed Dirichlet kernels *should* average to a nonnegative value. Perhaps we'd conjecture this nonnegativity to emerge *asymptotically*, but we're observing something stronger: for every N, $F_N(u) = \frac{1}{N+1} \sum_{n=0}^{N} D_N(u)$ is nonnegative.

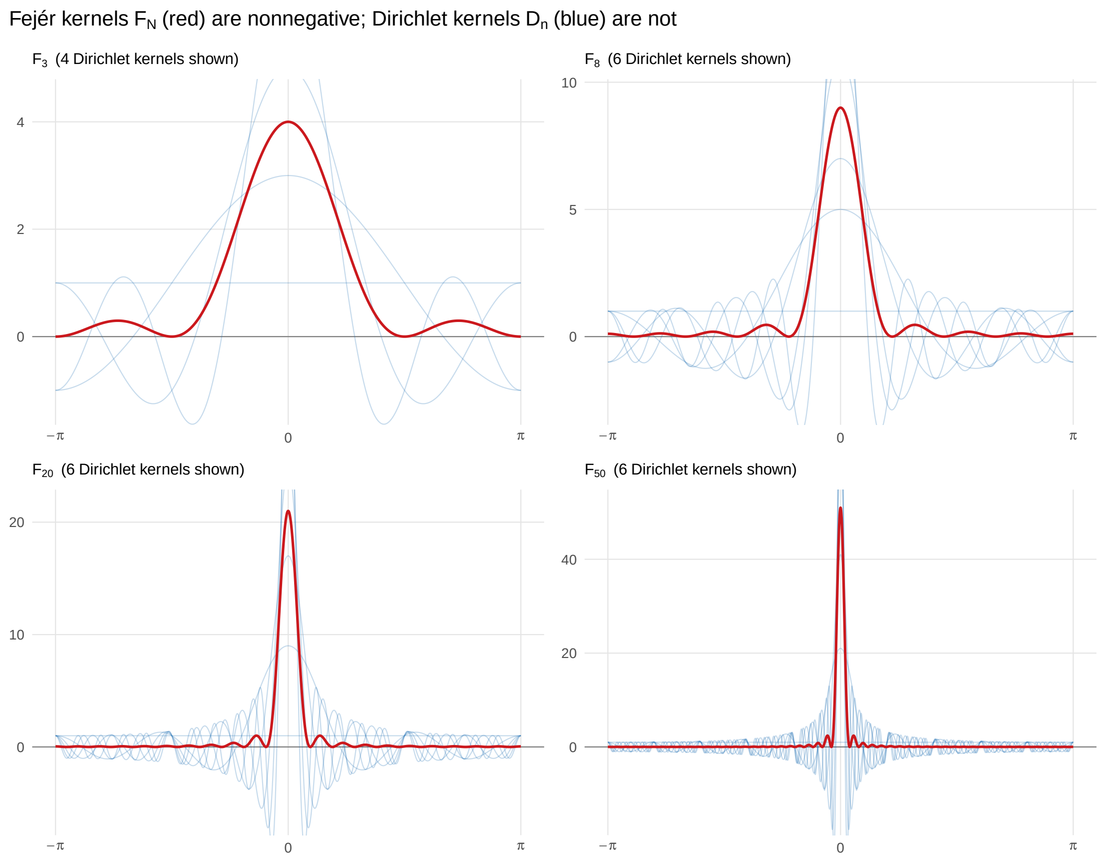

Illustration of the signed Dirichlet Kernels averaging to form the nonnegative Fejér Kernel.

A quick derivation can confirm these heuristic observations: recall we derived the closed form expression for D_N given by

$$D_N(u) = \begin{cases} \frac{\sin\left(\left(N+\frac{1}{2}\right)u\right)}{2\sin\left(\frac{u}{2}\right)} & u \neq 0, \\ N + \frac{1}{2} & u = 0. \end{cases}$$

from this, by applying the identity

$$\sum_{n=0}^{N} \sin((2n+1)\theta) = \frac{\sin^2((N+1)\theta)}{\sin(\theta)}$$

at $\theta = u/2$, we can confirm that a closed-form for $F_N(u)$ is

$$F_N(u) = \begin{cases} \dfrac{1}{2(N+1)} \left(\dfrac{\sin\left(\frac{(N+1)u}{2}\right)}{\sin\left(\frac{u}{2}\right)} \right)^2 & u \neq 0, \\ \dfrac{N+1}{2} & u = 0. \end{cases}$$

What this reveals, quite interestingly, is that the Fejér kernel *really is* strictly nonnegative: $F_N(u) \geq 0$ for any $u \in \mathbb{R}$ and $N \in \mathbb{N}$. Though this is certainly obvious on computing the closed-form, it really isn't so from the definition as an average of Dirichlet kernels — at least, not to us.

The derivation reveals that our observation from the plot is true — it really is that the Fejér kernels are nonnegative for any N. Nevertheless, what the plot also illustrates is a *concentration* effect: as N grows, the Dirichlet kernels average in such a way that the central peak at $u = 0$ grows taller. (We can verify the maximum stays at $u = 0$ by noting that F_N is even and continuous, so its derivative must be odd and continuous; this means that the plot of the derivative $F_N'(u)$ crosses the origin, $F_N'(0) = 0$, confirming that $u = 0$ is an extrema, and checking the second-derivative will reveal $F_N''(0) < 0$, so that $u = 0$ is a local maximum.)

Furthermore, we can observe that $F_N(0) = \frac{N+1}{2}$ for every N, and we can verify via the second-derivative test that this is a maximum over $u \in \mathbb{R}$ for every N. What this means is that the peak of F_N stays at $u = 0$ and grows linearly in N. In the limit as $N \to \infty$, we observe that $F_N(0) \to \infty$ while $F_N(u) \to 0$ for every other $u \neq 0$ as $N \to \infty$. In particular, the Fejér kernels converge to a Dirac delta mass at $u = 0$.

And notice what happens when we *integrate* $F_N(u)$ over $u \in [-\pi, \pi]$: using the definition $D_N(u) = \frac{1}{2} + \sum_{k=1}^{n} \cos(ku)$ directly, we have that

$$\int_{-\pi}^{\pi} F_N(u)\,du = \frac{1}{N+1} \int_{-\pi}^{\pi} \sum_{n=0}^{N} D_n(u)\,du = \frac{1}{N+1} \left[(N+1) \int_{-\pi}^{\pi} \frac{1}{2}\,du + \sum_{n=0}^{N} \sum_{k=1}^{n} \int_{-\pi}^{\pi} \cos(ku)\,du \right].$$

But each integral $\int_{-\pi}^{\pi} \cos(ku)\,du$ integrates a 2π-periodic cosine over a full period — in particular, each of these integrals vanishes, so the sum vanishes. This leaves only the constant

$$\int_{-\pi}^{\pi} F_N(u)\,du = \frac{1}{N+1} \left[(N+1) \int_{-\pi}^{\pi} \frac{1}{2}\,du \right] = \frac{(N+1)\pi}{(N+1)} = \pi.$$

Rearranging this, we observe that for every $N \in \mathbb{N}$,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} F_N(u)\,du = 1.$$

Together, what we've just discovered is that the Fejér kernels F_N are nonnegative and integrate to 1 on $[-\pi, \pi]$:

$$F_N(u) \geq 0 \quad \forall u \in \mathbb{R}, \quad \text{and} \quad \frac{1}{\pi} \int_{-\pi}^{\pi} F_N(u)\,du = 1.$$

We take this opportunity to present a couple of familiar definitions.

Definition 2.3.3 (Probability Density Function)

A function $p\colon [a, b] \to \mathbb{R}$ is a **probability density function** (PDF/pdf) on $[a, b]$ if it is both:

(i) **(Nonnegative)** We have that

$$p(x) \geq 0, \quad \forall x \in [a, b].$$

(ii) **(Normalized)** The function p integrates to one on $[a, b]$:

$$\int_a^b p(x)\,dx = 1.$$

What we observe, then, is that by defining

$$p_N(u) := \frac{1}{\pi} F_N(u), \quad \text{where} \quad F_N(u) = \frac{1}{N+1} \sum_{n=0}^{N} D_n(u),$$

the functions $(p_N)_{N \in \mathbb{N}}$ each satisfy the conditions we outlined above — in particular, each p_N is a valid probability density function on $[-\pi, \pi]$.

We can go further. Given a probability density function p on $[a, b]$, we can define the associated *cumulative distribution function*:

Definition 2.3.4 (Cumulative Distribution Function)

Let $p \colon [a, b] \to \mathbb{R}$ be a probability density function on $[a, b]$. The **cumulative distribution function** (CDF/cdf) associated to p is

$$P(t) := \int_a^t p(u)\, du, \qquad u \in [a, b].$$

As one can verify using the fundamental theorem of calculus and the properties of the pdf p, we note the following properties satisfied by any valid cdf:

(i) Right-continuous for any $u \in [a, b]$ (here, P is actually *fully* continuous),

(ii) Nondecreasing, meaning $u < v \implies P(u) \leq P(v)$ for any $u, v \in [a, b]$, and

(iii) Normalized at the endpoints, in the sense that at the endpoints, we have

$$P(a) = \int_a^a p(t)\, dt = 0, \qquad \text{and} \qquad P(b) = \int_a^b p(t)\, dt = 1.$$

In our setting, the CDF associated to the Nth Fejér kernel p_N is

$$P_N(t) := \frac{1}{\pi} \int_{-\pi}^{t} F_N(u)\, du, \qquad t \in [-\pi, \pi].$$

(Verify that the functions P_N satisfy the three properties of a cumulative distribution!)

But we're still not done — recall that we just found an integral representation for σ_N given by

$$\sigma_N f(x) = \frac{1}{\ell} \int_{-\ell}^{\ell} f(u) F_N \left(\frac{\pi(x-u)}{\ell} \right) du.$$

Setting $\ell = \pi$, this says that

$$\sigma_N f(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(u) F_N(x-u)\, du = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x-t) F_N(t)\, dt,$$

where we've substituted $t = x - u$ in the second equality.

But, if we pull the $1/\pi$ inside the integral, then observe that

$$\frac{1}{\pi} F_N(t) = p_N(t),$$

where p_N is a probability density function. Whence we have

$$\sigma_N f(x) = \int_{-\pi}^{\pi} f(x-t)p_N(t)\,dt.$$

But now, notice that since

$$P_N(t) := \frac{1}{\pi}\int_{-\pi}^{t} F_N(u)\,du, \quad t \in [-\pi, \pi],$$

we have that

$$\frac{d}{dt}P_N(t) = \frac{1}{\pi}\cdot\frac{d}{dt}\int_{-\pi}^{t} F_N(u)\,du = \frac{1}{\pi}F_N(t) = p_N(t),$$

or, equivalently, that

$$dP_N(t) = p_N(t)\,dt.$$

At this point, we write the Nth Cesàro mean as

$$\sigma_N f(x) = \int_{-\pi}^{\pi} f(x-t)\,dP_N(t),$$

which is structurally similar to the Riemann-Stieltjes integral $\int_a^b g\,d\alpha$ for integrators α that are of *bounded variation* on $[a,b]$.

Indeed, this isn't just a passing resemblance. Realize that, by construction, the cdfs P_N as we've constructed them are each continuous, hence they are automatically of bounded variation. Thus the notation $\int_{-\pi}^{\pi} f(x-t)\,dP_N(t)$ tells us that the Cesàro means σ_N are just operators mapping functions $f \in L^1[-\pi,\pi]$ to their convolutions by the Fejér CDFs P_N on $[-\pi,\pi]$.

But again, P_N isn't *just* of bounded variation — it is also *nondecreasing* and *normalized*. We might wonder if these properties endow the Riemann-Stieltjes integral $\int f\,dP_N$ with any additional interpretation beyond merely "computing the area under a curve." To answer this, let us retrace our steps to the Riemann-Stieltjes integral: recall that if $\alpha\colon [a,b] \to \mathbb{R}$ is of bounded variation — which, recall, means that if $\mathcal{P} = \{a = t_0, t_1, \dots, t_k = b\}$ denotes a partition of $[a,b]$, then we have that the total variation

$$V_a^b(\alpha) = \sup_{\mathcal{P}} \sum_{\mathcal{P}} |\Delta\alpha_i| < \infty,$$

Whenever α is of bounded variation and $g\colon [a,b] \to \mathbb{R}$ is continuous, we *define* the Riemann-Stieltjes integral $\int_a^b g\,d\alpha$ as the *common* limit of Riemann-Stieltjes sums taken over finer and finer partitions $\|\mathcal{P}\| := \sup|\Delta t_i| \downarrow 0$:

$$\int_a^b g(t)\,d\alpha(t) := \lim_{\|\mathcal{P}\|\downarrow 0} \sum_{\mathcal{P}} g(c_i)\Delta\alpha_i, \qquad \Delta\alpha_i := \alpha(t_i) - \alpha(t_{i-1}).$$

Again, nothing changes when we use $\alpha = P_N$, so on $[-\pi,\pi]$, we have

$$\sigma_N f(x) := \int_{-\pi}^{\pi} f(x-t)\,dP_N(t) = \lim_{\|\mathcal{P}\|\downarrow 0} \sum_{\mathcal{P}} f(x-c_i)[P_N(t_i) - P_N(t_{i-1})], \qquad c_i \in [t_{i-1}, t_i].$$

Here is where we can invoke the CDF properties of P_N. In particular, for the time being, let us work with the *constant function*

$$\mathbf{1}(t) = 1, \quad \forall t \in [-\pi,\pi],$$

so that for any $x \in [-\pi, \pi]$, the integrand reduces to $\mathbf{1}(x-t)\,dP_N(t) = dP_N(t)$, yielding

$$\sigma_N \mathbf{1}(x) = \int_{-\pi}^{\pi} dP_N.$$

We can directly solve this integral just by plugging in the endpoints and using the endpoint property of CDFs: since $P_N(-\pi) = 0$ and $P_N(\pi) = 1$, we find that

$$\sigma_N \mathbf{1}(x) = P_N(\pi) - P_N(-\pi) = 1 - 0 = 1, \qquad \forall x \in [-\pi, \pi].$$

What we've just verified is that the Cesàro means *preserve constants* — that is, if f is identically equal to a constant c on $[-\pi, \pi]$, then $\sigma_N f(x) = c$ for every x and every N, or

$$f(t) = c \quad \forall t \in [-\pi, \pi] \implies \sigma_N f(x) = \int_{-\pi}^{\pi} c\,dP_N(t) = c \quad \forall x \in [-\pi, \pi] \quad \text{and} \quad N \in \mathbb{N}. \tag{i}$$

This is the first property we record.

For the second property, let us take $g\colon [-\pi, \pi] \to \mathbb{R}$ is strictly nonnegative: $g(t) \geq 0$ for every $t \in [-\pi, \pi]$. Then, examining the Riemann-Stieltjes sum

$$\sum_{\mathcal{P}} g(c_i)\Delta P_N(t_i),$$

we observe that (a) $g(c_i) \geq 0$ by our assumption that g was nonnegative, and (b) each increment $\Delta P_N(t_i) = P_N(t_i) - P_N(t_{i-1}) \geq 0$, following *directly* from the nondecreasingness of the CDF P_N. In particular, since the partition $\mathcal{P}$ has been arbitrary, what we have is that *for any* partition $\mathcal{P} = \{a = t_0, t_1, \dots, t_k = b\}$ of $[a, b]$ and any sample points $c_i \in [t_i, t_{i-1}]$, the product

$$g(c_i)\Delta P_N(t_i) \geq 0 \quad \forall i;$$

so that, in particular, we have

$$\int_a^b g(t)\,dP_N(t) = \sup_{\mathcal{P}} \sum_{\mathcal{P}} \underbrace{g(c_i)\Delta P_N(t_i)}_{\geq 0\,\forall i} \geq 0.$$

What we've just verified is that taking the integrator $\alpha = P_N$ a CDF, the resulting Riemann-Stieltjes integral *preserves nonnegativity*:

$$g \geq 0 \text{ on } [-\pi, \pi] \implies \sigma_N g(x) := \int_{-\pi}^{\pi} g(x-t)\,dP_N(t) \geq 0. \tag{ii}$$

Now, from this preservation of constants (i) and nonnegativity (ii) alone, we can derive a striking consequence. Suppose that $g\colon [-\pi, \pi] \to \mathbb{R}$ is *bounded*, with

$$m \leq g(t) \leq M, \quad \forall t \in [-\pi, \pi].$$

Then

$$g(t) - m \geq 0 \quad \forall t \in [-\pi, \pi],$$

so that, by (ii),

$$\int_{-\pi}^{\pi} [g(t) - m]\,dP_N(t) \geq 0.$$

By linearity of the Riemann-Stieltjes integral and (i) with $c = m$,

$$\int_{-\pi}^{\pi} g(t)\, dP_N(t) - \int_{-\pi}^{\pi} m\, dP_N(t) \geq 0 \implies \int_{-\pi}^{\pi} g(t)\, dP_N(t) \geq m. \tag{iii}$$

The same argument applied to the other side of the inequality

$$M - g(t) \geq 0 \quad \forall t \in [-\pi, \pi]$$

gives us

$$\int_{-\pi}^{\pi} g(t)\, dP_N(t) \leq M.$$

Together, (iii) and (iv) say that

$$m \leq g(t) \leq M \quad \forall t \in [-\pi, \pi] \implies m \leq \int_{-\pi}^{\pi} g(t)\, dP_N(t) \leq M. \tag{v}$$

This is absolutely central to our probabilistic interpretation of integration in general. Let us reiterate: what (v) says is that integration by a continuous CDF P_N *preserves boundedness*. If we imagine $t \mapsto g(t)$ with $m \leq g(t) \leq M$ as mapping individuals in a class t to their test scores $g(t)$, what property (v) forces is that integration of g by a CDF P_N acts like a *weighted average*: we'll observe $m \leq \int g\, dP_N \leq M$, so that the "$P_N$-weighted average of the class's test scores $g(t)$ ($\int g\, dP_N$)" is trapped between "the lowest (m) and highest (M) scores in the class."

If we henceforth agree to "imagine the Riemann-Stieltjes integral $\int_a^b f\, dP_N$ as $\sum_{\mathcal{P}} f(c_i)\, \Delta P_N(t_i)$, a ΔP_N-weighted average of f at points c_i interspersed throughout $[a, b]$," we can make everything much more intuitively appealing. Property (i) forces the weights to sum to 1:

$$\sum_{\mathcal{P}} \Delta P_N(t_i) = P_N(\pi) - P_N(-\pi) = 1,$$

and property (ii) forces the weights to be nonnegative:

$$\Delta P_N(t_i) = P_N(t_i) - P_N(t_{i-1}) \geq 0, \quad \forall i.$$

In this way, the sum

$$\sum_{\mathcal{P}} f(c_i)\Delta P_N(t_i) = f(c_1)\Delta P_N(t_1) + f(c_2)\Delta P_N(t_2) + \cdots + f(c_k)\Delta P_N(t_k)$$

is nothing more than a **convex combination** of the points $f(c_1), \ldots f(c_k)$:

Definition 2.3.5 (Convex Combination)

If $\{c_1, \ldots, c_n\}$ are points in a vector space over scalar field F, then if $w_1, \ldots, w_n \in F$ are scalars such that

$$w_i \geq 0 \quad \forall i, \quad \text{and} \quad \sum_{i=1}^{n} w_i = 1,$$

we call

$$(c_1, \ldots, c_n) \mapsto \sum_{i=1}^{n} w_i c_i = w_1 c_1 + \cdots + w_n c_n$$

a **convex combination** of the points $\{c_1, \dots, c_n\}$.

This is where the story comes full circle (trigonometry pun intended). When we define a probability distribution on a finite set $\{x_1, \dots, x_k\}$, what we're doing is assigning to each x_i a number,

$$p_i := \mathbb{P}(x_i),$$

such that

$$p_i \geq 0 \quad \forall i, \quad \text{and} \quad \sum_{i=1}^{k} p_i = 1.$$

It follows that the sum

$$\sum_{i=1}^{k} p_i x_i = p_1 x_1 + \cdots + p_k x_k$$

is a convex combination. But, even more commonly, we identify this convex combination with a familiar statistical operator:

Definition 2.3.6 (Expected Value)

Suppose $X\colon \Omega \to \{x_1, \dots, x_n\}$ is a discrete random variable mapping **events** ω drawn from a **sample space** Ω to values $X(\omega)$ in a finite **support set** $\mathcal{X} = \{x_1, \dots, x_n\}$. If the probability distribution of X is described by

$$p_i := \mathbb{P}(X = x_i), \quad i = 1, \dots, k,$$

then the **expected value** of X is

$$\mathbb{E}[X] = \sum_{i=1}^{k} p_i x_i.$$

Similarly, if $g\colon \{c_1, \dots, c_k\} \to \mathbb{R}$ is a function of $X \mapsto g(X)$, then the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_{i=1}^{k} p_i g(x_i).$$

Again, take a moment to compare the convex combination

$$\sum_{i=1}^{k} w_i c_i; \quad w_i \geq 0, \quad \sum_{i=1}^{k} w_i = 1;$$

with the expected value

$$\sum_{i=1}^{k} p_i x_i; \quad p_i \geq 0, \quad \sum_{i=1}^{k} p_i = 1.$$

They're the same thing! They are not "analogous" or "similar" — they're *identical*. What this is to say is that an expectation *is* a convex combination, and a convex combination *is* an expectation. The *only difference* is the name we give the weights: in *convex geometry*, we call the w_i **convex coefficients**, and in *statistics* we call the p_i **probabilities**.

But this identification is not merely a notational curiosity — what it means is that *every theorem* about convex combinations is *automatically* a theorem about expected values, and vice versa. For instance, we showed in (v) that

$$m \leq g(t) \leq M, \quad t \in [-\pi, \pi] \implies m \leq \int_{-\pi}^{\pi} g(t)\, dP_N(t) \leq M.$$

In the language of convex geometry, this is the trivial fact that convex combinations cannot leave the **convex hull** of their inputs; in the language of statistics, this is the fact that the expected value of a **bounded random variable** lies between its minimum and maximum:

$$m \leq X(\omega) \leq M, \quad \omega \in \Omega \implies m \leq \mathbb{E}[X] \leq M,$$

and

$$m \leq g(X(\omega)) \leq M, \quad \omega \in \Omega \implies m \leq \mathbb{E}[g(X)] \leq M.$$

Again, these are *not* "new theorems"; they are all describing exactly the same phenomenon.

So, now, let us return once again to the Cesàro means. Recalling that

$$\sigma_N f(x) := \int_{-\pi}^{\pi} f(x-t)\, dP_N(t) = \lim_{\|\mathcal{P}\|\downarrow 0} \sum_{\mathcal{P}} f(x-c_i)\Delta P_N(t_i), \qquad c_i \in [t_{i-1}, t_i].$$

let us zoom in on the sum

$$\sum_{\mathcal{P}} f(x-c_i)\Delta P_N(t_i); \quad \Delta P_N(t_i) \geq 0, \quad \sum_{\mathcal{P}} \Delta P_N(t_i) = 1.$$

This is a convex combination — therefore an expected value — of the values $f(x-c_1), \ldots, f(x-c_k)$ with probabilities $\Delta P_N(t_1), \ldots, \Delta P_N(t_k)$ respectively.

But what exactly *are* "these probabilities" $\Delta P_N(t_i)$ we've been speaking of — what are they probabilities *of*? We've been saying that "$\Delta P_N(t_i)$ is the probability assigned to the ith bin $[t_{i-1}, t_i]$," but we haven't said *what's landing* in those bins. In the finite case, we had no trouble answering this question — for example, in the test score analogy, we map the jth student in the class to their test score $c_j \in \mathbb{R}$, and we say that $\Delta P_N(t_i)$ is the fraction of test scores c_j falling into the ith bin $[t_{i-1}, t_i]$.

But in our setting, there is no finite class of students — the domain $[-\pi, \pi]$ is a continuum, and the CDF P_N assigns probability to *every subinterval*, not just to finitely many points. What we recognize is that, in the finite case, a random variable X taking values in the support set $\mathcal{X} = \{x_1, \ldots, x_k\}$ was fully specified by listing its probabilities

$$p_i := \mathbb{P}(X = x_i), \quad i = 1, \ldots, k.$$

In the continuous case, the probability that a quantity takes on any *exact* value is typically zero — if we "draw a number at random from $[-\pi, \pi]$ according to a continuous P_N," then the probability we land on say, $t = 0.7$ exactly, is zero. Intuitively, we imagine that it isn't exact events which are meaningful to a continuous probability distribution, but *intervals* of values: in particular, if X is distributed according to P_N, then to any closed interval $[s, t] \subset [-\pi, \pi]$ we assign the probability

$$\mathbb{P}(X \in [s,t]) := P_N(t) - P_N(s).$$

The intuitive idea here is that, since $P_N(-\pi) = 0$, $P_N(\pi) = 1$, and since P_N is monotone nondecreasing and continuous, we observe the same telescoping phenomenon as before:

$$\mathbb{P}(X \in [s,t]) = P_N(t) - P_N(s) = [P_N(t) - P_N(a)] - [P_N(s) - P_N(a)] = \int_a^t dP_N - \int_a^s dP_N = \int_s^t dP_N.$$

This is the conceptual shift taking us from discrete to continuous probability: we stop asking "$\mathbb{P}(X = t)$" and start asking "$\mathbb{P}(X \in [s,t])$". Probabilities become Riemann-Stieltjes integrals, and whenever P_N is continuous, we associate to the CDF P_N a continuous density $p_N = P_N'$, and we have that

$$\mathbb{P}(X \in [s,t]) = \int_s^t dP_N(u) = \int_s^t p_N(u)\,du.$$

This is likely a familiar formula; it says that to find the probability that a continuous random variable X falls into $[s,t]$, we integrate its PDF p_N over $[s,t]$. But what we're careful to remind you of is that, again, the Riemann-Stieltjes notation $\int_s^t dP_N$ is *not* "just a shorthand" for $dP_N(u) = p_N\,du$ — if P_N fails to be continuously differentiable, then p_N may not exist, but it could be that $\int_s^t dP_N$ stays meaningful.

We summarize these findings in the following definition.

Definition 2.3.7 (Continuous Random Variable)

Let $P\colon [a,b] \to [0,1]$ be a cumulative distribution function on $[a,b]$ with a continuous density $P_N' = p$. A **continuous random variable** U with distribution function P (equivalently, with density p) is a quantity taking values in the continuous support set $\mathcal{U} = [a,b]$, whose probability of falling into any subinterval $[s,t] \subseteq [a,b]$ is given by

$$\mathbb{P}(U \in [s,t]) = \int_s^t dP = \int_s^t p(u)\,du = P(t) - P(s).$$

The **expected value** of a function $U \mapsto g(U)$ is then the Riemann-Stieltjes integral

$$\mathbb{E}[g(U)] = \int_a^b g(t)\,dP(t) = \int_a^b g(t)p(t)\,dt.$$

Finally, note that as shorthand, we write $U \sim P$ or $U \sim p$ as shorthand for "U is a random variable distributed according to P/p."

Finally, the language catches up with the mathematics. Henceforth, let us define a continuous random variable U_N distributed according to our Fejér CDF P_N — equivalently, distributed according to the Fejér density $p_N = F_N/\pi$ where F_N is the Nth Fejér kernel. Then, if f is a function on $[-\pi,\pi]$, the Nth Cesàro mean is

$$\sigma_N f(x) = \int_{-\pi}^{\pi} f(x-t)\,dP_N(t) = \int_{-\pi}^{\pi} f(x-t)p_N(t)\,dt = \mathbb{E}[f(x-U_N)].$$

The notation $\mathbb{E}[f(x-U_N)]$ can feel like a sleight of hand — it may seem like we've replaced a dummy variable of integration t in $\int f(x-t)\,dP_N(t)$ with a random variable U_N in $\mathbb{E}[f(x-U_N)]$. But the two objects t and U_N are playing very different roles. The dummy variable t is a label for the axis we integrate over — it has no identity of its own, just as the t in $\int_0^1 t^2\,dt$ is nothing more than a name for "the thing running from 0 to 1." The random variable U_N, on the other hand, is a name for *the entire ensemble* of possible values that could be drawn from P_N, weighted by how likely each one is. When we write $f(x-U_N)$, we do not mean "evaluate f at the number $x-U_N$" — we mean "consider the entire collection of values $\{f(x-t) : t \in [-\pi,\pi]\}$, one for each outcome that U_N could take, and understand that $\mathbb{E}[\cdot]$ will average them with weights $p_N(t)\,dt$."

Thus, while it may seem as if we've replaced the perfectly concrete CDF P_N in the integral definition with a mysterious quantity U_N that "takes values" according to P_N, what we've really done is introduced a *linguistic device* to aid in making the Riemann-Stieltjes integral's intuitive mechanism perfectly transparent. Compare the two notations:

$$\int_{-\pi}^{\pi} f(x-t)\, dP_N(t) \qquad \text{versus} \qquad \mathbb{E}[f(x-U_N)].$$

The left side is a Riemann-Stieltjes integral — technically precise, but intuitively opaque about what it's *really doing*. On the other hand, the expectation on the right *describes a process*: first, draw $U_N \sim P_N$, then shift $U_N \mapsto x - U_N$, then apply $x - U_N \mapsto f(x-U_N)$, and average the results $f(x-U_N) \mapsto \mathbb{E}[f(x-U_N)]$. We're *reorganizing* by "moving the weights from the differential $dP_N(t)$, describing how likely we are to draw $t \in [-\pi, \pi]$ under P_N" over, instead understanding that "the weights described by P_N are passed to U_N when we assert that $U_N \sim P_N$; we draw such a realization, label it $t \in [-\pi, \pi]$, and average, understanding it is the *sampling process* which provides the expectation's interpretation as a *weighted* average."

With this understanding in hand, take another look at the formula

$$\sigma_N f(x) = \mathbb{E}[f(x-U_N)], \qquad U_N \sim p_N.$$

What this says is that, to compute the Nth Cesàro mean $\sigma_N f(x)$, we "start by drawing U_N according to the Nth Fejér density $p_N = \frac{F_N}{\pi}$; we perturb U_N by a shift $x - U_N$; evaluate f at the perturbed shift $x - U_N$; and average over the support $\mathcal{U} = [-\pi, \pi]$." What we can understand, then, is that the Cesàro means characterize *randomly smoothed* versions of f — it asks not "what is $f(x)$", but rather "what is f *near* x, on average?"

What we might wonder, then, is why we bothered with all of this statistical terminology. Well, recall earlier we said that, "essentially," the Cesàro means operated analogously to the Laws of Large Numbers:

$$\frac{1}{N}\sum_{n=1}^{N} X_n \to \mathbb{E}[X] \quad \text{as} \quad N \to \infty.$$

But now, we can be a bit more precise about this comparison, because it isn't quite right for our current setting. In particular, the LLN describes what happens when you average many independent draws from the *same* distribution — it says that if $X_1, \dots, X_N \sim p$ are independent samples from a continuous density, then the average of the samples will converge to the population mean as N grows larger and larger. But here, we are *not* "averaging many draws from a fixed distribution" — we are taking a *single* expectation, $\mathbb{E}[f(x-U_N)]$, under a distribution $U_N \sim p_N$ which *itself* is changing with N. That is, the sequence $U_0, U_1, U_2, \dots$ are each drawn from *different distributions* $p_0, p_1, p_2, \dots$ — so the LLN, at least in its usual form, does not directly apply.

What *is* happening is something arguably simpler. We have a single expectation $\mathbb{E}[f(x-U_N)]$ under a distribution $U_N \sim P_N$ that is *concentrating near zero* — we observed this in the plots of the Fejér kernels F_N, and p_N is just the scaled F_N/π. The density p_N is piling more and more of its mass near the origin as N grows large. In this case, the appropriate tool to describe the phenomenon is not a law of large numbers, but rather the following.

Definition 2.3.8 (Convergence in Probability)

A sequence of random variables $U_1, U_2, \dots$ **converges in probability** to a constant c if, for every $\delta > 0$,

$$\lim_{N\to\infty} \mathbb{P}(|U_N - c| > \delta) = 0;$$

in such a case, we write

$$U_N \xrightarrow{\mathbb{P}} c.$$

This is exactly the language for what we've been observing in the Fejér kernel plots. The density $p_N = F_N/\pi$ concentrates at the origin: for any $\delta > 0$,

$$\mathbb{P}(|U_N| > \delta) = \int_{-\pi}^{-\delta} p_N(t)\,dt + \int_{\delta}^{\pi} p_N(t)\,dt \to 0 \quad \text{as} \quad N \to \infty.$$

In particular, this means that $\mathbb{P}(|U_N - 0| > \delta) \to 0$ as $N \to \infty$, so that

$$U_N \xrightarrow{\mathbb{P}} 0.$$

Again, this is not a statement about averaging many draws from a fixed distribution — it is a statement about a *sequence* of random variables U_N drawn from a distribution P_N indexed by N, and "what happens to the randomness" of the *individual* random variables U_N as $N \to \infty$.

We quickly summarize these results, and a few others of import, in the following lemma.

Lemma 2.3.9 (Expected Value and Convergence Properties)

Let P be a cumulative distribution function on $[a, b]$ with density $p = P'$, and let $U \sim P$ be a continuous random variable. Then, for any functions $g, h \colon [a, b] \to \mathbb{R}$ that are Riemann-Stieltjes integrable with respect to P, we have the following:

(i) **(Linearity)** For any constants $\alpha, \beta \in \mathbb{R}$,

$$\mathbb{E}[\alpha\, g(U) + \beta\, h(U)] = \alpha\, \mathbb{E}[g(U)] + \beta\, \mathbb{E}[h(U)].$$

This follows directly from the linearity of the Riemann-Stieltjes integral.

(ii) **(Monotonicity/Positivity)** If $g(t) \le h(t)$ for all $t \in [a, b]$, then

$$\mathbb{E}[g(U)] \le \mathbb{E}[h(U)].$$

In particular, $g \ge 0$ implies $\mathbb{E}[g(U)] \ge 0$. This is a direct consequence of $p \ge 0$: the Riemann-Stieltjes sums have nonnegative weights, so the ordering of the integrands is preserved.

(iii) **(Triangle Inequality)** We have that

$$|\mathbb{E}[g(U)]| \le \mathbb{E}[|g(U)|].$$

This follows from (i) and (ii): since $-|g| \le g \le |g|$, monotonicity gives $-\mathbb{E}[|g(U)|] \le \mathbb{E}[g(U)] \le \mathbb{E}[|g(U)|]$.

(iv) **(Boundedness)** If $m \le g(t) \le M$ for all $t \in [a, b]$, then

$$m \le \mathbb{E}[g(U)] \le M.$$

This is the convex combination property we derived earlier — we include it here for completeness.

(v) **(Continuous Mapping Theorem)** Let $U_1, U_2, \ldots$ be a sequence of random variables such that $U_N \xrightarrow{\mathbb{P}} c$ for some constant $c \in [a,b]$. If $g\colon [a,b] \to \mathbb{R}$ is continuous at c, then

$$g(U_N) \xrightarrow{\mathbb{P}} g(c).$$

That is, continuous functions preserve convergence in probability.

(vi) **(Bounded Convergence)** Let $U_1, U_2, \ldots$ be random variables on $[a,b]$ and suppose $g_N \xrightarrow{\mathbb{P}} g$ (meaning $g_N(U_N) \xrightarrow{\mathbb{P}} g$ for some constant or random variable g). If $|g_N| \le M$ uniformly for all N, then

$$\mathbb{E}[g_N] \to \mathbb{E}[g].$$

That is, convergence in probability plus a uniform bound implies convergence of expectations.

Proof:

Properties (i) through (iv) follow immediately from the corresponding properties of the Riemann-Stieltjes integral and the fact that $p \ge 0$ with $\int p = 1$; we have essentially already proved them above.

For (v), fix $\epsilon > 0$. Since g is continuous at c, there exists $\delta > 0$ such that $|t - c| < \delta$ implies $|g(t) - g(c)| < \epsilon$. Then

$$\mathbb{P}(|g(U_N) - g(c)| \ge \epsilon) \le \mathbb{P}(|U_N - c| \ge \delta) \to 0$$

by the assumption that $U_N \xrightarrow{\mathbb{P}} c$. Since $\epsilon > 0$ was arbitrary, $g(U_N) \xrightarrow{\mathbb{P}} g(c)$.

For (vi), fix $\epsilon > 0$ and write

$$\begin{aligned} |\mathbb{E}[g_N] - \mathbb{E}[g]| &= |\mathbb{E}[g_N - g]| \le \mathbb{E}[|g_N - g|] \\ &= \underbrace{\int_{|g_N - g| \le \epsilon} |g_N - g| \, dP}_{< \epsilon \cdot 1 = \epsilon} + \underbrace{\int_{|g_N - g| > \epsilon} |g_N - g| \, dP}_{\le 2M \cdot \mathbb{P}(|g_N - g| > \epsilon)}. \end{aligned}$$

Since $g_N \xrightarrow{\mathbb{P}} g$, the second term vanishes as $N \to \infty$. Thus $\limsup |\mathbb{E}[g_N] - \mathbb{E}[g]| \le \epsilon$ for every $\epsilon > 0$, and the result follows.

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Remark.

Of course, the LLNs *are* concentration results — they say that if we define the sequence of sample means

$$\overline{X}_N := \frac{1}{N} \sum_{n=1}^{N} X_n,$$

then each U_N follows a different distribution, but as $N \to \infty$, we observe that $\overline{X}_N \xrightarrow{\mathbb{P}} \mathbb{E}[X]$, where $\mathbb{E}[X]$ plays the role of the fixed constant c. This is precisely what the Weak Law of Large

Numbers guarantees. In general, then, concentration results occupy a *more general* role in the statistical landscape — we use them broadly whenever examining sequences of random variables, where those random variables may themselves be composed of random parts (like the sample mean above.)

So, at this point, what we have is that $U_N \sim p_N$ the Nth Fejér density $p_N = F_N/\pi$ converges to zero in probability. In particular, let us observe that *pointwise* convergence of $\sigma_N f(x) \to f(x)$ for $x \in [-\pi, \pi]$ follows almost immediately from three standard results. Since $U_N \xrightarrow{\mathbb{P}} 0$ and the map $t \mapsto f(x-t)$ is continuous at $t = 0$, the **continuous mapping theorem** gives

$$f(x - U_N) \xrightarrow{\mathbb{P}} f(x - 0) = f(x).$$

Moreover, since f is continuous on the compact set $[-\pi, \pi]$, it is bounded: $|f| \leq M$ for some $M > 0$. In particular, $|f(x - U_N)| \leq M$ for every realization of U_N.

Thus, by the **bounded convergence theorem** — which states that if $Y_N \xrightarrow{\mathbb{P}} Y$ and $|Y_N| \leq M$ uniformly, then $\mathbb{E}[Y_N] \to \mathbb{E}[Y]$ — we conclude

$$\sigma_N f(x) = \mathbb{E}[f(x - U_N)] \to f(x).$$

This really is all we needed; in particular, realize that this effectively guarantees us pointwise convergence — precisely what the Dirichlet partial sums $S_N f(x)$ took us around 30 pages to develop — in exactly three lines. It required a bit of probabilistic machinery — convergence in probability, the continuous mapping theorem, and the bounded convergence theorem — but, again, these aren't really "probabilistic devices" as much as they are *integration* devices.

Thus, we've shown the pointwise convergence of $\sigma_N f \to f$ in three lines. We might imagine we're done; but, recall that we claimed something stronger of σ_N: namely, that it converged *uniformly* to its target, meaning that a single N works for all $x \in [-\pi, \pi]$ simultaneously. The argument above does not deliver this, because the continuous mapping theorem guarantees convergence at a rate that depends on the local behavior of f *near* x. To upgrade from pointwise to uniform, we need to quantify the argument, which is what the following proof does. Again recall that what we're *hoping* to show precisely is that

$$\sigma_N f(x) \to f(x) \quad \text{uniformly on} \quad [-\pi, \pi].$$

which we abbreviate as

$$\sigma_N f \rightrightarrows f \quad \text{on} \quad [-\pi, \pi].$$

So, how can we convert our statement about U_N into one about σ_N? We present a theorem whose proof will completely put the topic to rest.

Theorem 2.3.10 (Fejér)

Let $f\colon [-\ell, \ell] \to \mathbb{R}$ be continuous with a valid 2ℓ-periodic extension. Then the Cesàro means $\sigma_N f(x)$ converge **uniformly** to f on $[-\ell, \ell]$:

$$\sigma_N f \rightrightarrows f \quad \text{on} \quad [-\pi, \pi].$$

More generally, if $f \in L^1[-\ell, \ell]$ and f is continuous at a point $x_0 \in [-\ell, \ell]$ (though possibly admitting finitely many discontinuities elsewhere), then at x_0, we observe

$$\sigma_N f(x_0) \to f(x_0).$$

Proof:

We prove the first statement; the second follows by a localization argument (i.e., simply treating each of the continuous parts of $[-\ell, \ell]$ individually according to the argument below.)

Setting $\ell = \pi$ without loss of generality (where the general case follows by rescaling), we need to show that

$$\sup_{x\in[-\pi,\pi]} |\sigma_N f(x) - f(x)| \to 0 \quad \text{as} \quad N \to \infty.$$

First of all, since f is continuous on the compact interval $[-\pi, \pi]$, it follows that f is *uniformly continuous* on $[-\pi, \pi]$: given $\epsilon > 0$, we can choose $\delta > 0$ based *purely off* ϵ such that

$$|s-t| < \delta \implies |f(s) - f(t)| < \epsilon, \quad \forall s, t \in [-\pi, \pi].$$

(The idea here is that absent uniform convergence, it could be that "points s/t and s'/t' separated by a common distance $|s-t| = |s'-t'| < \delta$ don't behave the same; i.e., what could *also* matter is where s and t live relative to $[-\ell, \ell]$." Uniform convergence guarantees that no matter where we search in $[-\pi, \pi]$ for s and t, the condition that $|s-t| < \delta$ alone is sufficient to ensure $|f(s) - f(t)| < \epsilon$.)

Letting $x \in [-\pi, \pi]$ be arbitrary, what this uniform convergence implies is that, at the points $s = x - u$ and $t = x$, we have

$$|u| < \delta \implies |f(x-u) - f(x)| < \epsilon, \qquad \forall x \in [-\pi, \pi].$$

Now, using this uniformity, let us return to the Cesàro means σ_N. Since

$$\sigma_N f(x) = \mathbb{E}[f(x - U_N)], \quad U_N \sim p_N \quad \text{where} \quad \int_{-\pi}^{\pi} p_N(u)\, du = 1,$$

what we can write is

$$f(x) = f(x)\cdot 1 = f(x)\int_{-\pi}^{\pi} p_N(u)\, du = \int_{-\pi}^{\pi} f(x) p_N(u)\, du = \mathbb{E}_{U_N}[f(x)].$$

Now, using the linearity of expectation,

$$\sigma_N f(x) - f(x) = \mathbb{E}_{U_N}[f(x-U_N)] - \mathbb{E}_{U_N}[f(x)] = \mathbb{E}_{U_N}[f(x-U_N) - f(x)].$$

Taking absolute values and applying the triangle inequality inside the expectation (which preserves the inequality since $p_N \geq 0$):

$$|\sigma_N f(x) - f(x)| \leq \mathbb{E}_{U_N}\big[|f(x-U_N) - f(x)|\big] = \int_{-\pi}^{\pi} |f(x-t) - f(x)|\, p_N(t)\, dt.$$

We split this integral into the region where $|t| \leq \delta$ and the region where $|t| > \delta$:

$$|\sigma_N f(x) - f(x)| \leq \underbrace{\int_{|t|\leq\delta} |f(x-t) - f(x)|\, p_N(t)\, dt}_{=:I_1} + \underbrace{\int_{|t|>\delta} |f(x-t) - f(x)|\, p_N(t)\, dt}_{=:I_2}.$$

For I_1, note that when $|t| \le \delta$, uniform continuity gives $|f(x-t) - f(x)| < \epsilon$. Since $p_N \ge 0$ and $\int_{-\pi}^{\pi} p_N = 1$, we have

$$I_1 < \epsilon \int_{|t|\le\delta} p_N(t)\, dt \le \epsilon \int_{-\pi}^{\pi} p_N(t)\, dt = \epsilon.$$

Now, for I_2, realize that since f is continuous on the compact set $[-\pi, \pi]$, it is bounded: there exists $M > 0$ such that $|f(x)| \le M$ for all $x \in [-\pi, \pi]$. By the triangle inequality, $|f(x-t) - f(x)| \le 2M$ for any x, t. Thus

$$I_2 \le 2M \int_{|t|>\delta} p_N(t)\, dt = 2M \cdot \mathbb{P}(|U_N| > \delta).$$

But $U_N \xrightarrow{\mathbb{P}} 0$, so for any $\eta > 0$, there exists $N_0 \in \mathbb{N}$ such that $\mathbb{P}(|U_N| > \delta) < \eta$ for all $N \ge N_0$. Choosing $\eta = \epsilon/(2M)$, we obtain

$$I_2 < 2M \cdot \frac{\epsilon}{2M} = \epsilon \qquad \forall N \ge N_0.$$

Combining our results for I_1 and I_2, what we have is that for all $N \ge N_0$ and all $x \in [-\pi, \pi]$,

$$|\sigma_N f(x) - f(x)| \le I_1 + I_2 < \epsilon + \epsilon = 2\epsilon.$$

Since the choice of δ depended only on ϵ (via uniform continuity), and the choice of N_0 depended only on δ and M (via convergence in probability), *neither choice depended on* x. In particular,

$$\sup_{x\in[-\pi,\pi]} |\sigma_N f(x) - f(x)| < 2\epsilon \qquad \forall N \ge N_0,$$

and since $\epsilon > 0$ was arbitrary, this is uniform convergence:

$$\sigma_N f \rightrightarrows f \quad \text{on} \quad [-\pi, \pi].$$

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The proof is worth reading twice, because its entire structure is visible from the probabilistic reformulation. Once we identified $\sigma_N f(x) = \mathbb{E}[f(x - U_N)]$ as an expected value under a concentrating distribution, the argument was forced: split the expectation into a "near" piece I_1 (where the shift U_N is small, so uniform continuity controls the integrand), and a "far" piece I_2 (where the integrand is bounded, and convergence in probability kills the weight). The nonnegativity of p_N entered where we bounded $\int_{|t|\le\delta} p_N \le 1$ — this step fails for the Dirichlet kernel, because D_N/π is not a density, and $\int_{|t|\le\delta} |D_N/\pi|$ can be much larger than 1. The Gibbs overshoot is, from this vantage, a failure of the "near" integral I_1: the signed kernel D_N assigns too much total weight to the near region, and the $\epsilon \cdot 1$ bound breaks.

2.3.2 Parseval and Variance

Parseval's Theorem

Let us now return to a promise we made earlier in the chapter. Recall that, when computing the Fourier series of

$$g(x) = |x|, \qquad x \in [-\pi, \pi],$$

we discovered the identity

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8},$$

and compared it with Leibniz's formula for π,

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4},$$

which arose from the Fourier series of the square-wave

$$f(x) = \begin{cases} -1 & -\pi \le x < 0, \\ 0 & x = 0, \\ 1 & 0 < x \le \pi. \end{cases}$$

What we found upon squaring the Leibniz series and rearranging was that

$$\underbrace{\frac{\pi^2}{16}}_{\text{cross-terms}} = \underbrace{\frac{\pi^2}{8}}_{\text{sum of squares}} - \underbrace{\frac{\pi^2}{16}}_{\text{square of sum}}.$$

We observed then that this was "quite analogous" to the variance identity,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

We promised at this time that we'd return to this observation when we discussed Parseval's theorem, and the time has come to reveal that the analogy with the variance was not an analogy at all.

To start, let us give the "official" definition of the variance of a random variable.

Definition 2.3.11 (Variance)

Let $X\colon \Omega \to [a,b]$ be a continuous random variable with CDF P, PDF $P' = p$, and mean $\mu_X := \mathbb{E}[X]$. Whenever it exists, we define the variance of X to be the quantity

$$\mathrm{Var}(X) := \mathbb{E}[(X-\mu_X)^2] = \int_a^b (x-\mu_X)^2 dP(x) = \int_a^b (x-\mu_X)^2 p(x)\, dx.$$

The variance measures the average (squared) distance that a realization from a random variable X falls from from its mean μ_X.

Now, to retrieve our proclaimed identity, let us expand the square in the definition and apply linearity:

$$\mathbb{E}[(X-\mu_X)^2] = \mathbb{E}[X^2 - 2\mu_X X + \mu_X^2] = \mathbb{E}[X^2] - 2\mu_X \underbrace{\mathbb{E}[X]}_{=\mu_X} + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2.$$

Thus, we obtained the desired expression

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu_X^2, \quad \text{where} \quad \mu_X = \mathbb{E}[X].$$

But notice what this formula demands: we require that both $\mu_X = \mathbb{E}[X]$ exist, and that $\mathbb{E}[X^2]$ exist. This brings us to a second definition.

Definition 2.3.12 (pth Moment)

If $X\colon \Omega \to [a,b]$ is a continuous random variable with CDF P and density $P' = p$, then we say the p**th moment of** X exists if we observe finiteness of the integral

$$\int_a^b |x|^p \, dP(x) = \int_a^b |x|^p p(x) \, dt < \infty,$$

in which case we define the pth moment to be

$$\mathbb{E}[X^p] := \int_a^b x^p \, dP(x) = \int_a^b x^p p(x) \, dx.$$

So, for the second moment $\mathbb{E}[X^2]$ to exist, we require that

$$\int_a^b |x|^2 p(x) \, dx < \infty,$$

in which case we define

$$\mathbb{E}[X^2] := \int_a^b x^2 p(x) \, dx.$$

We're curious at this point what qualities might render a function f to have *some* existence moments, but not others. On the one hand, it follows by an application of *Jensen's Inequality* that

$$\mathbb{E}[X^p] \text{ exists} \implies \mathbb{E}[X^q] \text{ exists for all } 1 \le q < p.$$

In general, Jensen's Inequality allows us to bound a (deterministic) convex function of a random variable's mean, $\phi(\mathbb{E}[X])$ for $\phi(\cdot)$ a convex function, by the mean of the (random) convex function, $\mathbb{E}[\phi(X)]$.

Definition 2.3.13 (Convex/Concave Functions)

A function $f\colon [a,b] \to \mathbb{R}$ is said to be **convex** if for any points in the interval $x, y \in [a,b]$ and scalar $\alpha \in [0,1]$,

$$f(\alpha x + (1-\alpha)y) \le \alpha f(x) + (1-\alpha) f(y).$$

Observe that the left-side is the function f applied to the **convex combination** $\alpha x + (1-\alpha)y \in [a,b]$, while the right-side is the convex combination of the individual evaluations. We say that g is **concave** if the reverse is true — i.e., if

$$g(\alpha x + (1-\alpha)y) \ge \alpha g(x) + (1-\alpha) g(y),$$

or, equivalently, if the function $-g$ is convex.

Theorem 2.3.14 (Jensen's Inequality)

If $f\colon [a,b] \to \mathbb{R}$ is convex and X is a random variable supported on $[a,b]$, and if

$$\mathbb{E}[f(X)] \quad \text{and} \quad g(\mathbb{E}[X]) \quad \text{exist,}$$

then

$$f(\mathbb{E}[X]) \le \mathbb{E}[f(X)], \quad f \text{ convex.}$$

If $g\colon [a,b] \to \mathbb{R}$ is concave on the same support, then the inequality flips:

$$g(\mathbb{E}[X]) \ge \mathbb{E}[g(X)], \quad g \text{ concave.}$$

Now, if $p > q \ge 1$, then $p/q > 1$, and it follows that $t \mapsto t^{p/q}$ is a convex function on $\mathbb{R}$ (check this using the definition!) By Jensen's inequality (noting that $q \cdot p/q = p$), we have that

$$\left(\mathbb{E}[|X|^q]\right)^{p/q} \le \mathbb{E}[|X|^p].$$

If the right side is finite, then so is the left; in particular, the existence of the pth moment implies the existence of the qth moment for any $1 \le q < p$.

Nevertheless, in spite of this observation, we're curious *why* some functions seem to "behave nicely to a point," then suddenly seem to "flip" to poorly behaved, diverging monsters. The most obvious example to us is illustrated in the plot below.

The plot illustrates the point concretely. If we define

$$f(t) = |t|^{-1/2}, \quad t \in [-1,1],$$

then f blows up near the origin: $f(0^+) = f(0^-) = \infty$. The blue curve shows that

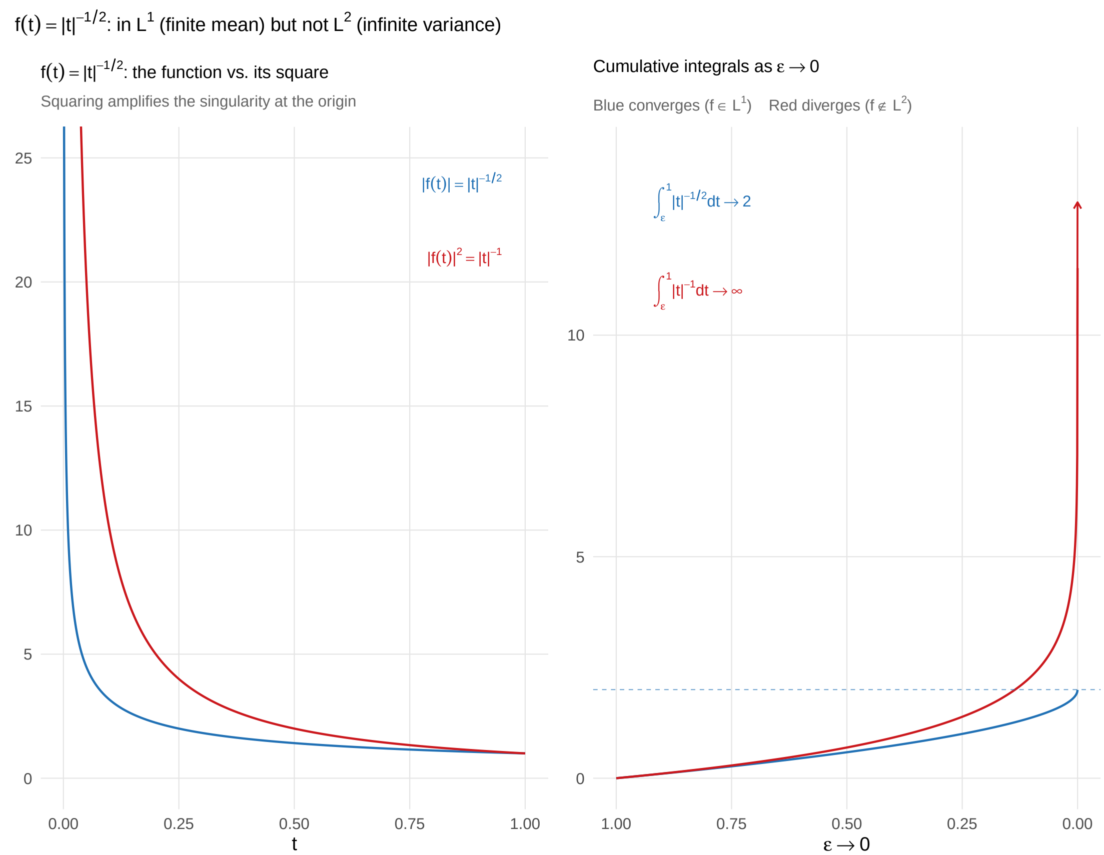
$$\int_\epsilon^1 |f(t)|\,dt = \int_\epsilon^1 |t|^{-1/2}\,dt \to 2 \quad \text{as} \quad \epsilon \downarrow 0,$$

while the red curve shows that

$$\int_\epsilon^1 |f(t)|^2\,dt = \int_\epsilon^1 |t|^{-1} \to \infty \quad \text{as} \quad \epsilon \downarrow 0.$$

That is, f is absolutely integrable on $[-1,1]$, but it is *not* square-integrable on $[-1,1]$. What we say in this case is that

$$f \in L^1[-1,1], \quad \text{but} \quad f \notin L^2[-1,1].$$



A function $f(t) = |t|^{-1/2}$, $t \in [-1, 1]$, whose first uniform moment exists but whose second uniform moment fails to exist.

Remark.

Notice that we defined f on $[-1, 1]$, but we asserted the convergence/divergence above after only checking the positive portion $(\epsilon, 1]$ as $\epsilon \downarrow 0$. This is because jere, $f(-x) = |-x|^{-1/2} = |x|^{-1/2} = f(x)$, so f is even. In particular then, notice that by construction, the integrand $|f(t)|^p$ is an *even function* whenever f is either even *or* odd (check this for odd functions too!) In particular, what we have is that

$$\int_\epsilon^1 |f(t)|^p \, dt = \int_{-1}^{-\epsilon} |f(t)|^p \, dt, \quad \forall p > 0,$$

so if the integral on the left converges (here, to 2 as $\epsilon \downarrow 0$), then the integral on the right converges to the same value. Conversely, if the integral on the left diverges, so too does the one on the right. In either case, it suffices to just check one half of the symmetric domain $[-1, 1]$.

At this point, you may be wondering what a *deterministic* function like $f(t) = |t|^{-1/2}$ has to do with the variance of a *random* quantity. The answer, of course, relies in changing our perception of *what "random" is allowed to mean.*

So, again consider the function $f(t) = |t|^{-1/2}$ on $[-1, 1]$. Again, there is osensibly "nothing random" about the function f — given $t = 0.3$, we compute $f(0.3) = 0.3^{-1/2} \approx 1.826$, and that's that. But the question here is really not about how f operates *given the input* $t = 0.3$, but *where* that input came from — i.e., the question is "why we were told to evaluate f at $t = 0.3$ in the first place."

In this light, suppose that, for example, we didn't choose $t = 0.3$ deliberately — instead, suppose we drew t at random from $[-1, 1]$ in such a way that every point was equally viable a priori, and we "just happened" to choose $t = 0.3$. We could characterize the situation by defining a *uniform* random variable $U \sim \text{Uniform}[-1, 1]$ with constant density

$$p_U(t) = \frac{1}{2}, \qquad t \in [-1, 1],$$

and asserting that we "draw a point $U = t$ from this distribution and evaluate the realization $f(t)$." In this case, we imagine that $f(t)$ is *still deterministic* — it is the value of f *given* (or *conditional on*) having drawn $U = t$.

On the other hand, if $U \sim \text{Uniform}[-1, 1]$, what would the notation $f(U)$ mean for the same function f? Well, clearly $f(U)$ is *no longer deterministic* — it is *random*, because its input U is random. Before we draw U, we don't know which $t \in [-1, 1]$ we'll get, and therefore we don't know which value $f(t)$ we'll end up with. The object $f(U)$ therefore serves to represent the *entire ensemble of possible outcomes*: it could be that

$$U = 0.3 \mapsto f(0.3) \approx 1.826, \quad U = 0.01 \mapsto f(0.01) = 10, \quad U = 0.0001 \mapsto f(0.0001) = 100,$$

and so on. But to fully characterize $f(U)$ as a random variable, we don't need to list every possible outcome — we just need to know *how likely* each outcome is. What we reckon with in this framework, then, is *not* in "re-characterizing how the function f behaves within $[-1, 1]$" — this is known, and is essentially the same whether or not we invoke any probability at all. Rather, what the probabilistic framework lets us do is reason about *what we're likely to see* when we don't get to choose where to look within $[-1, 1]$.

This is a subtle but important shift in perspective. The function $f(t) = |t|^{-1/2}$ is what it is — it blows up near the origin, it equals 1 at $t = \pm 1$, and none of that changes when we introduce a random variable U whose realizations fall in the same domain. What changes is *the questions we're able to ask*. Without probability, we ask "what is $f(0.01)$?" and receive a definite answer, $f(0.01) = 10$. But when we imagine that the input here manifests as the realization of a continuous random variable U, the question of "what is $f(0.01)$?" is hardly productive, since we're essentially guaranteed that we *never get to observe* $U = 0.01$ exactly. We can confirm this by computing the probability explicitly:

$$\mathbb{P}(U = 0.01) = \mathbb{P}(0.01 \le U \le 0.01) = \int_{0.01}^{0.01} \frac{1}{2}\, dt = 0.$$

Clearly the choice of $U = 0.01$ here is irrelevant; it is a consequence of the fact that integrals over singleton sets $\{x_0\} \subset \mathbb{R}$ *always* vanish. In particular, what we realize is that for a continuous random variable, questions about exact values are meaningless — the probability of any specific outcome is zero.

The questions that *are* productive, on the other hand, are those regarding *intervals*: "what is the probability that $f(U)$ exceeds 10?" or "what is the average value of $f(U)$?" or "typically, how spread out are the values of $f(U)$?" But again, realize that these are *not questions about* f — they are questions *about the sampling mechanism* of U. They are not solved by evaluating f at any particular point, but rather by *integrating f against the density of U*. What changes interpretatively is that, when we compute $\mathbb{E}[f(U)]$, we move from asking "what *do we see* when f is evaluated at a *specific* point" to "what *would we typically see* if f is evaluated at a *random* point." When we compute $\mathrm{Var}(f(U))$, we move from asking "how much does f change near a specific point t_0" to "how much does f *surprise us* relative to its average $\mathbb{E}[f(U)]$?" In general, this framework turns questions about *pointwise behavior* into questions about *average behavior*.

Remark.

It is worth emphasizing that this is how statisticians perceive randomness *in general*. When we declare "X is a random variable," we're *always* implying "there exist some deterministic function g and some random component U such that $X = g(U)$." This is nothing more than the usual distinction between the notations $y = f(x)$, but here, the distinction actually gives rise to *separate interpretations*: in particular, if we were observing realizations $X = x_0$ of the random variable X directly, then we can either imagine that (a) X is an "inherently random object" whose realizations x_0 we observe directly (more in line with *data analysis*), or (b) that $X = g(U)$ for some deterministic function g and some random input U, so that the realizations x_0 are actually $x_0 = g(u_0)$ for some realized $U = u_0$ (more in line with *statistical modeling*).

In our current setting, we operate firmly from within the second camp: the function f is known, the distribution of U is known (uniform on $[-\ell, \ell]$), and the random variable $f(U)$ is the composition. Everything we compute — $\mathbb{E}[f(U)]$, $\mathrm{Var}(f(U))$, and eventually the Fourier coefficients themselves — follows from this decomposition. The advantage of making the decomposition explicit is that it separates the two sources of structure in the problem: the *analytic* structure (what f looks like as a function) and the *probabilistic* structure (how U samples the domain). Different choices of distribution for U would produce different means, different variances, and different "coefficient" formulas — the Fourier setting is the special case where U is uniform, and it is the uniformity of U that gives the Fourier coefficients their clean orthogonality properties.

To summarize, when we write $f(U)$ for $U \sim \mathrm{Uniform}[-1, 1]$, we interpret the randomness as living not in the function f, but in its argument U; f is deterministic, but when we compose the two, the quantity $f(U)$ *becomes* random.

So, returning to $f(t) = |t|^{-1/2}$, $t \in [-1, 1]$, if $U \sim \mathrm{Uniform}[-1, 1]$, then we can ask questions about "how likely $f(U)$ is to take on large values" (say, $f(U) > 10$) by computing

$$\mathbb{P}(f(U) > 10) = \mathbb{P}(|U|^{-1/2} > 10) = \mathbb{P}(-0.01 < U < 0.01) = \int_{-0.01}^{0.01} \frac{1}{2}\, dt = 0.01.$$

This says that under a uniform sampling scheme on $[-1, 1]$, we're quite unlikely to observe values of $f(U) > 10$ (in particular, we observe them with probability 0.01). Heuristically, what we can imagine happening is that this probability is "low enough" to force a finite expectation:

$$\mathbb{E}[f(U)] = \int_{-1}^{1} f(t) p_U(t)\, dt = \frac{1}{2} \int_{-1}^{1} |t|^{-1/2}\, dt = \int_0^1 t^{-1/2}\, dt = \left.\frac{t^{1/2}}{1/2}\right|_0^1 = 2.$$

Nevertheless, this probability is still "too high" for the second moment to exist:

$$\mathbb{E}[|f(U)|^2] = \int_{-1}^{1} |f(t)|^2 p_U(t)\, dt = \frac{1}{2} \int_{-1}^{1} |t|^{-1}\, dt = \int_0^1 \frac{1}{t}\, dt = +\infty.$$

Since $\mathbb{E}[|f(U)|^2] = \infty$, the second moment $\mathbb{E}[f(U)^2]$ does not exist.

Returning to the variance identity, this means that

$$\mathrm{Var}(f(U)) = \underbrace{\mathbb{E}[f(U)^2]}_{=+\infty} - \underbrace{(\mathbb{E}[f(U)]^2)}_{=2^2} = +\infty.$$

So the variance of $f(U) = |U|^{-1/2}$, $U \sim \text{Uniform}[-1,1]$ is infinite — not because f is "badly behaved" in any dramatic sense (it is continuous away from the origin, absolutely integrable, and has finite mean), but because its singularity at $t = 0$ is severe enough that the *squared* deviations from the mean don't average out. The uniform searchlight U lands near $t = 0$ with small but positive probability, and when it does, $f(U) = |U|^{-1}$ is catastrophically large — large enough to render the second moment nonexistent.

At this point, we've seen what can go wrong: $f(t) = |t|^{-1/2}$ lives in $L^1[-1,1]$ (f is absolutely integrable, or just integrable), but not in $L^2[-1,1]$ (f is *not* square-integrable). What we've *also* seen is that we can reframe these statements about "the L^p-integrability of the deterministic $t \mapsto f(t)$ on the interval $t \in [-1,1]$" in terms of "the existence pth moment of the random $U \mapsto f(U)$ under uniform sampling, $U \sim \text{Uniform}[-1,1]$." In particular, what we observed was that if

$$U \sim \text{Uniform}[-1,1], \qquad \text{with density} \qquad p_U(t) = \frac{1}{2}, \quad t \in [-1,1],$$

then

$$f \in L^1[-1,1] \iff \int_{-1}^{1} |f(t)|\,dt < \infty \iff \mathbb{E}_U[|f(U)|] = \int_{-1}^{1} |f(t)| p_U(t)\,dt < \infty \iff \mathbb{E}_U[f(U)] \text{ exists,}$$

and

$$f \in L^2[-1,1] \iff \int_{-1}^{1} |f(t)|^2\,dt < \infty \iff \mathbb{E}_U[|f(U)|^2] = \int_{-1}^{1} |f(t)|^2 p_U(t)\,dt < \infty \iff \mathbb{E}_U[f(U)^2] \text{ exists.}$$

What we might also understand is that

$$dP_U(t) = p_U(t)\,dt = \frac{1}{2}dt \implies P_U(t) = \frac{t}{2},$$

so that

$$\mathbb{E}[|f(U)|] = \int_{-1}^{1} |f(t)| p_U(t)\,dt = \int_{-1}^{1} |f(t)|\,dP_U(t).$$

In particular, we observe that in this case, the Riemann-Stieltjes integral $\int |f|\,dP_U$ amounts to nothing more than a constant rescaling of the ordinary integral $\int |f|\,dt$. Again, this is obvious when we write $dP_U = p_U(t)dt$ where p_U is a constant — but the point we seek to emphasize is that writing dt or dP_U in this case are *essentially synonymous* — every infinitesimal strip dt is uniformly spaced, and each infinitesimal strip $dP_U = \frac{1}{2}\,dt$ is simply a *linear rescaling* of those uniformly spaced dt-strips. If we were working with a **standard uniform** $V \sim \text{Uniform}[0,1]$, we would have that $p_V(t) = 1$ for every $t \in [0,1]$, and that

$$dP_V = p_V(t)\,dt = 1 \cdot dt = dt.$$

That is, when we have a function f on $[0,1]$, it is precisely the same thing for us to "let $V \sim \text{Uniform}[0,1]$, assign each infinitesimal strip $[t, t+dt] \subset [0,1]$ the probability $dP_V(t)$ that V lands there, and integrate $\int_0^1 f(t)\,dP_V(t)$" as it is for us to "uniformly slice the interval $[0,1]$ into infinitesimal pieces dt and integrate $\int_0^1 f(t)\,dt$." It is abundantly clear that this is a consequence of the uniformity of the sampling scheme; when we sample uniformly from an interval, we assign equal probability to every infinitesimal strip. There is no internal bias on behalf of the "strip-producing mechanism" dP_V when V is a uniform random variable, and this is precisely how we define the differential dt to behave in the first place.

It should also be clear that there is really no distinction between $V \sim \text{Uniform}[0,1]$ and a general $U \sim \text{Uniform}[a,b]$ — what happens is that

$$dP_U(t) = \frac{1}{b-a}dt,$$

so everything reduces back to the ordinary Riemann integral with integrator dt — we're just rescaling by the length of the interval so that $dP_U(t)$ has a probabilistic interpretation: since

$$\int_a^b dP_U(t) = P_U(b) - P_U(a) = 1 - 0 = 1.$$

we force

$$p_U(t) = \frac{1}{b-a}, \quad t \in [a,b]$$

in order for

$$\int_a^b dP_U(t) := \int_a^b p_U(t)\,dt = \frac{1}{b-a}\int_a^b dt = \frac{1}{b-a}[b-a] = 1.$$

In particular, what all of this amounts to for *our* purposes (yes, we do remember them!) is the observation that, on any bounded interval, there is an *exact* correspondence between an integral of the form

$$\frac{1}{b-a}\int_a^b f(t)\,dt$$

and an expectation of the form

$$\mathbb{E}_U[f(U)], \qquad U \sim \text{Uniform}[a,b].$$

With this identification in hand, let us return to the Fourier setting on $[-\ell,\ell]$. So, if $f\colon [-\ell,\ell] \to \mathbb{R}$ is a function whose Fourier coefficients are

$$a_0 = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\,dt, \quad a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\cos\left(\frac{n\pi t}{\ell}\right)\,dt, \quad b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\sin\left(\frac{n\pi t}{\ell}\right)\,dt,$$

naturally, we want to define

$$U \sim \text{Uniform}[-\ell,\ell], \qquad \text{with density} \qquad p_U(t) = \frac{1}{2\ell}, \quad t \in [-\ell,\ell].$$

Immediately, we can observe that the common prefactor in the Fourier coefficients is

$$\frac{1}{\ell} = 2\cdot\frac{1}{2\ell} = 2p_U(t),$$

so we can write, say,

$$\frac{a_0}{2} = \frac{1}{2}\cdot\frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\,dt = \int_{-\ell}^{\ell} f(t)\,\frac{1}{2\ell}dt = \int_{-\ell}^{\ell} f(t)p_U(t)\,dt = \mathbb{E}_U[f(U)].$$

What's quite surprising about this is that, in the Fourier series

$$\mathcal{S}f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{\ell}\right) + b_n \sin\left(\frac{n\pi x}{\ell}\right) \right),$$

we can observe that exact same factor of $\frac{a_0}{2}$ playing the role of the leading coefficient. In particular, if $S_N f(x)$ is the Nth partial sum of the series above, then what we have is that

$$S_0 f(x) = \frac{a_0}{2} = \mathbb{E}_U[f(U)], \qquad U \sim \text{Uniform}[-\ell, \ell].$$

That is, the "zeroth-order Fourier approximation" to f — the best approximation we can obtain without including any of the oscillatory terms — is the mean of f under uniform sampling.

Now, we seek to reckon with the remaining oscillatory terms in the Fourier series,

$$\sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{\pi n x}{\ell}\right) + b_n \sin\left(\frac{\pi n x}{\ell}\right) \right).$$

Let us plug in the expectation $\mathbb{E}_U[f(U)]$, $U \sim \text{Uniform}[-\ell, \ell]$ to the constant $a_0/2$ in the Fourier series:

$$\mathcal{S}f(x) = \mathbb{E}_U[f(U)] + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{\pi n x}{\ell}\right) + b_n \sin\left(\frac{\pi n x}{\ell}\right) \right).$$

Rearranging, we find that

$$\mathcal{S}f(x) - \mathbb{E}_U[f(U)] = \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{\pi n x}{\ell}\right) + b_n \sin\left(\frac{\pi n x}{\ell}\right) \right).$$

So, let us observe something about the above: recalling the variance identity

$$\text{Var}_U(f(U)) = \mathbb{E}_U\left[\left(f(U) - \mathbb{E}_U[f(U)] \right)^2 \right]$$

,
what we might notice is that the argument inside the square of the expectation on the right,

$$f(U) - \mathbb{E}_U[f(U)],$$

looks quite similar to the left-side of our Fourier expansion,

$$\mathcal{S}f(x) - \mathbb{E}_U[f(U)].$$

Indeed, what the left-side of the Fourier expansion reveals is that the oscillatory terms $\sum a_n \cos(\cdots) + b_n \sin(\cdots)$ serve a very explicit purpose: to track the deviation of $\mathcal{S}f(x)$ from the function's grand mean $\mathbb{E}_U[f(U)]$ over all of $[-\ell, \ell]$ at the argument we supply, $x \in [-\ell, \ell]$. The two expressions above are functionally doing the same thing. The difference is that we're still supplying the argument $x \mapsto \mathcal{S}f(x)$ in the Fourier series; however, supposing we're interested in the *average* deviation of $\mathcal{S}f$ from its mean across all of $[-\ell, \ell]$, what we could do is write

$$\mathcal{S}f(U) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi U}{\ell}\right) + b_n \sin\left(\frac{n\pi U}{\ell}\right) \right), \qquad U \sim \text{Uniform}[-\ell, \ell],$$

with the understanding that the procedure for evaluating $\mathcal{S}f(U)$ is to first draw the uniform realization $U = u \in [-\ell, \ell]$, and to then evaluate $\mathcal{S}f(u)$ at the selected u. Nothing has functionally changed about our representation here; it encompasses the entirety of the deterministic case, where we evaluate $\mathcal{S}f(x)$ at a point $x \in [-\ell, \ell]$ we "choose ourselves"; the only difference is that, now, we're "not the ones choosing the point."

So, what we may write is

$$\mathcal{S}f(U) = \mathbb{E}_U[f(U)] + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi U}{\ell}\right) + b_n \sin\left(\frac{n\pi U}{\ell}\right)\right), \qquad U \sim \text{Uniform}[-\ell, \ell]$$

subtract to find

$$\mathcal{S}f(U) - \mathbb{E}_U[f(U)] = \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi U}{\ell}\right) + b_n \sin\left(\frac{n\pi U}{\ell}\right)\right),$$

and wonder whether we can explicitly connect this with the form of the variance identity, $\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]$. Indeed, let us square both sides to obtain

$$\Big(\mathcal{S}f(U) - \mathbb{E}_U[f(U)]\Big)^2 = \left[\sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi U}{\ell}\right) + b_n \sin\left(\frac{n\pi U}{\ell}\right)\right]^2,$$

and let us integrate both sides over $[-\ell, \ell]$ by the integrator $\alpha = P_U$, the CDF of $U \sim$ Uniform$[-\ell, \ell]$:

$$\int_{-\ell}^{\ell} \Big(\mathcal{S}f(t) - \mathbb{E}_U[f(U)]\Big)^2 \, dP_U(t) = \int_{-\ell}^{\ell} \left[\sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{\ell}\right) + b_n \sin\left(\frac{n\pi t}{\ell}\right)\right]^2 \, dP_U(t).$$

But the left side here is nothing other than

$$\int_{-\ell}^{\ell} \Big(\mathcal{S}f(t) - \mathbb{E}_U[f(U)]\Big)^2 \, dP_U(t) = \mathbb{E}_U\left[\Big(\mathcal{S}f(U) - \mathbb{E}_U[f(U)]\Big)^2\right] = \text{Var}(\mathcal{S}f(U)).$$

where we note that in establishing the equality, we used the fact that $\mathbb{E}_U[f(U)] = \mathbb{E}_U[\mathcal{S}f(U)]$.

According to this notation, what we have at this point is that

$$\text{Var}(\mathcal{S}f(U)) = \int_{-\ell}^{\ell} \left[\sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{\ell}\right) + b_n \sin\left(\frac{n\pi t}{\ell}\right)\right]^2 \, dP_U(t)$$

Take a look at the sum inside the integral:

$$\left[\sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{\ell}\right) + b_n \sin\left(\frac{n\pi t}{\ell}\right)\right]^2 = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi t}{\ell}\right)\right]^2 + \sum_{n=1}^{\infty} \left[b_n \sin\left(\frac{n\pi t}{\ell}\right)\right]^2$$

$$+ 2\sum_{n \neq m} \left[a_n a_m \cos\left(\frac{n\pi t}{\ell}\right) \cos\left(\frac{m\pi t}{\ell}\right)\right] + 2\sum_{n \neq m} \left[b_n b_m \sin\left(\frac{n\pi t}{\ell}\right) \sin\left(\frac{m\pi t}{\ell}\right)\right] + 2\sum_{n \neq m} \left[a_n b_m \cos\left(\frac{n\pi t}{\ell}\right) \sin\left(\cdots\right.\right.$$

Now, integrate both sides against $dP_U(t)$. We're working on bounded intervals, so the integral distributes just fine over the sums on the right; in particular, after factoring out the constants 2, a_n, and b_n, we wind up with integrals of the form

$$\int_{-\ell}^{\ell} \cos^2\left(\frac{n\pi t}{\ell}\right) \, dP_U(t), \quad \int_{-\ell}^{\ell} \sin^2\left(\frac{n\pi t}{\ell}\right),$$

$$\int_{-\ell}^{\ell} \cos\left(\frac{n\pi t}{\ell}\right)\cos\left(\frac{m\pi t}{\ell}\right)\,dP_U(t),\quad \int_{-\ell}^{\ell} \sin\left(\frac{n\pi t}{\ell}\right)\sin\left(\frac{m\pi t}{\ell}\right)\,dP_U(t),\quad \int_{-\ell}^{\ell} \cos\left(\frac{n\pi t}{\ell}\right)\sin\left(\frac{m\pi t}{\ell}\right)\,dP_U(t).$$

Under a uniform U, we have that $dP_U(t) = \frac{1}{2\ell}dt$. What we wind up with, then, is a collection of integrals we already know how to solve: they're just the orthogonality relation integrals, so

$$\frac{1}{2\ell}\int_{-\ell}^{\ell} \cos\left(\frac{n\pi t}{\ell}\right)\cos\left(\frac{m\pi t}{\ell}\right)\,dt = \begin{cases} \frac{1}{2} & n = m \\ 0 & n \neq m, \end{cases}$$

that

$$\frac{1}{2\ell}\int_{-\ell}^{\ell} \sin\left(\frac{n\pi t}{\ell}\right)\sin\left(\frac{m\pi t}{\ell}\right)\,dt = \begin{cases} \frac{1}{2} & n = m \\ 0 & n \neq m, \end{cases}$$

and that

$$\frac{1}{2\ell}\int_{-\ell}^{\ell} \cos\left(\frac{n\pi t}{\ell}\right)\sin\left(\frac{m\pi t}{\ell}\right)\,dt = 0, \quad \forall n, m \in \mathbb{N}.$$

In particular, of the five sums in the expansion above, the last three — all the cross-terms — are annihilated entirely. What's leftover is

$$\operatorname{Var}(\mathcal{S}f(U)) = \sum_{n=1}^{\infty} a_n^2 \underbrace{\left[\int_{-\ell}^{\ell} \cos^2\left(\frac{n\pi t}{\ell}\right)\,dP_U(t)\right]}_{=1/2\ \forall n\in\mathbb{N}} + \sum_{n=1}^{\infty} b_n^2 \underbrace{\left[\int_{-\ell}^{\ell} \sin^2\left(\frac{n\pi t}{\ell}\right)\,dP_U(t)\right]}_{=1/2\ \forall n\in\mathbb{N}} = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2+b_n^2).$$

What we've seemingly just established is that

$$\operatorname{Var}(\mathcal{S}f(U)) = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2).$$

We take a moment to emphasize the similarity to the variance identity

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).$$

What we observe here is that, in general, it is *not* the case that the variance of the sum $X+Y$ is equal to the sum of the variances of X and Y. For this to occur, we need

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \iff \operatorname{Cov}(X,Y) = 0.$$

Here, $\operatorname{Cov}(X,Y)$ is the **covariance** between the random variables X and Y. The covariance is closely related to the familiar **correlation**,

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} \in [-1,1].$$

Realize that

$$\operatorname{Cov}(X,Y) = 0 \iff \rho_{X,Y} = 0,$$

so that the condition

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \iff \rho_{X,Y} = 0 \iff X \text{ and } Y \text{ are uncorrelated.}$$

We bring this up now because, going back to

$$\mathrm{Var}(\mathcal{S}f(U)) = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2),$$

what we observe is that the "sum $X + Y$" is played by the role of $\mathcal{S}f(U)$, and the X and Y are played the a_n and b_n. What ties this all together is the notion that the Fourier trigonometric basis $\{\cos(nx), \sin(nx)\}_{n\in\mathbb{N}}$ is an **orthogonal basis** for the space of square-integrable functions $L^2[-\ell, \ell]$, and that orthogonality in this analytic sense is precisely equivalent to the statement that the basis functions are uncorrelated in the statistical sense. That is, the vanishing of the cross-term integrals

$$\frac{1}{2\ell}\int_{-\ell}^{\ell}\cos\left(\frac{n\pi t}{\ell}\right)\cos\left(\frac{m\pi t}{\ell}\right)\,dt = 0, \quad n \neq m,$$

is precisely equivalent to the statement that

$$\mathrm{Cov}_U\left(\cos\left(\frac{n\pi U}{\ell}\right), \cos\left(\frac{m\pi U}{\ell}\right)\right) = 0, \quad n \neq m,$$

and similarly for the sin-sin and cos-sin pairs.

Remark.

To verify this for the cos-cos pairs explicitly, we can use the covariance identity

$$\mathrm{Cov}(X, Y) := \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y; \qquad \mu_X = \mathbb{E}[X], \quad \mu_Y = \mathbb{E}[Y].$$

For

$$X = \cos\left(\frac{n\pi U}{\ell}\right), \quad Y = \cos\left(\frac{m\pi U}{\ell}\right), \quad n \neq m,$$

where

$$U \sim \mathrm{Uniform}[-\ell, \ell] \quad \text{with density} \quad p_U(t) = \frac{1}{2\ell}, \quad t \in [-\ell, \ell],$$

we have that

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0,$$

since the density p_U doesn't involve t, so the integral over a complete period vanishes: e.g.,

$$\mathbb{E}[X] = \frac{1}{2\ell}\int_{-\ell}^{\ell}\cos\left(\frac{n\pi t}{\ell}\right)\,dt = 0.$$

This renders the covariance identity

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - 0 = \mathbb{E}[XY],$$

which, since the source of randomness in X and Y comes from the common random variable U, gives us the desired equivalence,

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] = \frac{1}{2\ell}\int_{-\ell}^{\ell}\cos\left(\frac{n\pi t}{\ell}\right)\cos\left(\frac{m\pi t}{\ell}\right)\,dt = 0, \quad n \neq m,$$

(Observe that there's no joint density here, no two-dimensional integration, etc. The reason is, again, that U is the basis for X and Y, each of which is a deterministic function of U. When we imagine the procedure in practice, we'd begin by drawing $U = u$, then plug that in to obtain $X(u) = \cos(n\pi u/\ell) := x$ and $Y(u) = \cos(m\pi u/\ell) := y$.)

So the reason that $\mathrm{Var}(\mathcal{S}f(U))$ decomposes into a sum of individual contributions — one per harmonic with no interaction — is exactly the same reason that

$$\mathrm{Var}(X_1 + \cdots + X_N) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_N)$$

whenever the X_i are *mutually uncorrelated*. Each term $\frac{1}{2}(a_n^2 + b_n^2)$ is the variance contribution at frequency n — the amount of "surprise" relative to the mean $\mathbb{E}_U[f(U)]$ — that the nth harmonic resolves, and the orthogonality relations guarantee these contributions simply add with no interference.

So, again, what we have is that if $f \in L^2[-\ell, \ell]$ is square-integrable, then

$$\mathrm{Var}(\mathcal{S}f(U)) = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2),$$

where we've defined

$$\mathrm{Var}(\mathcal{S}f(U)) = \mathbb{E}_U\left[\Big(\mathcal{S}f(U) - \mathbb{E}_U[f(U)]\Big)^2\right] = \int_{-\ell}^{\ell}\Big(\mathcal{S}f(t) - \mathbb{E}_U[f(U)]\Big)^2\, dP_U(t).$$

Now, let us take a step back for a moment and consider, in general, what it *means* to "compute a variance." We've established a few equivalent formulas: in particular,

$$\mathrm{Var}(f(U)) = \mathbb{E}\left[\Big(f(U) - \mathbb{E}[f(U)]\Big)^2\right] = \mathbb{E}[f(U)^2] - (\mathbb{E}[f(U)])^2.$$

We'd like to justify our substituting

$$f(U) \quad \text{for} \quad \mathcal{S}f(U) \quad \text{in} \quad \mathrm{Var}(\mathcal{S}f(U)),$$

since, then, we'd be able to apply the formula we've been actively deriving and say

$$\mathrm{Var}(f(U)) \overset{?}{=} \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

What we need in order to justify this is for the integral

$$\frac{1}{2\ell}\int_{-\ell}^{\ell}|f(t) - S_N f(t)|^2\, dt \to 0 \quad \text{as} \quad N \to \infty,$$

which says that

$$S_N f \xrightarrow{L^2} f \quad \text{as} \quad N \to \infty.$$

This is justified whenever $f \in L^2[-\ell, \ell]$; in particular, *every* function of bounded variation on a bounded interval $[-\ell, \ell]$ is in $L^2[-\infty, \infty]$, and this has been our we *are* justified in our substituting $f(U)$ for its Fourier approximation $\mathcal{S}f(U)$ in the variance computation. What this gives is

$$\mathrm{Var}(f(U)) = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2).$$

Writing

$$\mathrm{Var}(f(U)) = \mathbb{E}[f(U)^2] - \underbrace{(\mathbb{E}[f(U)])^2}_{=a_0^2/4},$$

this says that

$$\mathbb{E}[f(U)^2] = \frac{a_0^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2).$$

This latter formula is most often expressed in the following way, where all we've done is rewritten the expectation on the left as an integral against the density of a Uniform$[-\ell, \ell]$ random variable (and trivially taken an absolute value, which is of greater import when we apply the formula for *complex*-valued functions, where the absolute value is replaced by the modulus).

Theorem 2.3.15 (Parseval's Theorem)

If $f \in L^2[-\ell, \ell]$ with Fourier coefficients a_0, a_n, b_n, then

$$\frac{1}{2\ell}\int_{-\ell}^{\ell} |f(t)|^2\, dt = \frac{a_0^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2).$$

2.3.3 Master Theorem

Fourier Master Theorem

We've outlined several different conditions useful in conducting Fourier analysis. We summarize these, and some other useful equivalent conditions, in the following theorem[a]

Theorem 2.3.16 (Dirichlet Master Theorem)

Let $f\colon [-\ell, \ell] \to \mathbb{R}$ be a function with 2ℓ-periodic extension, and denote by $\mathcal{S}f$ the Fourier series of the extension,

$$\mathcal{S}f(x) := \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{\ell}\right) + b_n \sin\left(\frac{n\pi x}{\ell}\right)\right),$$

where the Fourier coefficients are

$$a_0 = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\, dt, \quad a_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\cos\left(\frac{n\pi t}{\ell}\right)\, dt, \quad b_n = \frac{1}{\ell}\int_{-\ell}^{\ell} f(t)\sin\left(\frac{n\pi t}{\ell}\right)\, dt.$$

We denote by $S_N f$ the Nth Fourier partial sum

$$S_N f(x) := \frac{a_0}{2} + \sum_{n=1}^{N}\left(a_n \cos\frac{n\pi x}{\ell} + b_n \sin\frac{n\pi x}{\ell}\right),$$

and by $\sigma_N f$ the Nth Cesàro mean

$$\sigma_N f(x) := \frac{1}{N+1}\sum_{n=0}^{N} S_n f(x).$$

Then, we have the following results.

(a) Coefficient Behavior.

- **($f \in L^1$: Existence and decay)** The minimal assumption under which the Fourier coefficients a_n, b_n exist. Under this condition, the Riemann-Lebesgue lemma guarantees that

$$a_n \to 0 \quad \text{and} \quad b_n \to 0 \quad \text{as} \quad n \to \infty,$$

  but with *no control* over the rate of decay.

- **($f \in C^k$ with k-times continuously differentiable periodic extension: Decay rates)** Each additional continuous derivative buys us one additional power of decay:

$$a_n = O(n^{-k}) \quad \text{and} \quad b_n = O(n^{-k}).$$

  The smoother the function, the fewer terms needed for a good Fourier approximation.

(b) Pointwise Convergence of $S_N f$.

- **($f \in L^1$ and satisfies the Dini condition at x_0)** If the one-sided limits $f(x_0^+)$ and $f(x_0^-)$ exist and

$$\int_0^\delta \frac{|f(x_0 \pm t) - f(x_0^\pm)|}{t}\, dt < \infty \quad \text{for some} \quad \delta > 0, \qquad \text{(Dini condition)}$$

  then the Fourier partial sums converge pointwise at x_0:

$$S_N f(x_0) \to \frac{f(x_0^+) + f(x_0^-)}{2} \quad \text{as} \quad N \to \infty.$$

  This is a *local* condition delivering convergence at a *single* point.

- **(f of Bounded Variation: Dirichlet-Jordan)** If the *total variation* of f on $[-\ell, \ell]$,

$$V_{-\ell}^{\ell}(f) = \sup_{\mathcal{P}} \sum_{\mathcal{P}} |f(t_i) - f(t_{i-1})| < \infty, \qquad \mathcal{P} \text{ a partition of } [-\ell, \ell],$$

  then the sequence of Fourier partial sums converge pointwise on all of $[-\ell, \ell]$:

$$S_N f(x_0) \to \frac{f(x_0^+) + f(x_0^-)}{2}, \quad \forall x_0 \in [-\ell, \ell].$$

  This is a *global* condition delivering *global* pointwise convergence. However, convergence is *only* pointwise — the Gibbs phenomenon prevents uniform convergence near jump discontinuities.

- **(f piecewise smooth: Classical Dirichlet conditions)** If f and its first derivative f' are (i) piecewise continuous with (ii) finitely many jump discontinuities, then f is automatically of bounded variation, and the Dirichlet-Jordan theorem applies. This is often a quick check sufficing for most functions of interest; it is often presented as "the actor" in the Dirichlet-Jordan theorem, but we emphasize that it these "classical" Dirichlet conditions are merely vessels for ensuring the broader condition of Bounded Variation, which *is* the real actor.

(c) Cesàro Convergence of $\sigma_N f$.

- **($f \in L^1$ with one-sided limits at x_0)** The Cesàro means converge pointwise:

$$\sigma_N f(x_0) \to \frac{f(x_0^+) + f(x_0^-)}{2} \quad \text{as} \quad N \to \infty,$$

  with *no* Gibbs overshoot, and *no* Dini condition required. This is what averaging buys us: convergence under much weaker hypotheses. Kolmogorov (1923^b, 1926) constructed integrable functions whose Fourier partial sums $S_N f$ diverge almost everywhere, and later, everywhere. Yet the Cesàro means are immune to such pathologies: $\sigma_N f(x_0) \to f(x_0)$ whenever the one-sided limits exist at x_0. (We can equivalently say that $\sigma_N f \to f$ **almost everywhere** on $[-\ell, \ell]$ for *every* $f \in L^1$, without the stipulation on the one-sided limits — as it turns out, these one-sided limits exist almost everywhere in a precise sense due to Lebesgue, as we'll discuss in due course.)
- **(f continuous with continuous periodic extension: Fejér)** The Cesàro means converge **uniformly**:

$$\sigma_N f \rightrightarrows f \quad \text{on} \quad [-\ell, \ell].$$

  This is the strongest mode of convergence in the theorem, and it *only* requires f be continuous — no bounded variation, no Dini condition. We also emphasize that if all we're given is the continuity of f, then we're essentially *forced* into preferring the Cesàro means: du Bois-Reymond (1876) constructed a continuous function whose Fourier series diverged at a point. Again, Cesàro is immune: $\sigma_N f \rightrightarrows f$ on $[-\ell, \ell]$ for *every* f with continuous periodic extension.

(d) L^2 Convergence and Parseval.

- **($f \in L^2$: Mean-square convergence)** If f is square-integrable, so that

$$\int_{-\ell}^{\ell} |f(t)|^2 \, dt < \infty,$$

  then

$$S_N f \xrightarrow{L^2} f,$$

  meaning that

$$\|S_N - f\|_{L^2}^2 = \int_{-\ell}^{\ell} |S_N f(t) - f(t)|^2 \, dt \to 0 \quad \text{as} \quad N \to \infty.$$

  This is a genuinely different mode of convergence from both pointwise and uniform — neither implies the other.
- **($f \in L^2$: Parseval's equality)** Under the same condition, we have

$$\frac{1}{2\ell} \int_{-\ell}^{\ell} |f(t)|^2 \, dt = \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

  Equivalently, we can understand Parseval's equality from a statistical perspective: if

$$U \sim \text{Uniform}[-\ell, \ell] \qquad \text{with density} \qquad p_U(t) = \frac{1}{2\ell}, \quad t \in [-\ell, \ell],$$

then if $f \in L^2$, the variance of $f(U)$ under $U \sim \text{Uniform}[-\ell, \ell]$ exists, and is given by

$$\text{Var}_U(f(U)) = \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2).$$

We can recover the form we gave originally after noting that

$$\mathbb{E}_U[f(U)] = \frac{a_0}{2},$$

and applying the variance identity $\text{Var}(f(U)) = \mathbb{E}[f(U)^2] - (\mathbb{E}[f(U)])^2$

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[a]The theorem primarily serves as a reference; in particular, we mention notions like Hölder continuity which we've not formally defined. This is no cause for alarm — we'll never use a result without discussing its intuition, but we'd be remiss to not attempt to reconcile all of the seemingly competing frameworks for characterizing the behavior of Fourier series.

[b]https://people.math.osu.edu/sinnott.1/ReadingClassics/Kolmogorov1.pdf

2.4 Exercises

Exercise 1: Parseval and The Riemann Zeta Function

Parseval's equality is not merely a convergence theorem — it is a machine for evaluating infinite series in closed form. We've already seen hints of this: earlier in the chapter, the Fourier series of $|t|$ on $[-\pi, \pi]$ gave us the identity

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$$

seemingly for free. In this exercise, we continue our exploration by considering series of the form

$$\zeta(s) := \sum_{n=1}^{\infty}\frac{1}{n^s}, \quad s > 1.$$

We call $\zeta(\cdot)$ the **Riemann zeta function** — a deceptively simple function lying at the heart of perhaps the most famous open problem in mathematics, the *Riemann Hypothesis*. Our interest in ζ is more modest: we will use Parseval's equality to evaluate $\zeta(s)$ at $s = 2$ and $s = 4$, and derive several related identities that we will need in the exercises that follow.

(a) Show by comparison with $\int_1^{\infty} t^{-s}\,dt$ that

$$\zeta(s) \text{ converges for any } s > 1.$$

On the other hand, show that $\zeta(s)$ diverges at $s = 1$. (Hint: what's another name for the series $\zeta(1)$?)

(b) **(The Basel problem.)** Let

$$f(t) = t, \quad t \in [-\pi, \pi].$$

(i) Show that the Fourier coefficients of f are given by

$$a_0 = 0; \qquad a_n = 0, \quad n \geq 1; \qquad b_n = \frac{2(-1)^{n+1}}{n}, \quad n \geq 1.$$

(Hint: why is $a_n = 0$ for all n immediate by the symmetry of f?)

(ii) Apply Parseval's equality,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)|^2\,dt = \frac{a_0^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2),$$

to conclude that

$$\zeta(2) = \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}.$$

(This is the **Basel problem**: first posed in 1650, and solved by Euler in 1734. The problem resisted the efforts of Leibniz, the Bernoulli brothers, and virtually every major mathematician of the era for nearly a century. Euler's solution made him famous overnight; we have just reproduced it in a few lines.)

(c) (The quartic sum.) Now, let

$$g(t) = t^2, \quad t \in [-\pi, \pi].$$

(i) Compute the Fourier coefficients for g — in particular, show that

$$a_0 = \frac{2\pi^2}{3}; \qquad a_n = \frac{4(-1)^n}{n^2}, \quad n \geq 1; \qquad b_n = 0, \quad n \geq 1.$$

(Hint: why does $b_n = 0$ follow by the symmetry of g? For the cosine coefficients, integrate by parts twice or apply the tabular DI method.)

(ii) Apply Parseval's equality to conclude that

$$\zeta(4) = \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}.$$

(d) (Odd-index identities.) Observe that for any $s > 1$,

$$\zeta(s) = \sum_{\substack{n=1\\ n \text{ odd}}}^{\infty}\frac{1}{n^s} + \sum_{\substack{n=1\\ n \text{ even}}}^{\infty}\frac{1}{n^s}.$$

(i) Show that the even-index sum satisfies

$$\sum_{\substack{n=1\\ n \text{ even}}}^{\infty}\frac{1}{n^s} = \frac{1}{2^s}\zeta(s),$$

and solve for the odd-index sum to obtain

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^s} = \left(1 - \frac{1}{2^s}\right)\zeta(s).$$

(ii) Apply this at $s = 2$ and $s = 4$ to derive

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}, \qquad \sum_{k=0}^{\infty}\frac{1}{(2k+1)^4} = \frac{\pi^4}{96}.$$

Verify the first identity against the result we obtained earlier in the chapter from the Fourier series of $|t|$.

(e) (The general pattern.) The arguments of parts (b) and (c) generalize: applying Parseval to t^{2k-1} or t^{2k} on $[-\pi, \pi]$ produces a closed-form evaluation of $\zeta(2k)$ for every positive integer k. One can show that

$$\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k}}{2(2k)!} B_{2k},$$

where B_{2k} are the **Bernoulli numbers**. Every value $\zeta(2), \zeta(4), \zeta(6), \ldots$ is a rational multiple of the corresponding power of π.

(i) Examine the mechanism: $f(t) = t$ has coefficients $b_n \sim 1/n$; Parseval squares them to produce $\zeta(2)$. The function $g(t) = t^2$ has $a_n \sim 1/n^2$; squaring gives $\zeta(4)$. Explain why the squaring inherent in Parseval — a statement about $\|f\|_U^2$ — means that functions with integer-rate coefficient decay can only produce *even* zeta values. What decay rate would you need to produce $\zeta(3)$?

(ii) Suppose that, according to the observation at the last part, we defined the function

$$h(t) := \sum_{n=1}^{\infty} \frac{\sin(nt)}{n^{3/2}}.$$

Verify that Parseval applied to h yields

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |h(t)|^2 \, dt = \frac{1}{2} \zeta(3).$$

Explain why this is useless: h is *defined* as a Fourier series, so we have no independent expression for $|h(t)|^2$ that would let us evaluate the left-hand side.

The odd zeta values appear to be of a fundamentally different character. The irrationality of $\zeta(3)$ (**Apéry's constant**) was proved only in 1978; whether it is transcendental, whether it is algebraically independent of π, and whether *any* odd zeta value admits a closed form remain open.

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Exercise 2: The Wirtinger Inequality

In Exercise 1, we used Parseval's Theorem to evaluate infinite series. Here, we continue to explore its applications by deriving a fundamental inequality due to Wirtinger. Here, recall that we define the space $C^1[a, b]$ to contain the continuously differentiable functions on $[a, b]$ — i.e., for a differentiable function $f\colon [a, b] \to \mathbb{R}$, we have that

$$f \in C^1[a, b] \iff f \text{ and } f' \text{ are continuous everywhere on } [a, b].$$

(a) Let $f \in C^1[-\pi, \pi]$ with 2π-periodic extension and Fourier coefficients a_n, b_n. Show that the Fourier coefficients of the derivative f' are

$$a_n' = n b_n, \qquad b_n' = -n a_n, \qquad n \geq 1.$$

(Hint: integrate by parts in the defining integrals for the Fourier coefficients of f'.)

(b) Suppose that the constant coefficient in $S_N f$ is $a_0 = 0$, so that if $U \sim \text{Uniform}[-\pi, \pi]$, then

$$\mathbb{E}_U[f(U)] = 0.$$

(Recall: why are these conditions equivalent?) Apply Parseval's equality to both f and f' to show that

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f'(t)|^2\,dt = \frac{1}{2}\sum_{n=1}^{\infty} n^2(a_n^2+b_n^2),$$

and conclude the **Wirtinger inequality**:

$$\int_{-\pi}^{\pi} |f'(t)|^2\,dt \geq \int_{-\pi}^{\pi} |f(t)|^2\,dt.$$

When does Wirtinger hold with *equality*? Characterize the extremal functions completely.

(c) Reinterpret the Wirtinger inequality in the statistical language of this chapter. In particular, argue that it says: If f has zero mean under uniform sampling, then the variance of $f'(U)$ is at least as large as the variance of $f(U)$. Why should this be intuitive? (Hint: what does differentiation do to *oscillations*?)

Exercise 3: Spectral Variance

Let $f \in L^2[-\ell,\ell]$ be non-constant, with Fourier partial sums $S_N f$ and Fourier coefficients a_n, b_n, and suppose that

$$U \sim \text{Uniform}[-\ell,\ell].$$

(a) Show that

$$\langle g,h\rangle_U = \int_{-\ell}^{\ell} g(t)h(t)\,dP_U(t) = \frac{1}{2\ell}\int_{-\ell}^{\ell} g(t)h(t)\,dt$$

defines a valid inner product on the real function space $L^2[-\ell,\ell]$ by showing it satisfies:

(i) **(Nonnegative Definiteness)** If $g \in L^2[-\ell,\ell]$, then

$$\langle g,g\rangle_U \geq 0, \quad \text{and} \quad \langle g,g\rangle_U = 0 \iff g = \mathbf{0} \text{ on } [-\ell,\ell].$$

(ii) **(Homogeneity)** For any $\alpha \in \mathbb{R}$ and any $g,h \in L^2[-\ell,\ell]$,

$$\langle \alpha g,h\rangle_U = \alpha\langle g,h\rangle_U.$$

(iii) **(Bilinearity)** If $g_1, g_2, h \in L^2[-\ell,\ell]$,

$$\langle g_1+g_2,h\rangle_U = \langle g_1,h\rangle_U + \langle g_2,h\rangle_U$$

(b) Under this uniform inner product $\langle\cdot,\cdot\rangle_U$ on $L^2[-\ell,\ell]$, show that

$$\langle f - S_N f, S_N f\rangle_U = 0,$$

(Hint: it suffices to show that $\langle f - S_N f, \phi_n\rangle = 0$ for each basis function $\phi_n \in \{1, \cos(n\pi t/\ell), \sin(n\pi t/\ell)\}_{n=1}^N$ appearing in $S_N f$. Why does this follow from directly from the definition of the Fourier coefficients a_n and b_n?)

(c) Using the decomposition

$$f = S_N f + (f - S_N f),$$

apply the orthogonality shown in part (b) to establish **Bessel's Inequality**:

$$\|S_N f\|_U^2 \leq \|f\|_U^2,$$

where we've defined the uniform norm on $L^2[-\ell,\ell]$ to be

$$\|g\|_U^2 = \langle g,g\rangle_U, \quad g \in L^2[-\ell,\ell].$$

(d) Expand both sides of Bessel's inequality in terms of Fourier coefficients to obtain

$$\frac{a_0^2}{4} + \frac{1}{2}\sum_{n=1}^{N}(a_n^2 + b_n^2) \le \frac{1}{2\ell}\int_{-\ell}^{\ell} |f(t)|^2\, dt,$$

and use Parseval's equality to conclude that the **spectral coefficient of determination**,

$$R_N^2(f) := \frac{\sum_{n=1}^{N}(a_n^2 + b_n^2)}{\sum_{n=1}^{\infty}(a_n^2 + b_n^2)},$$

satisfies

$$R_N^2(f) \in [0, 1].$$

Why is the assumption that f be non-constant necessary for $R_N^2(f)$ to be well-defined? (Hint: consider the case when f is constant, so that $f(t) = \mathbb{E}_U[f(U)]$ for all $t \in [-\ell, \ell]$.) When is $R_N^2 = 0$?

(e) Conclude from Parseval's theorem that

$$R_N^2(f) \nearrow 1 \quad \text{as} \quad N \to \infty.$$

Exercise 4: Rates of Spectral Concentration

We now compute the spectral variance derived in Exercise 3,

$$R_N^2(f) := \frac{\sum_{n=1}^{N}(a_n^2 + b_n^2)}{\sum_{n=1}^{\infty}(a_n^2 + b_n^2)}$$

for three functions in $L^2[-\pi, \pi]$, two of whose Fourier coefficients we derived earlier in the chapter, and examine how their smoothness controls the rate at which $R_N^2 \to 1$ — in particular, the smoother the function, the fewer Fourier coefficients we need to compute to obtain a reasonable approximation.

(a) **(Square wave)** Let f be the square-wave on $[-\pi, \pi]$:

$$f(x) = \operatorname{sgn}(x) = \begin{cases} -1 & -\pi \le x < 0 \\ 0 & x = 0 \\ 1 & 0 < x \le \pi \end{cases}$$

Note that f is discontinuous at $x = 0$: $f \notin C[-\pi, \pi]$.
Earlier, we computed the Fourier coefficients of f to be

$$a_0 = 0, \qquad a_n = 0 \quad \forall n \ge 1, \qquad b_n = \begin{cases} \dfrac{4}{n\pi} & n \text{ odd}, \\ 0 & n \text{ even}. \end{cases}$$

Using the identity

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8},$$

(which we derived from the Fourier series of $|t|$ earlier in the chapter), show that

$$\sum_{n=1}^{\infty}(a_n^2 + b_n^2) = \frac{16}{\pi^2}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = 2.$$

Verify this against a direct computation: $f^2(t) = 1$ for all $t \neq 0$ on $[-\pi, \pi]$, so

$$\mathbb{E}_U[f(U)^2] = 1;$$

f is odd, so

$$\mathbb{E}_U[f(U)] = 0;$$

hence, we conclude

$$\mathrm{Var}_U(f(U)) = 1 - 0^2 = 1 = \frac{1}{2} \cdot 2 = \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

(b) Show that the spectral R_N^2 of the square wave is

$$R_N^2(f) = \frac{8}{\pi^2} \sum_{\substack{k=1 \\ k \text{ odd}}}^{N} \frac{1}{k^2},$$

and that the *unexplained* spectral variance (everything unaccounted for after taking the first N terms of the Fourier approximation) satisfies

$$1 - R_N^2 = \frac{8}{\pi^2} \sum_{\substack{k>N \\ k \text{ odd}}} \frac{1}{k^2} \sim \frac{4}{\pi^2 N} \quad \text{as} \quad N \to \infty.$$

(c) **(Triangle wave)** Now, let g be the triangle wave

$$g(x) = |x|, \quad x \in [-\pi, \pi].$$

Confirm that g is continuous on $[-\pi, \pi]$ but that its first derivative g' has a discontinuity at $x = 0$. Conclude that $g \in C[-\pi, \pi]$, but that $g \notin C^1[-\pi, \pi]$.

(d) Verify that

$$b_n = 0 \quad \forall n \geq 1, \qquad a_0 = \pi, \qquad a_n = \begin{cases} -\dfrac{4}{n^2 \pi} & n \text{ odd}, \\ 0 & n \text{ even}, \end{cases}$$

and, using the identity established in exercise 1, prove that

$$\sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{16}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^4} = \frac{\pi^2}{6}.$$

Verify this against a direct computation of $\mathrm{Var}_U(g(U))$: since $g(t) = |t|$ on $[-\pi, \pi]$,

$$\mathbb{E}_U[g(U)] = \frac{a_0}{2} = \frac{\pi}{2}, \quad \mathbb{E}_U[g(U)^2] = \frac{1}{2\pi} \int_{-\pi}^{\pi} t^2 \, dt = \frac{\pi^2}{3},$$

so that

$$\mathrm{Var}_U(g(U)) = \frac{\pi^2}{3} - \left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{12} = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

(e) Show that the spectral R_N^2 of the triangle wave g is

$$R_N^2(g) = \frac{96}{\pi^4} \sum_{\substack{k=1 \\ k \text{ odd}}}^{N} \frac{1}{k^4},$$

and that the unexplained spectral variance satisfies

$$1 - R_N^2(g) = \frac{96}{\pi^4} \sum_{\substack{k>N \\ k \text{ odd}}} \frac{1}{k^4} \sim \frac{16}{\pi^4 N^3} \quad \text{as} \quad N \to \infty.$$

(Hint: compare the tail sum with $\int_N^\infty t^{-4}\,dt$.)

(f) **(Parabolic arch)** Finally, consider

$$h(t) := t(\pi - |t|), \qquad t \in [-\pi, \pi],$$

which is equivalent to

$$h(t) = \begin{cases} t(\pi + t) & -\pi \le t \le 0, \\ t(\pi - t) & 0 \le t \le \pi. \end{cases}$$

Confirm that h is continuous on $[-\pi, \pi]$, that its first derivative h' is continuous on $[-\pi, \pi]$, but that its second derivative h'' has a discontinuity at $t = 0$. In particular, conclude that $h \in C^1[-\pi, \pi]$ but that $h \notin C^2[-\pi, \pi]$.

(g) Based on the pattern established in parts (b) and (e) — the square wave $f \notin C[-\pi, \pi]$ is discontinuous with $1 - R_N^2(f) \sim 1/N$, and the triangle wave $g \in C[-\pi, \pi]$ but $g \notin C^1[-\pi, \pi]$ with $1 - R_N^2(g) \sim 1/N^3$ — predict the rate of decay $1 - R_N^2(h)$ of the parabolic arch before computing it explicitly.

(h) Verify your prediction. In particular, since h is odd, $a_n = 0$ for all $n \ge 0$, and compute

$$b_n = \begin{cases} \dfrac{8}{\pi n^3} & n \text{ odd}, \\ 0 & n \text{ even}. \end{cases}$$

(Hint: h is odd, so simplify to $b_n = \frac{2}{\pi} \int_0^\pi t(\pi - t)\sin(nt)\,dt$ and apply the tabular DI method. Alternatively, try differentiating the Fourier series of the triangle wave term by term and applying the coefficient formulas from Exercise 2.)

(i) Show that for the parabolic arch h,

$$\sum_{n=1}^\infty b_n^2 = \frac{64}{\pi^2} \sum_{k=0}^\infty \frac{1}{(2k+1)^6},$$

and conclude that

$$1 - R_N^2(h) \sim \frac{C}{N^5},$$

for an explicit constant C. In particular, this means that $1 - R_N^2(h) = O(N^{-5})$.

(j) Compare the three rates of spectral concentration derived for the square wave f, the triangle wave g, and the parabolic arch h. Observe that the square wave is discontinuous — its coefficients decay like $1/n$, and the unexplained spectral variance decays like $1/N$. The triangle wave is continuous (but has corners, so it is not differentiable); its coefficients decay like $1/n^2$, and its unexplained variance decays like $1/N^3$. In both cases, we observe that coefficients decaying as $O(n^{-\alpha})$ produce an unexplained variance decaying like $O(N^{1-2\alpha})$. Extend this observation to the parabolic arch h, and explain why this pattern holds in general. (Hint: if $|c_n| = O(n^{-\alpha})$ so that $|c_n| \sim Cn^{-\alpha}$ for some constant C, what is the asymptotic behavior of the tail sum $\sum_{n>N} n^{-2\alpha}$) Conclude that each additional derivative of f — which, by the results of this chapter, buys one extra power of $1/n$ in the coefficient decay rate — gets us *two* extra powers of $1/N$ in the rate at which $R_N^2 \to 1$.

For each of the three functions, the coefficient decay rate α counts the *order of the first missing derivative*: the square wave fails at continuity itself ($\alpha = 1$), the triangle wave is continuous but breaks at the first derivative ($\alpha = 2$), and the parabolic arch is C^1 but breaks at the second derivative ($\alpha = 3$). The spectral concentration rate is then $1/N^{1-2\alpha}$ where the exponent $1 - 2\alpha = -1, -3, -5, \dots,$ — we imagine that $\alpha = 1$ corresponds with a function of bounded variation that isn't continuous, $\alpha = 2$ with a continuous function which is non-differentiable, $\alpha = 3$ with a once-differentiable function whose second derivative doesn't exist, and so on. This is the same parity constraint we encountered in Exercise 1: the reason Parseval could evaluate $\zeta(2)$ and $\zeta(4)$ but not $\zeta(3)$ was that the squaring inherent to $|f|^2$ turns integer decay rates into even powers. Here, the same squaring mechanism turns integer decay rates into odd spectral exponents. Both gaps would be filled by functions of *fractional regularity* — i.e., those with coefficients decaying like $O(n^{-3/2}), O(n^{-5/2}), \dots, O(n^{-(2k+1)/2})$. Such spaces exist — they're **Sobolev spaces** — and we'll return to their constructions in due course.

	$f = \operatorname{sgn}(x)$	$g = \lvert x\rvert$	$h = x(\pi - \lvert x\rvert)$
Regularity	BV, not C^0	C^0, not C^1	C^1, not C^2
Coefficient decay	$\sim \frac{1}{n}$	$\sim \frac{1}{n^2}$	$\sim \frac{1}{n^3}$
$1 - R_N^2$	$\sim \frac{1}{N}$	$\sim \frac{1}{N^3}$	$\sim \frac{1}{N^5}$